Linear Algebra I & II

Lecture Notes

Professor: Prof. Dr. Paul Biran

Author: Filip Jaskovic

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Chapter 1

Foundations

Lecture 4: Grundlagen

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1.1 Sets and Functions

1.1.1 Cardinality

Definition (Cardinality)**.** The CARDINALITY of a finite set X is the number of elements in X. For example, if $X = \{x_1, \ldots, x_n\}$, then $|X| = n$.

Exercise

Let X be a finite set. Show that

$$|\mathcal{P}(X)| = 2^{|X|}.$$

1.1.2 Functions: basic notions

4 **Definition** (Functions)**.** A function is written

$$f : X \longrightarrow Y,$$

where X is called the DOMAIN of f, and Y is called the TARGET of f.

Two functions

$$f : X \longrightarrow Y, \qquad g : X' \longrightarrow Y'$$

are equal if and only if $X = X'$, $Y = Y'$, and for all $x \in X$ we have $f(x) = g(x)$.

Example.
- $f(x) = x^2$.
- $X := \{x \in \mathbb{Z} \mid x \geq 0\}$.
- $Y := \{\text{all family names}\}$.

- $g : X \longrightarrow Y, \quad g(x) :=$ family name of the student at ETH whose ID number is x.

Here g is *not well-defined*: e.g. $g(0) = ?$, $g(10^{10}) = ?$.

But if we extend the target to

$$Y' := \{\text{family names}\} \cup \{\text{Error}\},$$

then $g : X \to Y'$ becomes well-defined.

Definition (Image). Let $f : X \longrightarrow Y$. The IMAGE of f is the set

$$\operatorname{im} f := \{\, y \in Y \mid \exists x \in X \text{ s.t. } f(x) = y \,\}.$$

Equivalently,

$$\operatorname{im}(f) = f(X) \subseteq Y.$$

Definition (Restriction). Let $f : X \to Y$ and let $A \subseteq X$. We define a new map $f|_A : A \to Y$, called the RESTRICTION of f to A, by

$$(f|_A)(x) := f(x) \quad \text{for } x \in A.$$

Example.
- The IDENTITY MAP: $\mathrm{id} : X \longrightarrow X$, $\mathrm{id}(x) := x$ for all $x \in X$.
- A CONSTANT FUNCTION: given $y \in Y$, define $f : X \to Y$, $f(x) := y$ for all
4 $x \in X$.
- The CHARACTERISTIC FUNCTION of $A \subseteq X$:

$$1_A : X \to \mathbb{R}, \qquad 1_A(x) = \begin{cases} 1 & x \in A, \\ 0 & x \notin A. \end{cases}$$

1.1.3 Injective, Surjective, Bijective Maps

Definition (Injective, Surjective, Bijective Maps). Let $f : X \longrightarrow Y$.

(1) f is called INJECTIVE (or an INJECTION) if

$$\forall x_1, x_2 \in X : \; f(x_1) = f(x_2) \;\Rightarrow\; x_1 = x_2.$$

(2) f is called SURJECTIVE (or an SURJECTION) if

$$\forall y \in Y \; \exists x \in X : \; f(x) = y.$$

Equivalently, $f(X) = Y$.

(3) f is called BIJECTIVE (or a BIJECTION) if it is both injective and surjective. Equivalently,

$$\forall y \in Y \; \exists!\, x \in X : \; f(x) = y.$$

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Example. Consider
$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$
This map is not injective and not surjective.

However,
$$f : \mathbb{R}_{\geq 0} \to \mathbb{R}, \qquad f(x) = x^2,$$
is injective, but not surjective.

1.1.4 Inverse Map

Definition (Inverse Map)**.** Let $f : X \longrightarrow Y$ be a bijection. The INVERSE MAP of f is the map
$$f^{-1} : Y \longrightarrow X,$$
which assigns to each $y \in Y$ the unique $x \in X$ such that $f(x) = y$.

1.1.5 Composition of Maps

Definition (Composition)**.** Let $f : X \longrightarrow Y$, $g : Y \longrightarrow Z$. The COMPOSITION $g \circ f : X \longrightarrow Z$ is defined by
$$(g \circ f)(x) := g(f(x)).$$
Diagrammatically,
$$X \xrightarrow{f} Y \xrightarrow{g} Z, \qquad X \xrightarrow{g \circ f} Z.$$

Example. We have
$$f \circ f^{-1} = \mathrm{id}_Y, \qquad f^{-1} \circ f = \mathrm{id}_X\,.$$
The order of composition matters: $f \circ g$ and $g \circ f$ are not the same map. It can even happen that $g \circ f$ is defined but $f \circ g$ is not.

Example. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x + 1$, and $g : \mathbb{R} \to \mathbb{R}$, $g(x) = x^2$. Then
$$(g \circ f)(x) = g(x+1) = (x+1)^2 = x^2 + 2x + 1,$$
$$(f \circ g)(x) = f(x^2) = (x^2) + 1 = x^2 + 1.$$
Thus $g \circ f \neq f \circ g$.

Example. The map
$$f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}, \qquad f(x) = x^2,$$
is bijective.

Lemma. Let $f : X \longrightarrow Y$, $g : Y \longrightarrow Z$.

(1) If f and g are injective, then $g \circ f$ is injective.

(2) If f and g are surjective, then $g \circ f$ is surjective.

(3) If f and g are bijective, then $g \circ f$ is bijective.

Proof. (1) We need to prove that for all $x_1, x_2 \in X$, if

$$(g \circ f)(x_1) = (g \circ f)(x_2),$$

then $x_1 = x_2$. So let $x_1, x_2 \in X$ with $(g \circ f)(x_1) = (g \circ f)(x_2)$. Since g is injective, this implies $f(x_1) = f(x_2)$. Since f is injective, we then get $x_1 = x_2$, as required.

(2) We need to show that for all $z \in Z$, there exists $x \in X$ with $(g \circ f)(x) = z$. So let $z \in Z$. Since g is surjective, there exists $y \in Y$ such that $g(y) = z$. Since f is surjective, there exists $x \in X$ such that $f(x) = y$. Then

$$(g \circ f)(x) = g(f(x)) = g(y) = z,$$

as desired.

(3) If both f and g are bijective, then they are both injective and surjective. By (1) and (2), it follows that $g \circ f$ is both injective and surjective, hence bijective. □

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1.1.6 Image and Inverse Image

Definition (Image of a Subset). Let $f : X \longrightarrow Y$ and $A \subseteq X$. The IMAGE of A under f is

$$f(A) := \{\, y \in Y \mid \exists x \in A \text{ s.t. } f(x) = y \,\}.$$

Equivalently,

$$f(A) = \{\, f(x) \mid x \in A \,\}.$$

Definition (Inverse Image). Let $f : X \longrightarrow Y$ and $B \subseteq Y$. The INVERSE IMAGE of B under f is

$$f^{-1}(B) := \{\, x \in X \mid f(x) \in B \,\} \subseteq X.$$

Example. For $f : \mathbb{R} \to \mathbb{R},\ f(x) = x^2$,

$$\begin{aligned} f^{-1}(\{4\}) &= \{-2, 2\}, & f^{-1}(\varnothing) &= \varnothing, \\ f^{-1}(\{1, 9\}) &= \{-3, -1, 1, 3\}, & f^{-1}(\{-1\}) &= \varnothing, \\ f^{-1}(\{0\}) &= \{0\}, & f^{-1}(\{-1, 0\}) &= \{0\}. \end{aligned}$$

Chapter 2

Vector Spaces

Lecture 5: beginning of Linear Algebra

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2.1 Graph of a Function

2.2 Bases

Definition. Let $f\colon X \longrightarrow Y$ be a map. The graph of f is the subset $\mathrm{Graph}(f) \subseteq X \times Y$ defined by

$$\mathrm{Graph}(f) \coloneqq \{(x, y) \in X \times Y \mid y = f(x)\}.$$

2.3 Fields

Definition (Field)**.** A field is a tuple $(K, +, \cdot, 0, 1)$ consisting of a set K with two maps

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$$+\colon K \times K \to K, \quad (x, y) \mapsto x + y, \qquad \cdot\colon K \times K \to K, \quad (x, y) \mapsto x \cdot y,$$

and distinguished elements $0, 1 \in K$, such that the following field axioms hold:

Additive structure

$$\begin{aligned}
&\forall x, y, z \in K: && x + (y + z) = (x + y) + z && \text{(associativity of addition)} && \text{(K1)}\\
&\forall x, y \in K: && x + y = y + x && \text{(commutativity of addition)} && \text{(K2)}\\
&\forall x \in K: && 0 + x = x && \text{(additive identity)} && \text{(K3)}\\
&\forall x \in K: && \exists\,(-x) \in K:\ x + (-x) = 0 && \text{(additive inverse)} && \text{(K4)}
\end{aligned}$$

Multiplicative structure

$$\begin{array}{lll} \forall x,y,z \in K: & x\cdot(y\cdot z) = (x\cdot y)\cdot z & \text{(associativity of multiplication)} \quad \text{(K5)} \\ \forall x \in K: & 1\cdot x = x & \text{(multiplicative identity)} \quad \text{(K6)} \\ \forall x \in K\setminus\{0\}: & \exists\, x^{-1}\in K:\ x\cdot x^{-1} = 1 & \text{(multiplicative inverse)} \quad \text{(K7)} \end{array}$$

Linking laws and nontriviality

$$\begin{array}{lll} \forall x,y,z \in K: & x\cdot(y+z) = x\cdot y + x\cdot z, & \text{(distributivity)} \quad \text{(K8)} \\ & (y+z)\cdot x = y\cdot x + z\cdot x & \\ & 1 \neq 0 & \text{(nontriviality)} \quad \text{(K9)} \\ \forall x,y \in K: & x\cdot y = y\cdot x & \text{(commutativity of multiplication)} \quad \text{(K10)} \end{array}$$

Example. $\mathbb{Q} = \{a | a = \frac{m}{n}, m,n \in \mathbb{Z}, n \neq 0\}$
$1 = 1/1, \quad 0 = 0/1, \quad +,\ \times$ are the usual definitions. $\mathbb{R}$ =the real numbers.

Example. $\mathbb{C} = \{z \mid z = a + ib,\ a,b \in \mathbb{R},\ i^2 = -1\}$.

Example. $\mathbb{F}_2 = \{0,1\}$ with $1+1=0$.

Example. $F_3 = \{0,1,2\}$ operations modulo 3.

5 **Example.** $\mathbb{Z} \longrightarrow$ not a field.
Let $x = 2 \in \mathbb{Z}$. Does there exist $x' \in \mathbb{Z}$ such that $2x' = 1$? NO!

2.4 Systems of Linear Equations

We start with the system

$$\begin{cases} x + 3y + 4z = -1 \\ 2x - y + z = 5 \end{cases}.$$

2.4.1 Row reduction

$$\begin{cases} x + 3y + 4z = -1 \\ 2x - y + z = 5 \end{cases} \xrightarrow{-2E_1+E_2 \ \to\ E_2} \begin{cases} x + 3y + 4z = -1 \\ -7y - 7z = 7 \end{cases}$$

$$\xrightarrow{-\frac{1}{7}E_2 \ \to\ E_2} \begin{cases} x + 3y + 4z = -1 \\ y + z = -1 \end{cases} \xrightarrow{-3E_2+E_1 \ \to\ E_1} \begin{cases} x + z = 2 \\ y + z = -1 \end{cases}$$

2.4.2 Parametric form of the solution

Set of all solutions (x,y,z) : choose z arbitrarily, $z = t \in \mathbb{R}, \quad y = -1-t, \quad x = 2-t$.

$$\{(x,y,z) \in \mathbb{R}^3 \mid z = t \in \mathbb{R},\ y = -1-t,\ x = 2-t\}.$$

Definition (Systems of linear equations and matrix notation). Fix a field F. A system of m equations in n unknowns over F:
Unknowns: $x_1, x_2, \ldots, x_n$. Coefficients $a_{ij} \in F$, right-hand sides $b_i \in F$.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n = b_m \end{cases} .$$

a_{ij} — coefficient of x_j in equation i.
$1 \le i \le m$. $1 \le j \le n$.
b_i — right-hand side of equation i.

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \ldots & a_{1n} & b_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & b_m \end{array}\right] \quad \text{(augmented) } m \times (n+1) \text{ matrix over } F.$$

Sometimes we write

$$A = (a_{ij})_{1 \le i \le m,\ 1 \le j \le n} \in \mathrm{M}_{m \times n}(F).$$

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$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

$$S\colon \quad A\mathbf{x} = \mathbf{b}.$$

$$L(S) = \{\mathbf{x} \in F^n \mid A\mathbf{x} = \mathbf{b}\} = \left\{ (x_1, \ldots, x_n) \in F^n \;\middle|\; A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{b} \right\}.$$

Matrix–vector product via row sums

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_i \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^{n} a_{1j}x_j \\ \vdots \\ \sum_{j=1}^{n} a_{ij}x_j \\ \vdots \\ \sum_{j=1}^{n} a_{mj}x_j \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ \vdots \\ a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}.$$

Component-wise formula

For $1 \le i \le m$,

$$(A\mathbf{x})_i = \sum_{j=1}^{n} a_{ij}\, x_j \quad \Longrightarrow \quad A\mathbf{x} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}.$$

Example.

$$\begin{cases} x + 3y + 4z = -1 \\ 2x - y + z = 5 \end{cases} \quad <=> \begin{cases} x_1 + 3x_2 + 4x_3 = -1 \\ 2x_1 - x_2 + x_3 = 5 \end{cases}.$$

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$$A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & -1 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}.$$

$$\text{Extended matrix:} \qquad \left[\begin{array}{c|c} A & b \end{array}\right]$$

$$\text{Exp.} \quad \begin{cases} 2x_1 - x_2 + 3x_3 + 2x_4 = 0, \\ x_1 + 4x_2 - x_4 = 3, \\ 2x_1 + 6x_2 - x_3 + 5x_4 = 9. \end{cases}$$

2.4.3 Elementary row operations

- choose $0 \neq c \in F$;multiply row i by c $(c \times R_i \longrightarrow R_i)$
- choose $c \in F$, choose $1 \le ij \le m$

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2.5 Elementary Row Operations and Row Equivalence

(1) choose $0 \neq c \in F$, multiply eq./row number i by c. $(c \times R_i \longrightarrow R_i)$

(2) choose $c \in F$, choose $1 \le i, j \le m,\ i \neq j$, replace eq./row number j by eq./row number
6 $j + c$ times eq./row number i. $(R_j + c \times R_i \longrightarrow R_j)$

(3) Interchange eq./row number i and eq./row number j. $(R_i \longleftrightarrow R_j)$

Definition. Let C and C' be $r \times s$ matrices with entries in F. We say that C' is *row equivalent* to C if C' can be obtained from C by a finite sequence of elementary row operations.

$$C = C_0 \longrightarrow C_1 \longrightarrow C_2 \longrightarrow \ldots \longrightarrow C_k = C'$$

Theorem. Let $S\colon A\mathbf{x} = \mathbf{b}$ and $S'\colon A'\mathbf{x} = \mathbf{b}'$ be two systems of m linear equations in n unknowns over F. Suppose that the augmented matrices $[A \mid b]$ and $[A' \mid b']$ are row equivalent. Then $L(S') = L(S)$.

Proof. We begin with the following auxiliary statements.

Claim. If S' is obtained from S by one elementary row operation, then $L(S) \subseteq L(S')$.

Proof of claim. Let $(x_1, \ldots, x_n) \in L(S)$. We need to show that $(x_1, \ldots, x_n) \in L(S')$. Consider the three types of elementary operations:

(1) $c \times R_i \to R_i$ with $c \neq 0$. In this case, all equations of S' coincide with those of S except equation $\#i$. If

$$a_{i1}x_1 + a_{i2}x_2 + \ldots + a_{in}x_n = b_i,$$

then multiplying both sides by c gives

$$ca_{i1}x_1 + ca_{i2}x_2 + \ldots + ca_{in}x_n = cb_i.$$

Thus $(x_1, \ldots, x_n)$ satisfies equation $\#i$ of S' as well (since $c \neq 0$).

(2) $R_j + c \times R_i \to R_j$ with $i \neq j$. Again, all equations coincide except equation $\#j$. From

$$a_{j1}x_1 + \ldots + a_{jn}x_n = b_j$$

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and $c(a_{i1}x_1 + \ldots + a_{in}x_n) = cb_i$, we obtain

$$a_{j1}x_1 + \ldots + a_{jn}x_n + c(a_{i1}x_1 + \ldots + a_{in}x_n) = b_j + cb_i,$$

so $(x_1, \ldots, x_n)$ satisfies equation $\#j$ of S'.

(3) $R_i \leftrightarrow R_j$. Trivially, any solution of S is a solution of S'.

Claim. If S' is obtained from S by one elementary row operation, then S can be obtained from S' by one elementary row operation.

Proof of claim. (1) If $c \times R_i \to R_i$ transforms S to S', then $(1/c) \times R_i \to R_i$ transforms S' back to S.

(2) If $R_j + c \times R_i \to R_j$ (with $i \neq j$) transforms S to S', then $R_j - c \times R_i \to R_j$ transforms S' back to S (row $\#i$ is unchanged).

(3) If $R_i \leftrightarrow R_j$ transforms S to S', then the same swap transforms S' back to S.

Claim. If S' is obtained from S by one elementary row operation, then $L(S) = L(S')$.

Proof of claim. By Claim 1, $L(S) \subseteq L(S')$. By Claim 2, S can be obtained from S' by one elementary row operation; applying Claim 1 with the roles of S and S' interchanged yields $L(S') \subseteq L(S)$. Hence $L(S) = L(S')$.

We are now in a position to prove the theorem. By assumption, there is a finite sequence of elementary row operations

$$S = S_0 \longrightarrow S_1 \longrightarrow \ldots \longrightarrow S_k = S'.$$

By Claim 3, we have

$$L(S) = L(S_0) = L(S_1) = \ldots = L(S_{k-1}) = L(S_k) = L(S').$$

$\square$

Definition. A $m \times n$ matrix A is called ROW REDUCED if the following two conditions hold:

1. The first non-zero entry in each non-zero row of A is 1 (the leading entry or pivot).
2. Each column of A which contains the leading non-zero entry of some row has all its other entries equal to 0.

Examples.

$$\begin{pmatrix} 0 & 0 & 0 & 1 & 2 \\ 0 & 1 & -3 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

is row reduced.

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$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

is not row reduced, because the second column contains a leading 1 (in row 2) and also a nonzero entry (in row 3).

$$\begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & -3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 5 \end{pmatrix}.$$

is now reduced.

Theorem. Every $m \times n$ matrix A is row equivalent to a row-reduced matrix.

Proof. Let $A = (a_{ij})$ with $1 \leq i \leq m$, $1 \leq j \leq n$, and entries in F. If all entries in the first row of A are zero, then that row already satisfies the pivot condition. Otherwise, let k be the smallest index with $a_{1k} \neq 0$. Multiply row #1 by $1/a_{1k}$; now the pivot in row #1 equals 1. For each $2 \leq i \leq m$, add $-a_{ik}$ times row #1 to row #i (that is, $R_i + (-a_{ik})\, R_1 \to R_i$). As a result, the entries in column k below the pivot become zero. Moreover, entries to the left of column k in each row remain unchanged because the pivot in row #1 is the leftmost nonzero entry.

We now turn to row #2. If every entry in row #2 is zero, we leave the row unchanged. Otherwise, let the pivot be in column k' with $k' \neq k$. Divide row #2 by its pivot so
6 that the leading entry becomes 1, and then eliminate the other entries in column k' (except the pivot itself) by adding suitable multiples of row #2 to the other rows. □

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2.6 Row-Reduced Echelon Form

Definition. An $m \times n$ matrix A is said to be in row-reduced echelon form if the following holds:

1. A is row-reduced.
2. Every row of A that has only 0 entries appears below every row which has a non-zero entry.
3. If rows $1, \ldots, r$ are the non-zero rows of A, and if the pivot of row i is in column k_i, $i = 1, \ldots, r$, then $k_1 < k_2 < \cdots < k_r$.

Theorem. Every $m \times n$ matrix A is row equivalent to a row-reduced echelon matrix.

Proof. Apply the previous theorem and, if necessary, perform permutations of the rows. □

Remark. Useful for solving

$$Ax = b.$$

We can replace $[A \mid b]$ by a row-reduced echelon matrix row equivalent to it:

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$$[A' \mid b'].$$

2.6.1 Worked example

$$\left(\begin{array}{cccc|c} 0 & -9 & 3 & 4 & 19 \\ 1 & 4 & 0 & -1 & 5 \\ 2 & 6 & -1 & 5 & -5 \end{array}\right) \xrightarrow{-\frac{1}{3}R_1 \to R_1} \left(\begin{array}{cccc|c} 0 & 1 & -\frac{1}{3} & -\frac{4}{3} & -1 \\ 1 & 4 & 0 & -1 & 5 \\ 2 & 6 & -1 & 5 & -5 \end{array}\right)$$

$$R_2 - 4R_1 \to R_2 \quad , \quad R_3 - 6R_1 \to R_3$$

$$\left(\begin{array}{cccc|c} 1 & 1 & -\frac{1}{3} & -\frac{4}{3} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{3} & 9 \\ 2 & 0 & 1 & \frac{23}{3} & 1 \end{array}\right) \xrightarrow{R_3 - 2R_2 \to R_3} \left(\begin{array}{cccc|c} 1 & 1 & -\frac{1}{3} & -\frac{4}{3} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{3} & 9 \\ 0 & 0 & -\frac{5}{3} & \frac{55}{9} & -17 \end{array}\right)$$

$$\xrightarrow{-\frac{3}{5}R_3 \to R_3} \left(\begin{array}{cccc|c} 1 & 1 & -\frac{1}{3} & -\frac{4}{3} & -1 \\ 1 & 0 & \frac{4}{3} & \frac{7}{3} & 9 \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array}\right)$$

$$\underbrace{R_2 - \tfrac{4}{3}R_3 \to R_2, \qquad R_1 + \tfrac{1}{3}R_3 \to R_1}_{\text{eliminate the 3rd column above the pivot}} \left(\begin{array}{cccc|c} 0 & -\frac{5}{3} & \frac{12}{5} & & \\ 1 & 0 & 0 & \frac{17}{3} & -\frac{23}{5} \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & \frac{17}{3} & -\frac{23}{5} \\ 0 & 1 & 0 & -\frac{5}{3} & \frac{12}{5} \\ 0 & 0 & 1 & -\frac{11}{3} & \frac{51}{5} \end{array}\right)$$

2.6.2 Another worked example

2.7 Definition of Vector Spaces

Definition (Vector space)**.** A vector space over a field K is a set V (the elements of K are called scalars, the elements of V are called vectors) endowed with two operations

$$+ : V \times V \to V, \quad (u, v) \mapsto u + v, \qquad \cdot : K \times V \to V, \quad (\lambda, v) \mapsto \lambda v,$$

such that the following axioms hold:

(V1) For all $v_1, v_2, v_3 \in V$, we have $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$ (associativity of vector addition).

(V2) There exists an element $0 \in V$ such that for all $v \in V$, $0 + v = v$ (additive identity).

(V3) For every $v \in V$, there exists $v' \in V$ such that $v + v' = 0$ (additive inverse).

7 (V4) For all $v_1, v_2 \in V$, we have $v_1 + v_2 = v_2 + v_1$ (commutativity of vector addition).

(V5) For all $a, b \in K$ and all $v \in V$, we have $(ab)v = a(bv)$ (associativity of scalar multiplication).

(V6) For all $v \in V$, $1 \cdot v = v$ (multiplicative identity).

(V7) For all $a \in K$ and all $v_1, v_2 \in V$, we have $a(v_1 + v_2) = av_1 + av_2$ (distributivity over vector addition).

(V8) For all $a_1, a_2 \in K$ and all $v \in V$, we have $(a_1 + a_2)v = a_1 v + a_2 v$ (distributivity over field addition).

Example. Some Examples: Example 1.: The coordinate space K^n. We have

$$n \in \mathbb{N}, \qquad K^n = \{\, (a_1, \dots, a_n) \mid a_i \in K,\ 1 \le i \le n \,\}.$$

Equivalently,

$$\underbrace{K \times K \times \cdots \times K}_{n \text{ times}} \cong K^n.$$

We turn $V = K^n$ into a vector space by defining, for

$$v = (a_1, \dots, a_n), \quad w = (b_1, \dots, b_n) \in K^n,$$

the operations

$$v + w = (a_1 + b_1, \dots, a_n + b_n) \in K^n, \qquad a \in K, \quad a \cdot v := (a\,a_1, \dots, a\,a_n) \in K^n,$$

and the zero vector

$$0 := (0, \dots, 0) \in K^n.$$

EXC. Show that K^n with these operations is a vector space.

Lemma. Let V be a vector space over K. Then:

(a) The zero vector is unique, i.e., if $0' \in V$ also satisfies $\forall v \in V: \ 0' + v = v$, then $0' = 0$.

(b) Let $v \in V$. The element $v' \in V$ from (V3) is unique, i.e., if $\tilde{v} \in V$ satisfies $v + \tilde{v} = 0$, then $\tilde{v} = v'$.

(c) $\forall v \in V: \ 0 \cdot v = 0$ (the zero vector).

(d) $\forall v \in V, \ \forall a \in K: \ a \cdot 0 = 0$.

(e) $\forall v \in V: \ (-1) \cdot v = -v$.

(f) $\forall v \in V: \ -(-v) = v$.

(g) If $a \cdot v = 0$ for some $a \in K$, $v \in V$, then $a = 0$ or $v = 0$.

(h) Associativity for $+$ holds also for n elements, i.e., $v_1 + v_2 + \cdots + v_n$ makes sense no matter where we put the parentheses; for example,

$$(v_1 + v_2) + ((v_3 + v_4) + v_5) = v_1 + (v_2 + (v_3 + (v_4 + v_5)))$$

or

$$((v_1 + v_2) + v_3) + (v_4 + v_5).$$

7 **Proof.** (a) Uniqueness of 0. Suppose $e \in V$ satisfies $\forall v \in V: \ e + v = v$. We need to prove that $e = 0$. Indeed, $e + 0_V = 0_V$ by assumption. At the same time $0_V + e = e$ by the definition of 0_V. From these two equations we get $e = 0_V$.

(b) Uniqueness of v'. We need to prove that $v = v' and u \in V$ satisfy $v + u = 0$ and $v + v' = 0$. We have to show that $u = v'$. So:

$$u = u + 0 = (v + v') + u = v + (v' + v) + u = v' + (v + u) = v' + 0 = v'.$$

This proves statement b).

(c) Let $v \in V$. We have to how that $0 \cdot v = 0$. Indeed,

$$0 \cdot v = (0 + 0) \cdot v = 0 \cdot v + 0 \cdot v.$$

so we have $0 \cdot v = 0 \times 0 + 0 \times 0$
Add $-0 \times v$ to both sides.

$$\Rightarrow 0 = 0 \times v + 0 \times v - 0 \times v = 0 \times v + (0 \times v - 0 \times v) = 0 \times v + 0 = 0 \times v$$

(d) Let $a \in K$, $v \in V$ and assume that $a \cdot v = 0$. We need to prove that either $a = 0$ or $v = 0$. Indeed, assume that $a \neq 0$. By the axioms of a filed ,$\exists$ an element $a^{-1} \in K$

such that $a^{-1}a = 1$. Now: $v = 1 \cdot v = (a^{-1} \cdot a) \cdot v = a^{-1}(a \cdot v) = 0$by assumption.

$$= a^{-1}(a \cdot v) = 0 \text{ statement d of the lemma.}$$

□

Definition. Let V be a vector space over a field K. A subset $W \subseteq V$ is called a *linear subspace* of V if the following holds:
LSS1: $W \neq \emptyset$
LSS2: $\forall w_1, w_2 \in W$, we have $w_1 + w_2 \in W$.
LSS3: $\forall a \in K, \forall w \in W$ we have $a \cdot w \in W$.

Lemma. Let V be a vector space over K, and let $W \subseteq V$ be a subspace. Then W is a vector space in its own right when endowed with the operations coming from V.

Proof. By (LSS2) and (LSS3) the operations $+$ and $\cdot$ from V indeed define corresponding operations on W. It remains to check the axioms (V1)–(V8).

7 Exercise: Do that!

We check axiom (V2) for W. Pick any element $u \in W$ (in particular $W \neq \emptyset$). Since $0_V = 0_K \cdot u$ and W is a subspace, we have $0_V \in W$ by (LSS3). Define $0_W := 0_V$.

Clearly, for all $w \in W$ we have $0_W + w = 0_V + w = w$. □

Lemma (Criteron for a subset to be a subspace)**.** Let V be a V.S. over a field K, and $W \subseteq V$ a subset. Then W is a subspace of V iff the following holds:

1. $0 \in W$.
2. $\forall a_1, a_2 \in K$, $\forall w_1, w_2 \in W$ we have $a_1w_1 + a_2w_2 \in W$.

Proof. "$W \subseteq V$ is a subspace $\Rightarrow$ (1)+(2)": This follows at one from the fact that W is in itself a vect. sp..

(1)+(2)$\Rightarrow$ W is a subspace:
clearly $W \neq \emptyset$ b.c. $0 \in W$.(LSS1)
(LSS2): Let $w_1, w_2 \in W$. By (2), $1 \cdot w_1 + 1 \cdot w_2 \in W$ So (LLSS2) holds.
(LSS3): By (2), if $a \in K, w \in W$, then $a \cdot w + 0 \cdot 0 \in W$ So $a \cdot n \in W$. □

Lecture 8: Linear Subspaces

10 Oct 2025

2.8 Linear Subspaces

1) Let V be a v.s. over K. Then $\{0_v\} \subseteq V$ is a subspace called the 0-subspace of V or
8 the trivial subspace of V.
2) Fix $b \in K$. Consider the subset

$$U_b := \{(x_1, x_2, x_3) \in K^3 \mid x_1 + x_2 + x_3 = b\} \subseteq K^3.$$

Claim. U_b is a subspace iff $b = 0$.

Proof. $\Rightarrow$: Suppose U_b is a linear subspace. If

$$(x_1, x_2, x_3),\ (y_1, y_2, y_3) \in U_b,$$

then

$$x_1 + x_2 + x_3 = b, \quad y_1 + y_2 + y_3 = b. \tag{*}$$

Since U_b is a subspace,

$$\begin{aligned} &(x_1 + y_1, x_2 + y_2, x_3 + y_3) \in U_b \\ \Rightarrow\quad &(x_1 + y_1) + (x_2 + y_2) + (x_3 + y_3) = b. \end{aligned}$$

But from $(*)$ we get

$$(x_1 + y_1) + (x_2 + y_2) + (x_3 + y_3) = b + b = 2b,$$

so $b = 2b$, hence $b = 0$. $\square$

Exec. 1) Use the fact that $O \in U_b$ to deduce that $b = 0$. 2) Prove "$\Leftarrow$"

Let A be an $m \times n$ matrix with entries in K.

$$\big(A \in \mathrm{Mat}_{m\times n}(K) \text{ or } A \in M_{m\times n}(K)\big).$$

Fix $b \in K^m$, $b = (b_1, \dots, b_m)$ viewed as a column vector. Consider the set

8

$$L \coloneqq \{x \in K^n \mid Ax = b\} \subseteq K^n.$$

Claim: L is a linear subspace iff $b = 0 \in K^n$. For the proof, we need preparations.

Lemma. $\forall u, v \in K^n, \forall a \in K$ we have: 1) $A(u + v) = Au + Av$ 2) $A(au) = a(Au)$

Proof. Write $A = (a_{ij})$, $u = (u_1, \dots, u_n)^T$, $v = (v_1, \dots, v_n)^T$.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \quad u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}, \quad v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}.$$

$$Au + Av = \begin{pmatrix} a_{11}u_1 + a_{12}u_2 + \ldots + a_{1n}u_n \\ \vdots \\ a_{i1}u_1 + a_{i2}u_2 + \ldots + a_{in}u_n \\ \vdots \\ a_{m1}u_1 + a_{m2}u_2 + \ldots + a_{mn}u_n \end{pmatrix} + \begin{pmatrix} a_{11}v_1 + a_{12}v_2 + \ldots + a_{1n}v_n \\ \vdots \\ a_{i1}v_1 + a_{i2}v_2 + \ldots + a_{in}v_n \\ \vdots \\ a_{m1}v_1 + a_{m2}v_2 + \ldots + a_{mn}v_n \end{pmatrix}$$

$$= \begin{pmatrix} a_{11}(u_1+v_1)+a_{12}(u_2+v_2)+\ldots+a_{1n}(u_n+v_n) \\ \vdots \\ a_{i1}(u_1+v_1)+a_{i2}(u_2+v_2)+\ldots+a_{in}(u_n+v_n) \\ \vdots \\ a_{m1}(u_1+v_1)+a_{m2}(u_2+v_2)+\ldots+a_{mn}(u_n+v_n) \end{pmatrix} = A(u+v).$$

The scalar multiplication property is shown similarly:

8

$$A(au) = \begin{pmatrix} a_{11}(au_1)+\ldots+a_{1n}(au_n) \\ \vdots \\ a_{m1}(au_1)+\ldots+a_{mn}(au_n) \end{pmatrix} = a \begin{pmatrix} a_{11}u_1+\ldots+a_{1n}u_n \\ \vdots \\ a_{m1}u_1+\ldots+a_{mn}u_n \end{pmatrix} = a(Au).$$

□

Proof of claim: $\Leftarrow$: Assume $b=0$, show that L is a subspace. Indeed:

$$0 \in L \text{ b.c. } A0 = 0 = b.$$

.

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Proof. Proof of the claim continued:

$\Leftarrow$: Assume $b=0$, and we will prove that L is a subspace. Clearly $0 \in L$, because $A0 = 0 = b$. If $x = (x_1,\ldots,x_n) \in L$ and $y = (y_1,\ldots,y_n) \in L$, then

$$A\cdot(x+y) = A\cdot x + A\cdot y = 0+0 = 0,$$

hence $x+y \in L$. Let $x \in L$ and $a \in K$. Then

$$A(ax) = a(Ax) = a\cdot 0 = 0,$$

so $ax \in L$. It follows that $L \subseteq K^n$ is a subspace.

$\Rightarrow$: Suppose L is a subspace, and we will show that $b=0$. Since $L \subseteq K^n$ is a subspace, $0 \in L$. Hence $A\cdot 0 = b$, which implies $b=0$. □

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2.8.1 More examples of vector spaces and subspaces

Example. Let K be a field.

Define $K^\infty := \{(a_1,a_2,a_3,\ldots) \mid a_i \in K\ \forall i\}$. Addition and scalar multiplication on K^∞ are given by

$$(a_1,a_2,a_3,\ldots)+(b_1,b_2,b_3,\ldots) = (a_1+b_1,a_2+b_2,a_3+b_3,\ldots),$$

and for $a \in K$,

$$a\cdot(a_1,a_2,a_3,\ldots) = (a\cdot a_1, a\cdot a_2, a\cdot a_3,\ldots).$$

Exercise. K^∞ with $+$ and $\cdot$ as defined here is a vector space.

Example. Take $k = \mathbb{R}$.

$$\mathrm{Fib} := \{(a_1, a_2, a_3, \dots) \mid a_k = a_{k-1} + a_{k-2} \ \forall k \geq 3\}.$$

Exercise. Show that

$$\mathrm{Fib} \subseteq \mathbb{R}^\infty$$

is a linear subspace.

More generally, fix $\alpha, \beta \in \mathbb{R}$.
Then

$$U_{\alpha,\beta} = \{(a_1, a_2, \dots, a_k, \dots) \in \mathbb{R}^\infty \mid a_k = \alpha a_{k-1} + \beta a_{k-2} \ \forall k \geq 3\}.$$

Exercise. Show that $U_{\alpha,\beta} \subseteq \mathbb{R}^\infty$ is a subspace.

2.9 Examples of Vector Spaces

2.9.1 Spaces of Functions

9 Let K be a field, and let S be a non-empty set.

Define

$$K^S = \{\text{all functions } f : S \to K\}.$$

Define $+$ and $\cdot$ on K^S:

Let $f, g \in K^S$, $a \in K$.

$$(f : S \to K),$$
$$(g : S \to K)$$

Define $(f + g) : S \to K$ by

$$(f + g)(x) := f(x) + g(x) \quad \forall x \in S$$

(where the addition is in K).

Also define $af : S \to K$ by

$$(af)(x) := a\,f(x) \quad \forall x \in S.$$

Exercise. K^S with these operations is a vector space over K.

Example. Another example. Let $S = [0, 1] := \{x \in \mathbb{R} | 0 \leq x \leq 1\}$. Define $C(S) := \left\{f \in \mathbb{R}^S | f \text{ is continuous in } [0, 1]\right\}$

Claim. $C(S) \subseteq \mathbb{R}^S$ is a linear subspace. (exercise)

2.9.2 Polynomials

Let K be a field. Let x be a formal variable. A polynomial over K (or a polynomial with coefficients in K) in the variable x is an expression of the form:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0.$$

$$\text{where } n \in \mathbb{Z}_{\geq 0}, a_i \in K \forall 0 \leq i \leq n.$$

The zero polynomial is the polynomial with all coefficients equal to zero.

$$f(x) = 0 \text{ i.e. } a_0 = 0, a_1 = 0, a_2 = 0, \ldots, a_n = 0.$$

Remark.

$$K^3 \leftrightarrow K^{\{1,2,3\}} \longrightarrow a : \{1, 2, 3\} \to K \longrightarrow (a(1), a(2), a(3)) = (a_1, a_2, a_3) \in K^3.$$

Back to polynomials. We allow to omit o terms like $0 \cdot x^i$.

9 $$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots + a_n x^n, \qquad g(x) = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \ldots + b_m x^m.$$

$$\text{If} n < m, \text{we set} a_{n+1} = a_{n+2} = \ldots = a_m = 0.$$
$$\text{If} n > m, \text{we set} b_{m+1} = b_{m+2} = \ldots = b_n = 0.$$

Put $d = \max(n, m)$ and now we can write:

$$f(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_r x^r,$$
$$g(x) = b_0 + b_1 x + b_2 x^2 + \ldots + b_r x^r.$$

Define

$$f(x) + g(x) := (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \ldots + (a_r + b_r)x^r, .$$

If $c \in K, f(x)$ a polynomial:

$$c \cdot f(x) := (c \cdot a_0) + (c \cdot a_1)x + (c \cdot a_2)x^2 + \ldots + (c \cdot a_r)x^r.$$

Notation: $K[x]$ is the set of all polynomials with coefficients in K. Claim. $K[x]$ together with the operations $+$ and $\cdot$ defined above is a vector space over K. (exercise)

Every $f(x) \in K[x]$ defines also a function $f : K \to K$. The subset of K^K which comes from polynomials is a linear subspace of K^K isomorphic to $K[x]$. (exercise)

But, from $\tilde{f}$ we cannot always recover f.

Example.

$$K = \mathbb{F}_2 = \{0, 1\}, \quad f(x) = x^2 + x + 1, \quad \tilde{f}(0) = 1, \quad \tilde{f}(1) = 1.$$

$$\begin{gathered} f(x) = x \\ g(x) = x^2 f(x) \neq g(x) \text{or polynomials} \end{gathered}.$$

but they give rise to the same functions $\mathbb{F}_2 \longrightarrow \mathbb{F}_2$ This is b.c. $\forall x \in \mathbb{F}_2, x^2 = x$.

Let $f(x) = a_0 = a_1 x + a_2 x^2 + \ldots a_n x^n$ be a polynomial with $a_n \neq 0$. We say that the degree of f is n and write $\deg(f) := n \in \mathbb{Z}_{\geq 0}$.

If $f(x) = 0$, i.e. $f(x)$is the zero polynomial, we define $\deg(f(x)) := -\infty$.
(So $\deg f(x) = -\infty <=> f(x)$is the 0-polynomial)

$$\deg f(x) := \max \{k \in \mathbb{Z}_{\geq 0} \mid a_k \neq 0\}$$

Fix $d \in \mathbb{Z}_{\geq 0} \cup \{-\infty\}$.

Define

$$K[x]_{\leq d} := \{\, f \in K[x] \mid \deg(f) \leq d\,\}.$$

(We agree that $-\infty \leq k$ for all $k \in \mathbb{Z}$.)

Claim. $K[x]_{\leq d} \subseteq K[x]$ is a linear subspace. (exercise)

9

Q. Is $\{\, f \in K[x] \mid \deg f = d\,\}$ also a subspace of $K[x]$?

Q. Consider $\Big\{\, p \in K[x]_5 \;\Big|\; p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5,$
$a_0 + a_1 + a_2 + a_3 + a_4 + a_5 = 0 \Big\}.$

Is this a subspace of $K[x]$?

2.9.3 Spaces of Matrices

Consider $M_{m\times n}(K)$, the set of all $m \times n$ matrices with entries in K. Define $+$ and $\cdot$: $A = (a_{ij}), B = (b_{ij}) \alpha \in K$.

$$A + B = (a_{ij} + b_{ij}), \quad \alpha A = (\alpha a_{ij})..$$

Claim. With these operations, $M_{m\times n}(K)$ is a vector space over K. (exercise)

$$M_{3\times 2}(K) \longleftrightarrow K^6,$$

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} \longmapsto \big(a_{11},\, a_{12},\, a_{21},\, a_{22},\, a_{31},\, a_{32}\big).$$

$$M_{m\times n}(K) \cong K^{mn},$$

$$A = (a_{ij})_{1\leq i\leq m,\ 1\leq j\leq n} \longmapsto \big(a_{11}, a_{12}, \ldots, a_{1n},\ a_{21}, \ldots, a_{2n},\ \ldots,\ a_{m1}, \ldots, a_{mn}\big).$$

> **Remark.** The map above is a linear isomorphism (vector-space bijection) if K is a field.

9

Let V be a vector space. Let $S \subseteq V$ be a subspace Question: What is the smallest linear subspace of V that contains S'?ad

Lecture 10: Span of a set

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2.10 Span of a Set

Continuation in the next lecture.

Lecture 11

16.10.2025

2.11 Span of a Set (Continued)

Sol. Take the col. vectors $v_1 \dots v_n$ and make a $m \times n$ matrix out of them:

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{pmatrix}.$$

If

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in K^n, \text{then.}$$

$$A \cdot x = x \cdot v_1 + x_2 \cdot v_2 + \dots + x_n \cdot v_n = sum_{i=1}^n x_i \cdot v_i.$$

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Given $w \in K^n$, the question is whether or not the system of equations

$$Ax = w$$

has a solution.

If a solution exists, then $w \in \text{Sp}(v_1, \dots, v_n)$.

If a solution does **not** exist, then $w \notin \text{Sp}(v_1, \dots, v_n)$.

In the first case, any solution x will give a choice of coefficients $x_1, \dots, x_n$ such that $w = \sum_{i=1}^n x_i \cdot v_i$.

$$(A|w) = \begin{pmatrix} | & | & & | & | \\ v_1 & v_2 & \dots & v_n & w \\ | & | & & | & | \end{pmatrix}_{x \neq 0}.$$

Examples of spaces and how to spon them.

1. $V = K^n$. $e_1 = (1, 0, \ldots, 0)$, $e_2 = (0, 1, 0, \ldots, 0) \ldots e_n = (0, 0, \ldots, 1)$ Then $\mathrm{Sp}(e_1, e_2, \ldots, e_n) = K^n$.
2. $W = K[x]_d$ (polynomials in x of degree $\leq d$ with coeffic. in K). $W = \mathrm{Sp}(1, x, x^2, \ldots, x^d)$. (exercise)

   2') $V = K[x]$ (all polynomials in x). $V = \mathrm{Sp}(1, x, x^2, x^3, \ldots)$ infinite set.
3. $V = M_{m \times n}(K)$. For every $1 \leq k \leq m, 1 \leq l \leq n$ let E_{kl} be the matrix which has all entries $= 0$, except entry in row k and column l which is $= 1$.

$$\begin{array}{cc} & \\ k \rightarrow & \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} \\ & \uparrow \ell \end{array} = E_{k,\ell} \qquad a_{ij} = \begin{cases} 1, & \text{if } i = k \text{ and } j = \ell, \\ 0, & \text{otherwise.} \end{cases}$$

$$\boxed{E_{k,\ell} = (a_{ij})}$$

Then $M = (\mathrm{Sp}(\{E_{kl} \mid 1 \leq k \leq m, 1 \leq l \leq n\}))$. e.g.

$$M_{2\times 2}(K) = \mathrm{Sp}\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

Remark. K^n $n \in \mathbb{Z}_{\geq 0}$, $M_{m\times n}$, $K[x]_d$ are all finite-dimensional vector spaces. Exc: Show that $K[x]$ is not finite-dimensional.

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2.12 Linear Independence

2.12.1 Definition

Let V be a vector space over a field K.

Definition. Let $v_1, v_2, \ldots, v_n \in V$ be a list of n vectors in V. We say that the vectors $v_1, v_2, \ldots, v_n$ are *linearly independent* if the *only* linear combination of $v_1, v_2, \ldots, v_n$ that equals $0 \in V$ is the one with 0 coefficients.

Alternatively, $v_1, v_2, \ldots, v_n$ are linearly independent if $\forall a_1, a_2, \ldots, a_n \in K$ for which $a_1 v_1 + a_2 v_2 + \ldots + a_n v_n = 0$ we must have that $a_1 = a_2 = \ldots = a_n = 0$. Lets agree that the empty list is linearly independent.

If the vectors $v_1, v_2, \ldots, v_n$ are *not* linearly independent, we say that it is *linearly dependent*.

Remark. 1) 0 can always be written as a linear combination of any list of vectors. $0 = v_1 \cdot 0 + v_2 \cdot 0 + \ldots + v_n \cdot 0$.

We call this the trivial linear combination.

Remark. 2) Suppose that in our list $v_1, v_2, \ldots, v_n$ there is a repetition, i.e. $v_i = v_\ell$ for $1 \leq k < \ell \leq n$. Then $v_1, \ldots, v_n$ is *not* linearly independent $v_k + (-1) \cdot v_\ell = 0$

Remark. 3) The order of the elements in the list $v_1, \dots, v_n$ does not matter for linear independence.

Remark. 4) If 0 is one of the elements in the list $v_1, \dots, v_n$, then this list cannot be linearly independent. $1 \cdot 0 + 0 \cdot v_2 + \dots + 0 \cdot v_n = 0$

2.12.2 Generalization to families

Definition. Let $F = \{v_i\}_{i \in I}$ be a family of vectors in a vector space V over a field K. We say that F is linearly independent if $\forall n \in \mathbb{Z}_{\geq 1}$ and any sequence of *distinct* indices $i_1, i_2, \dots, i_n \in I$, the list of vectors $v_{i_1}, v_{i_2}, \dots, v_{i_n}$ is linearly independent.

Definition. continued Let $\emptyset \neq S \subseteq V$ be a subset. We say that S is lin. independent if every finite list of distinct vectors $v_1, \dots, v_n \in S$ is linearly independent ($n \in \mathbb{Z}_{\geq 1}$).

Lemma. If $S \subseteq V$ is linearly independent, then every subset $S' \subseteq S$ is also linearly independent.

Example. 1) $e_1, \dots, e_n \in K^n$ is linearly independent.

Proof. Suppose that $a_1 e_1 + a_2 e_2 + \dots + a_n e_n = 0$.

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$$a_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + a_n \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

□

Example. 2 The set $e_1, e_2, \dots, e_n, \dots$ in K^∞.

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix}, \quad \dots e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \end{pmatrix} \quad \dots.$$

is linearly independent. (exercise)

Example. 3 The set $\{1, x, x^2, x^3, \dots, x^d\} \subseteq K[x]_d$ is linearly independent. $\{1, x, x^2, x^3, \dots, x^d\} \subseteq K[x]$ is linearly independent. (exercise)

Example. 4) Let V be a vector space. Let $v \in V$. Show that $\{v\}$ is linearly independent iff $v \neq 0$.

Definition. A subset $S \subseteq V$ is called a basis of V if the following holds:

11 **1)** S is linearly independent.
2) S spans V. ($\mathrm{Sp}(S) = V$).

Lecture 12

26.10.2025

Proposition. A subset $S \subseteq V$ is a basis of V iff every $v \in V$ can be written in a unique way as a linear combination of vectors from S (up to order of summation). Meaning of "in a unique way..." The meaning is: suppose $v \in V$ is written as
$v =$ linear combination #1 (with vectors from S)
$v =$ linear combination #2 (with vectors from S)

Define

$$C_1 = \Big\{\text{all the vectors that appear in lin. combin. \#1 with non-zero coeff.} \quad .$$

$$C_2 = \Big\{\text{same as above but for lin. combin. \#2} \quad ,$$

then $C_1 = C_2$ and $\forall u \in C_1 = C_2$ the coeff. of u in lin. combin. #1 is the same as the coeff. of u in lin. combin. #2.

In case S is a finite set, say $S = \{v_1, v_2, \ldots, v_n\}$. The uniqueness means that if $v = a_1v_1 + a_2v_2 + \ldots + a_nv_n$ and $v = b_1v_1 + b_2v_2 + \ldots + b_nv_n$, with $a_i, b_i \in K$, then $a_1 = b_1, a_2 = b_2, \ldots, a_n = b_n$.

12 **Proof.** For simplicity, assume $S =$ finite. (S = infinite <– exc.)
$\Leftarrow$: Assume every $v \in V$ can be written in a unique way as a linear combination of vectors from S.
$\Rightarrow sp(S) = V$.
We will show now that S is linearly independent. Write $S = \{v_1, v_2, \ldots, v_n\}$. Consider the $0 \in V$. Clearly $0 = 0 \cdot v_1 + 0 \cdot v_2 + \ldots + 0 \cdot v_n$. But, by assumption there is a unique linear combination of $v_1, \ldots, v_n$ that gives 0. It follows that if $a_1, a_2, \ldots, a_n = 0$ then $a_1 = 0, \ldots, a_n = 0$. $\Rightarrow$ S is linearly independent.
$\Rightarrow$: Assume that S is a basis.

$$Sp(S) = V. \Rightarrow \text{every } v \in V \text{ can be written as a linear combination of } S.$$

It remains to show uniqueness. Assume by contradiction that $\exists v \in V$ that can be written as $v = a_1v_1 + \ldots + a_nv_n$ and $v = b_1v_1 + \ldots + b_nv_n$ and $a_k \neq b_k$ for some $1 \leq k \leq n$. $\Rightarrow 0 = (a_1 - b_1)v_1 + (a_2 - b_2)v_2 + \ldots + (a_k - b_k)v_k + \ldots + (a_n - b_n)v_n$.
$\Rightarrow$ 0 can be written as a non-trivial linear combination of $v_1, \ldots, v_n$.

$\Rightarrow$ S is NOT linearly independent. Contradiction to the assumption that S is a basis. $\square$

Exps. $\mathbb{K}^n$, $\quad e_1 = (1, 0, \ldots, 0), \quad e_2 = (0, 1, 0, \ldots, 0), \quad \ldots, \quad e_n = (0, \ldots, 0, 1).$

$$S = \{e_1, \ldots, e_n\} \text{ is a basis for } \mathbb{K}^n \text{ (a.k.a. the standard basis).}$$

$$1) \quad V = K[x]_{\leq d}, \quad S = \{1, x, x^2, \ldots, x^d\} \text{ is a basis.}$$

$$2') \quad V = K[x], \quad S = \{1, x, x^2, \ldots, x^n, \ldots\} \text{ is a basis.}$$

$$3) \quad V = M_{m\times n}(K), \quad E_{ij} = \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix}$$

where the 1 is in row i and column j.

Then

$$S = \{E_{ij} \mid 1 \leq i \leq m,\ 1 \leq j \leq n\}$$

is a basis for $M_{m\times n}(K)$.

Lemma (A). Let $v_1, \ldots, v_m \in V$ be a linearly dependent list of vectors. Then there exists an index j with $1 \leq j \leq m$ such that:
1) $v_j \in \mathrm{Sp}(v_1, \ldots, v_{j-1})$
2) $\mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_m) = \mathrm{Sp}(v_1, \ldots, v_m)$

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Proof. Since the list is linearly dependent, $\exists a_1, \ldots, a_m \in K$, not all of them 0, s.t.

$$a_1v_1 + a_2v_2 + \ldots + a_mv_m = 0.$$

Define $j \coloneqq \max\{i \mid a_i \neq 0\}$. (i.e. $a_j \neq 0$ and $a_{j+1} = a_{j+2} = \ldots = a_m = 0$).

$$\Rightarrow a_1v_1 + a_2v_2 + \ldots + a_jv_j = 0.$$

$$\Rightarrow v_j = -\frac{a_1}{a_j}v_1 - \frac{a_2}{a_j}v_2 - \ldots - \frac{a_{j-1}}{a_j}v_{j-1}.$$

This is possible because $a_j \neq 0$. To prove2): let $v \in \mathrm{Sp}(v_1, \ldots, v_m)$.

$$\Rightarrow v = c_1v_1 + c_2v_2 + \ldots + c_jv_j + \ldots + c_mv_m.$$

for some $c_1, \ldots, c_m \in K$. Substitute (*) in to (v) and we obtain that $v =$ a linear combination of $v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_m$.

$$\Rightarrow v \in \mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_m).$$

$$\Rightarrow \mathrm{Sp}(v_1, \ldots, v_m) \subseteq \mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_m).$$

But clearly we also have

$$\mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_m) \subseteq \mathrm{Sp}(v_1, \ldots, v_m).$$

$$\Rightarrow (2)\text{holds.}$$

□

Lemma (B). Under the same assumptions as in Lemma A, assume in addition that the first k items $v_1, \dots, v_k$ in the list $v_1, \dots, v_n$ are linearly independent, where $1 \le k < m$. Then the index j from Lemma A satisfies $k < j$.

Proof. Assume by contradiction that $j \le k$. We have:

$$v_j = a_1 v_1 + \dots + a_{j-1} v_{j-1}$$

for some$a_1, \dots, a_{j-1} \in K$.

$$\Rightarrow 0 = a_1 v_1 + \dots + a_{j-1} v_{j-1} - 1 \cdot v_j.$$

This is a non-trivial linear combination of $v_1, \dots, v_j$ that gives 0. But $v_1, \dots, v_k$ are linearly independent and $j \le k$. $\Rightarrow v_1, \dots, v_j$ are linearly independent. Contradiction. □

Lemma (C). Let $w_1, \dots, w_n \in V$ such that $\mathrm{Sp}(w_1, \dots, w_n) = V$, and let $v \in V$. Then the list $v, w_1, \dots, w_n$ is linearly dependent.

Proof. Write $v = a_1 w_1 + a_2 w_2 + \dots + a_n w_n$ for some $a_1, \dots, a_n \in K$.

$$\Rightarrow (-1) \cdot v + a_1 w_1 + a_2 w_2 + \dots + a_n w_n = 0.$$

This is a non-trivial linear combination that gives 0, proving dependence. □

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Lemma (D). Let $v_1, \dots, v_n \in V$ s.t. $\mathrm{Sp}(v_1, \dots, v_n) = V$. Let $v_1, \dots, v_m$ be a list of linearly independent vectors in V.
Then $m \le n$.

Proof. The proof will have m steps.
Step $\ell = 1$: We will replace u_1 by one of the vectors $v_1, \dots, v_n$. How? Consider $u_1, v_1 \dots, v_n$. By Lemma C, this list is linearly dependent. By Lemma A one of the vectors in this list is in the span of the vectors that appear before it in the list and if we drop this vector from the list the overall span of the list is still V. By Lemma B that vector that we can drop cannot be u_1 (because u_1 standalone is linearly independent.) We now drop that vector and we now get a list of n vectors that still spans V and the first vector in this list is u_1.
Step $2 \le \ell \le m$:
From step $\ell - 1$ we should have a list of n vectors that spans V and the first $\ell - 1$ vectors in this list are $u_1, \dots, u_{\ell-1}$ and the remaining $n - (\ell - 1)$ vectors are taken from the list $v_1, \dots, v_n$. Let's write this new list of n vectors as $u_1, \dots, u_{\ell-1}, w_1, \dots, w_{n-(\ell-1)}$. Consider now the list $u_1, \dots, u_{\ell-1}, u_\ell, w_1, \dots, w_{n-(\ell-1)}$. This list $u_1, \dots, u_{\ell-1}, u_\ell$ spans V, and by Lemma C the larger list $u_1, \dots, u_{\ell-1}, u_\ell, w_1, \dots, w_{n-(\ell-1)}$ must be linearly dependent. By Lemma A, we can omit one of the vectors in this list and still span V, and by Lemma B, the vector that we omit can be taken from the $w_1, \dots, w_{n-(\ell-1)}$ and not from the u's. Moreover the vector we omit must be in the span of those lying before it. The new list (for step ℓ) is:

$$u_1, \dots, u_\ell, \boxed{n - \ell \text{ vectors all taken from } v_1, \dots, v_n}. \longleftarrow \text{ list of length n}$$

Assume now by contradiction that $m > n$. Then we can perform step $\ell = 1, 2, \dots, n$ and obtain a list $\hat{t}$ that looks like $u_1, u_2, \dots, u_n$ and this list spans V. Since $m > n$,
12 we still have the vectors $u_{n+1}, \dots, u_m$. Consider $u_1, u_2, \dots, u_n, u_{n+1}$. By Lemma C this list is linearly dependent. Contradiction! (because we assumed $u_1, \dots, u_m$ is linearly independent). □

Lecture 13

29.10.2025

2.13 Dimension

In fact we proved a bit more than Lemma D, namely: we can remove m vectors from the list $v_1, \dots, v_n$, s.t. the remaining $n - m$ vectors $v_{i_1}, v_{i_2}, \dots, v_{i_{n-m}}$ $(1 \le i_1 < i_2 < \dots < i_{n-m} \le n)$ when put together with the list $u_1, \dots, u_m$ spans together V.

In particular *(means has a special case a less general case)*, this shows if $m = n$ (i.e. the u-list and the v-list have the same size) then $\mathrm{Sp}(u_1, \dots, u_n) = V$.

Theorem. Let V be a finite-dimensional vector space over a field K. Then V has a basis with finitely many elements. Moreover, every basis of V is finite and has the same number of elements.

For the proof, we need: *Lemma E*

Lemma (E). Let $w_1, \dots, w_\ell \in V$ and assume $\forall 1 \le j \le \ell$, $w_j \notin \mathrm{Sp}(w_1, \dots, w_{j-1})$. Then $w_1, \dots, w_\ell$ are linearly independent.

Proof. If $w_1, \dots, w_\ell$ were linearly dependent, then by Lemma A, there would exist anindex j with $1 \le j \le \ell$ such that $w_j \in \mathrm{Sp}(w_1, \dots, w_{j-1})$, contradicting our
13 assumption. □

Lemma (F). Assume $\mathrm{Sp}(v_1, \dots, v_n) = V$. Then $\exists$ a subset of $v_1, \dots, v_n$ which is a basis of V.

Proof. We will have n steps, and in each step we will consider one of the vectors from the list $v_1, \dots, v_n$, and decide whether or not to drop it from the list.

Step 1: If $v_1 = 0$, we drop it.
If $v_1 \ne 0$, we keep it.

Step $2 \le j \le n$: If $v_j \in \mathrm{Sp}(v_1, \dots, v_{j-1})$, we drop it, otherwise we keep it.

After performing n steps as above, we get a possibly shorter list $v_{i_1}, v_{i_2}, \dots, v_{i_m}$ with $1 \le i_1 < i_2 < \dots < i_m \le n$. We claim that $\mathrm{Sp}(v_{i_1}, v_{i_2}, \dots, v_{i_m}) = V$. Indeed, in any of the steps $1 \le j \le n$ we dropped the vector v_j only if $v_j \in \mathrm{Sp}(v_1, \dots, v_{j-1}) \Rightarrow \mathrm{Sp}(v_{i_1}, v_{i_2}, \dots, v_{i_m}) = V$.

We now claim that $v_{i_1}, v_{i_2}, \dots, v_{i_m}$ is linearly independent. Indeed, $\forall 1 \le k \le m$, we have

$$v_{i_k} \notin \mathrm{Sp}(v_{i_1}, v_{i_2}, \dots, v_{i_{k-1}}). \text{ In particular}$$
$$v_{i_k} \notin \mathrm{Sp}(v_1, v_2, \dots, v_{i_k - 1}).$$

By Lemma E, applied to the list $v_{i_1}, v_{i_2}, \ldots, v_{i_m}$, we deduce that $v_{i_1}, v_{i_2}, \ldots, v_{i_m}$ are linearly independent. $\Rightarrow v_{i_1}, v_{i_2}, \ldots, v_{i_m}$ both spans V and is linearly independent. $\square$

Proof (Proof of the Theorem). We need to prove:

1. V has a basis with finitely many elements.
2. Every basis of V has finitely many elements.
3. For every two bases A, B of V, we have $|A| = |B|$.

1.: Take any finite set $S \subseteq V$ s.t. $\mathrm{Sp}(S) = V$ (possible because V is finite-dimensional). By Lemma F, $\exists S' \subseteq S$ which is a basis of V.

2.: Let $\mathcal{C}$ be any basis for V. We will show that $\mathcal{C}$ is a finite set.

Suppose by contradiction that $\mathcal{C}$ is not finite. Also, take any finite basis, say S, for V (exists by 1.), say

$$S = \{v_1, v_2, \ldots, v_n\}.$$

Choose $n + 2025$ vectors $u_1, u_2, \ldots, u_{n+2025} \in \mathcal{C}$. $\mathrm{Sp}(v_1, v_2, \ldots, v_n) = V$. and $u_1, u_2, \ldots, u_{n+2025}$ are linearly independent. By Lemma D, we have that $n + 2025 \leq n$. Contradiction.

3.: By statement 2., A and B are both finite. Now $\mathrm{Sp}(B) = V$ and A is linearly independent. By Lemma D, $|A| \leq |B|$. We also have that $\mathrm{Sp}(A) = V$ and B is linearly independent. By Lemma D again, $|B| \leq |A|$. Thus, $|A| = |B|$.
13 (applause) $\square$

Note. From every good theorem we get a definition:

Definition. Let V be a finite-dimensional vector space over a field K. We define the dimension of V to be the number $n \in \mathbb{Z}_{\geq 0}$ of elements in (any!) a basis of V. We write $\dim(V) = n$.

$$\dim_K(V) = n.$$

(*)dim V is well defined. !

$$\dim\{0\} = 0 \quad \text{because a basis for } \{0\} \text{ is the empty set } \emptyset,$$

$$\text{and } |\emptyset| = 0.$$

Example (Examples). 1) $\dim K^n = n$ because the standard basis has n elements.
2) $\dim K[x]_d = d + 1$
3) $\dim M_{m\times n}(K) = m \cdot n$ 4) Let S be a set. $K^S = \{f : S \to K \text{ is a function}\}$. This is a vector space over K.

Exercise: If S is a finite set, say $S = \{s_1, s_2, \ldots, s_n\}$. Then $\dim(K^S) = |S|$. Find a basis.

Summary: Let V be a finite vector space. Then
1) Every finite list of vectors that spans V contains a sublist which is a basis for V.
2) Every linearly independent list of vectors $u_1, \ldots, u_m$ can be extended to a basis of V.

13 1) $v_1, \ldots, v_n \quad > v_{i_1}, \ldots, v_{i_k}$ basis.
2) $u_1, \ldots, u_m \quad > u_1, \ldots, u_m, w_1, \ldots, w_{d-m}$ basis.

Lecture 14

31.10.2025

Theorem (1). Let V be a finite-dimensional vector space of dimension $n := \dim V$. The following statements are equivalent:

1. $v_1, \ldots, v_n$ are linearly independent.
2. $v_1, \ldots, v_n$ spans V.
3. $v_1, \ldots, v_n$ form a basis for V.

Proof. $1 \Rightarrow 3$. By what was done last time, we can extend the list $v_1, \ldots, v_n$ and get a basis for V. But this list already has n ($= \dim V$) vectors, $\Rightarrow$ there is no extension possible, hence $v_1, \ldots, v_n$ is already a basis for V.

$3 \Rightarrow 1\&2.$: By definition of "basis".

$2 \Rightarrow 3$: By what we did last time, we can possibly drop some of the vectors from the list $v_1, \ldots, v_n$ and get a basis. But the length of the list $v_1, \ldots, v_n$ is already n ($= \dim V$), so dropping is impossible. $\Rightarrow v_1, \ldots, v_n$ was a basis from the beginning. □

Theorem (2). Suppose V is finite-dimensional and $n := \dim V$. Let $v_1, \ldots, v_k \in V$ be a list. Then

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1. If $k < n$, then $v_1, \ldots, v_k$ do *not* span V.
2. If $k > n$, then $v_1, \ldots, v_k$ are can*not* be linearly independent.

Proof. 1) Assume k < n. Suppose by contradiction that $\mathrm{Sp}(v_1, \ldots, v_k) = V$. By point 1 on the very left backboard :) we can thin this list to a sublist $v_{i_1}, \ldots, v_{i_k}$ which is a basis for V. But a basis for V has n elements, so $k = n$. ($\Rightarrow d \leq k < n$). Contradiction.

2) Assume $k > n$. Suppose by contradiction that $v_1, \ldots, v_k$ are linearly independent. By point 2 on the famous very left backboard ;) we can extend this list to a basis of V. ($\Rightarrow \dim V \geq k > n$). Quod est absurdum. □

2.13.1 Dimension of subspaces

Proposition. Let V be a finite-dimensional vector space. Then every subspace $U \subseteq V$ is also finite-dimensional and we have $\dim(U) \leq \dim(V)$. Moreover we have equality iff $U = V$.

Note (Lemma E). Let $w_1, \ldots, w_\ell \in V$ and assume that $\forall 1 \leq j \leq \ell$, $w_j \notin \mathrm{Sp}(w_1, \ldots, w_{j-1})$. Then $w_1, \ldots, w_\ell$ are linearly independent. This was proved last time.

Proof (Proof of Proposition). Put $n := \dim(V)$. We will show first that $\mathcal{U}$ has a finite basis. If $\mathcal{U} = \{0\}$, then $\emptyset$ is a basis for $\mathcal{U}$. So assume $\mathcal{U} \neq \{0\}$. Take $u_1 \in \mathcal{U}, u_1 \neq 0$. If $\mathcal{U} = \mathrm{Sp}(u_1)$ the u_1 is a basis for $\mathcal{U}$ and we are done. If $\mathcal{U} \neq \mathrm{Sp}(u_1)$, choose $u_2 \in \mathcal{U} \setminus \mathrm{Sp}(u_1)$. ($\subsetneq$) By Lemma E, u_1, u_2 are linearly independent. We continue this process and after j steps we obtain a list $u_1, u_2, \ldots, u_j \in \mathcal{U}$ that is linearly independent. If $\mathrm{Sp}(u_1, \ldots, u_j) = \mathcal{U}$ we are done. Otherwise, choose $u_{j+1} \in \mathcal{U} \setminus \mathrm{Sp}(u_1, \ldots, u_j)$. By Lemma E, $u_1, \ldots, u_j, u_{j+1}$ are linearly independent. However, in V a list of linearly independent vectors can have at most length $n := \dim(V)$. $\Rightarrow$ the procedure we are describing must end (i.e. $\mathrm{Sp}(u_1, \ldots, u_j) = \mathcal{U}$) after not more than n steps. Say that $k \leq n$ steps. So we obtain a list $u_1, \ldots, u_k$ $(k \leq n)$ in $\mathcal{U}$ which is linearly independent and spans $\mathcal{U}$. In other words, $u_1, \ldots, u_k$ is a basis for $\mathcal{U}$.

$\Rightarrow \dim(\mathcal{U}) = k \leq n = \dim(V)$.

Equality $(\dim(\mathcal{U}) = \dim(V))$ holds iff $\mathcal{U} = V$: obviously if $\mathcal{U} = V$ then $\dim \mathcal{U} = \dim V$.

Conversely, suppose $k = n$. $\Rightarrow$ the list of vectors $u_1, \ldots, u_k$ is a basis for $\mathcal{U}$, hence these are linearly independent vectors and they continue to be linearly independent in V. By Theorem 1 (applied to $u_1, \ldots, u_k \in V$), $u_1, \ldots, u_n$ is a basis for V. $\Rightarrow \mathcal{U} = \mathrm{Sp}(u_1, \ldots, u_n) = V$. ☐

Example. Consider K^m. Let $v_1, \ldots, v_n \in K^m$.

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Q1. Let $w \in K^m$. How to determine whether or not $w \in \mathrm{Sp}(v_1, \ldots, v_n)$?

Q2. Describe $\mathrm{Sp}(v_1, \ldots, v_n)$.

Q3. How to determine if $v_1, \ldots, v_n$ are linearly independent?

A1. $w \in \mathrm{Sp}(v_1, \ldots, v_n) \Leftrightarrow \exists x_1, \ldots, x_n \in K$ s.t. $w = x_1 v_1 + \cdots + x_n v_n$.

$$\Leftrightarrow \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = w$$

So the question is whether or not the linear system with unknowns $x_1, \ldots, x_n$ has a solution.

A2. Apply A1 to a general w. Perform elimination on $(A \mid w)$ until we obtain a reduced row echelon matrix $(A' \mid w')$. The equation $A'x = w'$ has a solution if and only if $Ax = w$ has a solution.

If A' has no zero rows (i.e. each row of A' contains a pivot), then $A'x = w'$ always has a solution. Hence
$$\mathrm{Sp}(v_1, \ldots, v_n) = K^m.$$

If A' has some zero rows, say the last $m-r$ rows are zero, then:

$$\text{Does } A'x = w' \text{ have a solution?}$$

This system has a solution iff the last entries satisfy

$$w'_{r+1} = \cdots = w'_m = 0.$$

Conclusion:

$$\operatorname{Sp}(v_1, \ldots, v_n) = \{\, b \in K^m \mid b'_{r+1} = \cdots = b'_m = 0 \,\}.$$

Example.

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \in \mathbb{R}^3. \qquad \text{Find } \operatorname{Sp}(v_1, v_2).$$

$$(A|b) = \begin{pmatrix} 1 & 1 & | & b_1 \\ 1 & 0 & | & b_2 \\ 1 & -1 & | & b_3 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & | & b_2 \\ 0 & 1 & | & b_1 - b_2 \\ 0 & -2 & | & b_3 - b_1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & | & b_1 \\ 0 & 1 & | & b_1 - b_2 \\ 0 & 0 & | & b_1 - 2b_2 + b_3 \end{pmatrix}.$$

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$$\Rightarrow \operatorname{Sp}(v_1, v_2) = \left\{ \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \in \mathbb{R}^3 \mid b_1 - 2b_2 + b_3 = 0 \right\}.$$

Example (A3). $v_1, \ldots, v_n \in K^m$ are linearly independent iff the system of equations

$$\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \ldots & v_n \\ | & | & & | \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

has only the trivial solution $\begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$. Take

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \ldots & v_n \\ | & | & & | \end{pmatrix}.$$

Do elimination to get A'. Then $v_1, \ldots, v_n$ are linearly independent iff $A'x = 0$ has only the trivial solution. This system of equations has only the trivial solution $\Leftrightarrow$

the reduced row echelon matrix A' has precisely n pivots

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$$\text{i.e.} \quad A' = \begin{pmatrix} 1 & 0 & \cdots & 0 & * & * & \cdots & * \\ 0 & 1 & \cdots & 0 & * & * & \cdots & * \\ \vdots & \vdots & ots & \vdots & \vdots & \vdots & ots & \vdots \\ 0 & 0 & \cdots & 1 & * & * & \cdots & * \end{pmatrix} \quad (\Leftrightarrow \text{ no free variables}).$$

Lecture 15

05.11.2025

Theorem (2 revisited)**.** Take a list $v_1, \dots, v_k \in K^m$ where n<m.

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_k \\ | & | & & | \end{pmatrix} \quad \text{we can have} \leq n \text{ pivots but } m > n.$$

$$\text{When trying to solve} \qquad A \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \qquad \text{we'll get}$$

$$\left(\begin{array}{ccc|c} 1 & \cdots & 0 & b' \\ & \ddots & \vdots & \vdots \\ 0 & \cdots & 1 & \\ \hline 0 & \cdots & 0 & b'' \\ \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & \end{array}\right)$$

after elimination. Each entry in b' and in b'' is of the type $\alpha_1 b_1 + \cdots + \alpha_m b_m$ with
15 $\alpha_i \in K$.

$$\mathrm{Sp}(v_1, \dots, v_n) = \{b \in K^m \mid b'' = 0\}'.$$

2) Assume now $v_1, \dots, v_k \in K^m$ and $n > m$. ($m = \dim K^m$). Why is $v_1, \dots, v_k$ linearly dependent? Well, we need to check if the system

$$\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_k \\ | & | & & | \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

has non-trivial solutions. After elimination we get

$$A' = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

and A' can have at most m pivots (b.c. every pivot is in a different row). But # of variables <-n > m $\Rightarrow$we must have a free variable $\Rightarrow \exists$ non-trivial solutions.

2.14 Row and Column Spaces

Definition. Let $A \in M_{m\times n}(K)$, and denote by $u_1, \dots, u_m \in K^n$ the rows of A, and by $v_1, \dots, v_n \in K^m$ the columns of A.

$$A = \begin{pmatrix} - & u_1 & - \\ & \vdots & \\ - & u_m & - \end{pmatrix} = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}.$$

Define two spaces:

- $\mathrm{RowS(A)} := \mathrm{Sp}(u_1, \dots, u_m) \subseteq K^n$
- $\mathrm{ColS(A)} := \mathrm{Sp}(v_1, \dots, v_n) \subseteq K^m$ Column Space of A

Lemma. If $A, B \in M_{m\times n}(K)$ are row equivalent, then $\mathrm{RowS}(A) = \mathrm{RowS}(B)$.

Proof. Let $M \in M_{m\times m}(K)$.

1. If we exchange two rows of M, the span of the rows does not change. So $\mathrm{RowS}(A)$ does not change under this operation.
2. The operation $\lambda R_i \to R_i,\ \lambda \in K \setminus \{0\}$. Here too, the span of the rows does NOT change.

$$\alpha_1 R_1 + \cdots + \alpha_i(\lambda R_i) + \cdots + \alpha_m R_m = \alpha_1 R_1 + \cdots + (\alpha_i\lambda) R_i + \cdots + \alpha_m R_m,$$

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$$\beta_1 R_1 + \cdots + \beta_m R_m = \beta_1 R_1 + \cdots + \left(\frac{\beta_i}{\lambda}\right)(\lambda R_i) + \cdots + \beta_m R_m.$$

3. The operation $R_i + \lambda R_j \to R_i,\ i \neq j$. We need to show

$$\mathrm{Sp}(R_1, \dots, R_m) = \mathrm{Sp}(R_1, \dots, R_i + \lambda R_j, \dots, R_m).$$

Indeed,

$$\alpha_1 R_1 + \cdots + \alpha_i(R_i + \lambda R_j) + \cdots + \alpha_m R_m = \alpha_1 R_1 + \cdots + \alpha_i R_i + \cdots + \alpha_m R_m + (\alpha_i\lambda) R_j,$$

so the span is contained in $\mathrm{Sp}(R_1, \dots, R_m)$.
Conversely,

$$R_i = (R_i + \lambda R_j) - \lambda R_j,$$

so $\mathrm{Sp}(R_1, \dots, R_m) \subseteq \mathrm{Sp}(R_1, \dots, R_i + \lambda R_j, \dots, R_m)$.

Summary: $\mathrm{Sp}(R_1, \dots, R_i + \lambda R_j, \dots, R_m) = \mathrm{Sp}(R_1, \dots, R_m)$.
Conclusion: After one elementary row operation on M, the $\mathrm{RowS}(M)$ is UNchanged.
If A and B are row equivalent, then $\exists$ a finite sequence $A = A_0, A_1, \dots, A_r = B$ such that A_{i+1} is obtained from A_i by one elementary row operation. By what we just proved,

$$\mathrm{RowS}(A) = \mathrm{RowS}(A_0) = \cdots = \mathrm{RowS}(A_r) = \mathrm{RowS}(B).$$

$\square$

Remark. It is NOT trhue that $\mathrm{ColS}(A) = \mathrm{ColS}(B)$ (exc. find an example).

Definition. Let $A \in M_{m\times n}(K)$. Define

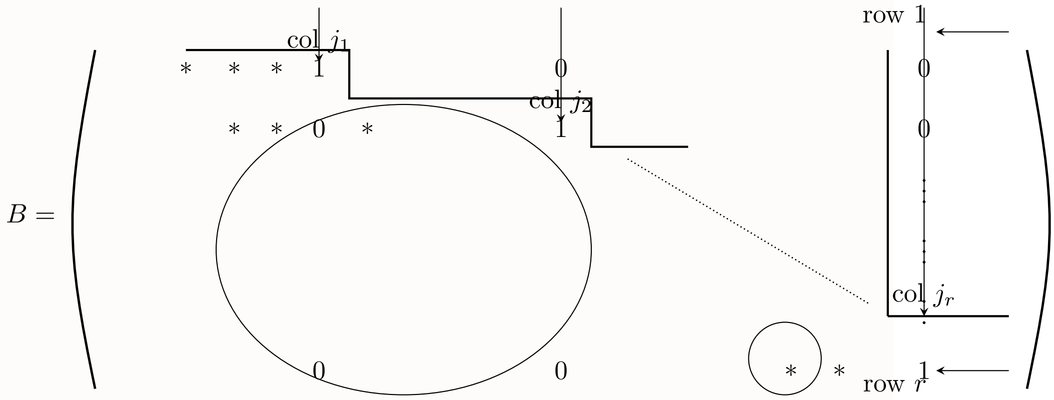
$$\text{row-rank}(A) := \dim(\mathrm{RowS}(A))$$

$$\text{col-rank}(A) := \dim(\mathrm{ColS}(A)).$$

Lemma. Let $B \in M_{m\times n}(K)$ be in row reduced echelon form. Then the rows of B that are not totally 0 form a basis of $\mathrm{RowS}(B)$. In particular, $\text{row-rank}(B)$ equals the number of pivots in B. Also the pivot columns of B form a basis of $\mathrm{ColS}(B)$. In particular, for matrices B that are in row reduced echelon form, we have

$$\text{row-rank}(B) = \text{col-rank}(B).$$

Proof.



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Denote by $j_1 < j_2 < \cdots < j_r$ the column numbers of the pivots. Denote by the $u_i, \ldots, u_r$ the non-zero rows of B. We claim that $u_1, \ldots, u_r$ are linearly independent. Indeed, assume that $\lambda_1 u_1 + \cdots + \lambda_r u_r = 0$ for some $\lambda_1, \ldots, \lambda_r \in K$. This is a vector in K^n. Entry number j_k in this vector is precisely λ_k i.e.

$$\lambda_1 u_1 + \cdots + \ldots + \lambda_r u_r = (\ldots, \lambda_1, \ldots, \lambda_2, \ldots, \lambda_k, \ldots, \lambda_r, \ldots).$$

But if we assume that this vector is 0, then $\lambda_1 = \lambda_2 = \cdots = \lambda_r = 0$. This shows our claim (that $u_1, \ldots, u_r$ are linearly independent).
But $\mathrm{RowS}(B) = \mathrm{Sp}(u_1, \ldots, u_r)$ By Thm 1, $u_1, \ldots, u_r$ is a basis of $\mathrm{RowS}(B)$. Also we get that $\text{row-rank}(B) = r$.

<u>ColS(B):</u> clearly $\mathrm{ColS}(B) \subseteq \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} \in K^m \mid x_{r+1} = \cdots = x_m = 0 \right\} = K^r \times \{0\} \leftarrow$ 0-vector of K^{m-r}.

At the same time, the pivot columns of B are of the form:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad e_r = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \text{(the } r\text{-th entry is 1).}$$

15

$$\text{So } e_1, \dots, e_r \text{ form a basis for } \mathbb{K}^r \times \{0\}.$$

$$\Rightarrow \quad \mathbb{K}^r \times \{0\} = \operatorname{span}(e_1, \dots, e_r) \subseteq \operatorname{ColS}(B) \subseteq \mathbb{K}^r \times \{0\},$$

$$\Rightarrow \quad \operatorname{ColS}(B) = \mathbb{K}^r \times \{0\} \quad \text{and} \quad e_1, \dots, e_r \text{ form a basis for it.}$$

(the pivot columns)

□

Lecture 16: Sums of vector spaces

12.11.2025

Last time we established the meaning of the pivots in the row reduced matrix. In particular, we found that if r is the amount of pivots, then $\text{row-rank}(B) = \text{col-rank}(B) = r$.

Let's quickly discuss an algorithm to find a basis for the span of a list of vectors $u_1, \dots, u_m \in K^n$. Just put them in a matrix

$$A = \begin{pmatrix} - & u_1 & - \\ - & u_2 & - \\ & \vdots & \\ - & u_m & - \end{pmatrix}$$

Apply elimination to get a matrix B in RREF. The non-zero rows of B give a basis for
16 the span of $\operatorname{Sp}(u_1, \dots, u_m)$, literally just citing the last Lemma.

Remark. The number of pivots one gets after elimination does *not* depend on the specific elimination process. Why? Because the number of pivots equal $\text{row-rank}(A)$, which is invariant.

Definition (Transpose of a matrix)**.** Let $A = (a_{ij}) \in \mathrm{M}_{m\times n}(K)$. Define the transposed matrix of A

$$A^\top := (b_{ij}) \in \mathrm{M}_{n\times m}(K),$$

where $b_{ij} = a_{ji}$.

We have a few simple properties of the transpose. $\forall A, B \in M_{m\times n}(K) : \forall \lambda \in K$:

$$\begin{aligned}
(A+B)^\top &= A^\top + B^\top \\
(\lambda A)^\top &= \lambda A^\top \\
\left(A^\top\right)^\top &= A \\
\mathrm{RowS}\left(A^\top\right) &= \mathrm{ColS}(A) \\
\mathrm{ColS}\left(A^\top\right) &= \mathrm{RowS}(A)
\end{aligned}$$

2.15 Sums of Vector Spaces

Let's maybe look at what happens when you sum vector spaces.

Definition (Sum of subspaces)**.** Let V be a VS and $U, W \subseteq V$ linear subspaces of V. We define the sum

$$U + W = \{u + w \mid u \in U, w \in W\}.$$

You can extend this to more that just two subspaces. For $U_1, \ldots, U_r \subseteq V$, we define

$$U_1 + \cdots + U_r = \{u_1 + \cdots + u_r \mid u_i \in U_i \; \forall 1 \le i \le r\}.$$

Proposition. Let $U, W \subseteq V$ be subspaces of V. Then,

16 1) $U + W = \mathrm{Sp}(U \cup W)$. In particular, $U + W \subseteq V$ is therefore a subspace.

2) If U, W are finite dimensional, then $U + W$ is also finite dimensional. One can write a basis for $U + W$ as follows:

- Choose a basis $p_1, \ldots, p_k$ for $U \cap W$ ($k := \dim(U \cap W)$).
- Extend $p_1, \ldots, p_k$ to a basis $p_1, \ldots, p_k, u_1, \ldots, u_{l-k}$ of U with $l := \dim U$.
- Extend $p_1, \ldots, p_k$ to a basis of W, $p_1, \ldots, p_k, w_1, \ldots, w_{m-k}$ with $m = \dim W$.

Then, $p_1, \ldots, p_k, u_1, \ldots, u_{l-k}, w_1, \ldots, w_{m-k}$ is a basis for $U + W$.

3) In particular, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Proof. 1) Clearly, $U + W \subseteq \mathrm{Sp}(U \cup W)$. We'll show also that $\mathrm{Sp}(U \cup W) \subseteq U + W$. Indeed, let $v \in \mathrm{Sp}(U \cup W)$. Then,

$$v = \sum_{i=1}^{s} a_i u_i + \sum_{j=1}^{r} b_j w_j = \bar{u} + \bar{w},$$

because those sums remain in the subspaces they originated from, hence $\bar{u} \in U$ and $\bar{w} \in W$. This proves the equality.

2) We first show that $p_1, \ldots, p_k, u_1, \ldots, u_{l-k}, w_1, \ldots, w_{m-k}$ are linearly independent. Indeed, assume for some $a_i, b_i, c_i \in K$

$$a_1 p_1 + \cdots + a_k p_k + b_1 u_1 + \cdots + b_{l-k} u_{l-k} + c_1 w_1 + \cdots + c_{m-k} w_{m-k} = 0.$$

Write

$$v = c_1 w_1 + \cdots + c_{m-k} w_{m-k} \in W.$$

Note that from the way we defined v, we have

$$v = -(a_1 p_1 + \cdots + a_k p_k + b_1 u_1 + \cdots + b_{l-k} u_{l-k}) \in U.$$

Hence $v \in U \cap W$. Since $p_1, \ldots, p_k$ form a basis for $U \cap W$, $\exists$ coefficients $\alpha_1, \ldots, \alpha_k \in K$, s.t.

$$v = \alpha_1 p_1 + \cdots + \alpha_k p_k.$$

But

$$v = c_1 w_1 + \cdots + c_{m-k} w_{m-k}$$

and

$$\alpha_1 p_1 + \cdots + \alpha_k p_k + (-c_1) w_1 + \cdots + (-c_{m-k}) w_{m-k} = 0.$$

But $p_1, \ldots, p_k, w_1, .., w_{m-k}$ are a basis for W, hence $\alpha_1 = \cdots = \alpha_k = 0$ and $c_1 = \cdots = c_{m-k} = 0$, hence $v = 0$. But this means

$$0 = v = -(a_1 p_1 + \cdots + a_k p_k + b_1 u_1 + \cdots + b_{l-k} u_k).$$

16 But $p_1, \ldots, p_k, u_1, \ldots, u_k$ are a basis (for U), hence $a_1 = \cdots = a_k = 0$ and $b_1 = \cdots = b_{l-k} = 0$. This shows that $p_1, \ldots, p_k, u_1, \ldots, u_{l-k}, \ldots, w_1, \ldots, w_{m-k}$ are linearly independent.

It remains to show that this list of vectors span $U + W$. Indeed, let $z \in U + W$. Write $z = u + w$ for some $u \in U, w \in W$. We can write

$$u = \alpha_1 p_1 + \cdots + \alpha_k p_k + a_1 u_1 + \cdots + a_{l-k} u_{l-k}$$

for some $\alpha_i, a_j \in K$. Then write

$$w = \beta_1 p_1 + \cdots + \beta_k p_k + b_1 u_1 + \cdots + b_{m-k} w_k$$

for some $\beta_i, b_j \in K$. Then,

$$z = (\alpha_1 + \beta_1) p_1 + \cdots + (\alpha_k + \beta_k) p_k + a_1 u_1 + \cdots + a_k u_{l-k} + b_1 w_1 + \cdots + b_{m-k} w_{m-k} \in \mathrm{Sp}(U \cup W).$$

3) We have the length of the basis we just constructed given by

$$\dim(U + W) = k + (l - k) + (m - k) = j + m - k = \dim U + \dim W - \dim(U \cap W).$$

$\square$

Corollary. Let $U, W \subseteq V$ be finite dimensional subspaces. Then the following five statements are equivalent:

1. $\dim(U + W) = \dim U + \dim V$.

2. $\dim(U \cap W) = 0$.
3. $U \cap W = \{0\}$.
4. Every $v \in U + W$ *can* be written in a *unique* way as $v = u + w$ with $u \in U, w \in W$.
5. If $u + w = v$, where $u \in U, w \in W \Rightarrow u = w = 0$.

Proof. $1 \Leftrightarrow 2$: Follows by the previous proposition (3), the dimension formula.

$2 \Leftrightarrow 3$: This is immediate by the fact that there is only one space which has $\dim = 0$, the trivial space $\{0\}$.

16 $3 \Rightarrow 4$: By definition, $\forall v \in V$ (or $V + W$?) can be written as $v = u + w$ with $u \in U, w \in W$. We need to show the uniqueness. Assume $v = u + w = u' + w'$ where $u, u' \in U$ and $w, w' \in W$. So $u - u' = w' - w$, so $u - u'$, $w' - w \in U \cap W = \{0\}$, hence $u' = u, w' = w$.

$4 \Rightarrow 5$: $0 = 0 + 0$, so if $0 = u + v$ and 4 holds, then $u = 0, v = 0$.

$5 \Rightarrow 3$: Let $v \in U \cap W$. Clearly $0 = v + (-v)$, with $v \in U$ and $-v \in W$. If 5 holds, then $v = 0$, hence $U \cap W = \{0\}$. □

Definition (Completement of a subspace)**.** Let V be a vector space and $U \subseteq V$ a subspace. A subspace $W \subseteq V$ is called a complement to U if $U + W = V$ and $U \cap W = \{0\}$.

Lecture 17

14.11.2025

Proposition (14.5)**.** Let V be a finite dimensional vector space and $U \subseteq V$ a subspace. Then there exists a complement $W \subseteq V$ which is a complement of U.

Proof. Choose a basis $u_1, \dots, u_\ell$ of U ($\ell := \dim U$). Extend this basis to a basis of V: $u_1, \dots, u_\ell, w_1, \dots, w_m$ ($sol + m = \dim V$). Take $W := \mathrm{Sp}(w_1, \dots, w_m)$. Clearly $U + W = V$, b.c. $U + W = \mathrm{Sp}(U \cup W) \subseteq \mathrm{Sp}(u_1, \dots, u_\ell, w_1, \dots, w_m) = V$. Also $UcapW = \{0\}$, b.c. if

$$\sum_{i=1}^{\ell} a_i u_i = \sum_{j=1}^{m} b_j w_j, \text{ then}$$

17 $$\sum_{i=1}^{\ell} a_i u_i + \sum_{j=1}^{m} (-b_j) w_j = 0.$$

But $u_1, \dots, u_\ell, w_1, \dots, w_m$ are linearly independent, so all coefficients are $0 \Rightarrow a_i = 0 \ \forall i$ and $b_j = 0 \ \forall j$. Alternatively:

$$\dim(UcapW) = \dim(U + W) - \dim U - \dim W = (\ell + m) - \ell - m = 0.$$

□

2.16 Homomorphisms of Vector Spaces

Maps bring intererest into vector spaces. A vector space is a set with some structure, and we are gonna study maps that respect this structure. Linear maps are also called linear transformations and homomorphisms of vector spaces.

Definition (15.a.1). Let V, W be vector spaces over the same field K. A map $T : V \to W$ is called linear if:

1. $\forall v_1, v_2 \in V : T(v_1 + v_2) = T(v_1) + T(v_2)$ (additivity)
2. $\forall a \in K, \forall v \in V : T(av) = aT(v)$ (homogeneity)

Somethimes $Tv = T(v)$ is used instead of $T(v)$.
The set of all linear maps $T : V \to W$ is denoted by $\mathrm{Hom}_K(V, W)$ or simply $\mathrm{Hom}(V, W)$ [a]

[a] sometimes $\mathrm{hom}(V, W)$

Exercise: – Show that $T : V \to W$ is linear $\Leftrightarrow T(a_1v_1+a_2v_2) = a_1T(v_1)+a_2T(v_2)\ \forall v_1, v_2 \in V, \forall a_1, a_2 \in K$

Example (1).

$$V = \text{vector space, } id : V \to V, \text{is a linear map.}$$

17 **Example** (2).

$$V, W = \text{vector spaces,} \quad O : V \to W, \quad v \in V \mapsto 0 \in W$$

the zero map is linear.

Example (3). $D : K[x] \to K[x]$, the (formal) derivative.

$$D\left(a_0 + a_1x + \ldots + a_nk^k \ldots a_nx^n\right) = a_1 + 2a_2x + \ldots + ka_kx^{k-1} + \ldots + na_nx^{n-1}.$$

$$\sum_{i=0}^{n} a_ix^i \mapsto \sum_{i=1}^{n} ia_ix^{i-1}.$$

This is a linear map.

$$D\left(ap(x) + bq(x)\right) = a \cdot p'(x) + b \cdot q'(x) = aD(p(x)) + bD(q(x)).$$

$a, b \in K\ \ p(x), q(x) \in K[x]$

Example (4).

$$C([a, b]) := \text{contin. functions } f : [a, b] \to \mathbb{R},$$

$$T : C([a, b]) \to \mathbb{R},$$

$$T(f) := \int_a^b f(x)\, dx.$$

Example (5).

$$S : K^\infty \to K^\infty$$

$$S(a_1, a_2, \ldots, a_n, \ldots) := (a_2, a_3, \ldots, a_n, \ldots)$$

The shift map, it is linear.

Example (6). Important example. Let $A \in M_{m\times n}(K)$.

Define

$$T_A : K^n_{\text{col}} \to K^m_{\text{col}}$$

$$T_A\left(\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}\right) := A \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in K^m_{\text{col}}, \qquad A \in K^{m\times n}.$$

We showed that $\quad A \cdot (a \cdot x) = a \cdot (Ax) \quad \forall\, a \in K,\ x \in K^n_{\text{col}}$

and

$$A(x+y) = Ax + Ay, \qquad \forall\, x, y \in K^n_{\text{col}}.$$

$$\Rightarrow \ T_A \text{ is a linear map.}$$

17 **Proposition (15.a.2).** Let V, W be vector spaces over K and $T : V \to W$ a linear map. Then:

1. $\forall n \in \mathbb{Z}_{\geq 0}, \forall v_1, \ldots, v_n \in V, \forall a_1, \ldots, a_n \in K :$

$$T\left(\sum_{i=1}^n a_i v_i\right) = \sum_{i=1}^n a_i T(v_i).$$

2. $T(0_V) = 0_W$

Proof. 1. Follows from the definition of linear maps (15.a.1) by induction on n.

2. $T(0_W) = T(0_V) - T(0_V) = T(0_V - 0_V) = T(0_V) \Rightarrow T(0_V) = 0_W$.

□

Theorem. Let V be a finite dimensional VS and $v_1, \ldots, v_n \in V$ a *basis* for V. Let W be any vector space and $w_1, \ldots, w_n \in W$ a list of arbitrary vectors. Then $\exists!\ T : V \to W$ linear s.t.

$$T(v_i) = w_i \quad \forall 1 \leq i \leq n.$$

Proof. Existence of T: Let $v \in V$. We want to define $T(v) \in W$. Since $v_1, \ldots, v_n$ form a basis of V, $\exists a_1, \ldots, a_n \in K$ s.t. $v = \sum_{i=1}^n a_i v_i$. Define $T(v) := a_1 w_1 +$

$\ldots + a_n w_n = \sum_{i=1}^{n} a_i w_i \in W(*)$. Not that T is well-defined, because $\forall v \in V$ the coefficients $a_1, \ldots, a_n \in K$ s.t. $v = a_1 v_1 + \ldots + a_n v_n = \sum_{i=1}^{n} a_i v_i$ are unique. (because $v_1, \ldots, v_n$ is a basis of V).

We claim that T is linear: Indeed, let $v', v'' \in V \exists a_1', \ldots, a_n' \in K, a_1'', \ldots, a_n'' \in K$ s.t. $v' = \sum_{i=1}^{n} a_i' v_i = a_1' v_1 + \ldots + a_n' v_n$, $v'' = \sum_{i=1}^{n} a_i'' v_i = a_1'' v_1 + \ldots + a_n'' v_n$.

$$v' + v'' = \sum_{i=1}^{n} (a_i' + a_i'') v_i = (a_1' + a_1'') v_1 + \ldots + (a_n' + a_n'') v_n$$

By *our* definition. (*),
$$\begin{aligned} T(v' + v'') &= \sum_{i=1}^{n} (a_i' + a_i'') w_i = (a_1' + a_1'') w_1 + \ldots + (a_n' + a_n'') w_n \\ &= \sum_{i=1}^{n} a_i' w_i + \sum_{i=1}^{n} a_i'' w_i = a_1' w_1 + \ldots + a_n' w_n + a_1'' w_1 + \ldots + a_n'' w_n \\ &= T(v') + T(v''). \end{aligned}$$

Let $\alpha \in K$, $v \in V$. Write $v = a_1 v_1 + \cdots + a_n v_n \quad (a_i \in K)$.

$$T(\alpha v) = T(\alpha a_1 v_1 + \cdots + \alpha a_n v_n) \overset{(*)}{=} \alpha a_1 w_1 + \cdots + \alpha a_n w_n = \alpha(a_1 w_1 + \cdots + a_n w_n) \overset{(*)}{=} \alpha T(v).$$

17 This proves that T is linear. Note that $T(v_i) = w_i$, because $v_i = 0 \cdot v_1 + \cdots + 1 \cdot v_i + \cdots + 0 \cdot v_n$.

Uniqueness of T. Suppose $T, S : V \to W$ are two linear maps with $T(v_i) = w_i = S(v_i)$ for all $i \leq n$. We'll show $T = S$. Let $v \in V$. We will show that $T(v) = S(v)$. Write $v = a_1 v_1 + \ldots + a_n v_n = \sum_{i=1}^{n} a_i v_i$ $(a_i \in K \forall i)$.

$$\begin{aligned} T(v) = T(a_1 v_1 + \ldots + a_n v_n) &= T(\sum_{i=1}^{n} a_i v_i) = a_1 T(v_1) + \ldots + a_n T(v_n) \\ &= \sum_{i=1}^{n} a_i T(v_i) = a_1 w_1 + \ldots + a_n w_n = \sum_{i=1}^{n} a_i w_i \\ &= S(a_1 v_1 + \ldots + a_n v_n) = S\left(\sum_{i=1}^{n} a_i v_i\right) = S(v). \end{aligned}$$

□

Example (Important example 15.a.4). Consider K^n with the standard basis $e_1, \ldots, e_n$. Let $w_1, \ldots, w_n \in K^m$ be n arbitrary vectors. By the previous theorem (15.a.3), $\exists!$ linear map $T : K^n \to K^m$ s.t. $T(e_i) = w_i$. We can describe T as $T = T_A$, where

$$A = \begin{pmatrix} | & & | \\ w_1 & \ldots & w_n \\ | & & | \end{pmatrix} \in K^{m \times n}.$$

Recall:

$$T_A : K^n \to K^m, \text{ is linear.}$$

$$T_A(v) = A \cdot v.$$

$$T_A(e_i) = \begin{pmatrix} | \\ w_i \\ | \end{pmatrix} \quad \forall i \qquad e_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \text{ (1 at the i-th position).}$$

17

Lemma. For every linear map $T : K^n \to K^m$, there exists a unique matrix $A \in M_{m\times n}(K)$ s.t. $T_A = T$. That matrix is

$$A = \begin{pmatrix} | & & | \\ Te_1 & \dots & Te_n \\ | & & | \end{pmatrix}.$$

Proof. This is immediate, as demonstrated in the last example. □

Lecture 18

19.11.2025

Kernel and Image of a linear map

Definition (Prop. + Def. 15.b.1)**.** Let V, W be vector spaces and $T : V \to W$ a linear map.

1. The set
$$\ker(T) := \{ v \in V \mid T(v) = 0_W \} \subseteq V$$
is a subspace of V. It is called the *kernel* of T.
2. The set
$$\operatorname{Im}(T) := \{ T(v) \mid v \in V \} \subseteq W$$
is a subspace of W. It is called the *image* of T.

18

Hence,
$$\ker(T) = T^{-1}(\{0_W\}), \qquad \operatorname{Im}(T) = T(V).$$

Proof.

1. $T(0_V) = 0_W$, hence $0_V \in \ker(T)$. Let $v_1, v_2 \in \ker(T)$ and $a, b \in K$. We have
$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) = 0 \Rightarrow av_1 + bv_2 \in \ker(T).$$
This shows that $\ker(T)$ is a subspace of V.
2. $T(0_V) = 0_W \Rightarrow 0_W \in \operatorname{Im}(T)$. Let $w_1, w_2 \in \operatorname{Im}(T)$ and $a, b \in K$. Then

$\exists v_1, v_2 \in V$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$. We have

$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) = aw_1 + bw_2.$$

So

$$aw_1 + bw_2 = T(\text{something}) \Rightarrow aw_1 + bw_2 \in \text{Im}(T).$$

□

Example. Let $A \in M_{m\times n}(K)$. Consider $T_A : K^n \to K^m$. $(T_A(x) := Ax)$

$$\ker(T_A) = \{x \in K^n \mid Ax = 0\}.$$

Proposition (15.b.2). Let $T : V \to W$ be a linear map.

1. T is injective $\Leftrightarrow \ker(T) = \{0_V\}$.
2. T is surjective $\Leftrightarrow \text{Im}(T) = W$.

Proof. (2) is a triviality.

(1) $\Rightarrow$: Clearly $\{0_V\} \subseteq \ker(T)$ because $T(0_V) = 0_W$. Since T is injective, $T^{-1}(\{0_W\})$ consists of at most one element. But we know that $0_V \in T^{-1}(\{0_W\})$. So

$$\ker(T) = T^{-1}(\{0_W\}) = \{0_V\}.$$

18

$\Leftarrow$: Assume $\ker(T) = \{0_V\}$. Let $v_1, v_2 \in V$ and assume $T(v_1) = T(v_2)$. We need to show that $v_1 = v_2$. Indeed,

$$\begin{aligned} T(v_1) - T(v_2) &= 0_W, \\ T(v_1 - v_2) &= 0_W. \end{aligned}$$

So,

$$\begin{aligned} v_1 - v_2 \in \ker(T) &\Rightarrow v_1 - v_2 = 0_V \\ &\Rightarrow v_1 = v_2. \end{aligned}$$

□

Example. Let $T : V \to W$ be a linear map.

1. If $V' \subseteq V$ is a linear subspace, then $T(V') \subseteq W$ is a linear subspace.
2. If $W' \subseteq W$ is a linear subspace, then $T^{-1}(W') \subseteq V$ is a linear subspace.

$$f : A \to B \cup B', \qquad B' \subseteq B$$

$$f^{-1}(B') := \{a \in A \mid f(a) \in B'\}.$$

Definition (bijective maps). A map $f : X \to Y$ is called bijective if it is both injective & surjective. In this case $\exists!$ map $g : Y \to X$ s.t. $g \circ f = id_X$ and $f \circ g = id_Y$. This map g is called the inverse of f. It's denoted by f^{-1}.

Definition (Isomorphism). A linear map $T : V \to W$ is called an isomorphism if $\exists$ a linear map $S : W \to V$ s.t. $S \circ T = id_V$ and $T \circ S = id_W$. If such a T exists, we say that V & W are *isomorphic* or that V is isomorphic to W. We write $V \cong W$(somethimes $V \simeq W$).

Remark. If T is unique, then T is bijective (b.c. it has an inverse S).

Lemma (15.b.4). Let $T : V \to W$ be a linear map. Assume T is bijective. Then its inverse map $T^{-1} : W \to V$ must be linear.

Proof. Let $w_1, w_2 \in W$ and $a, b \in K$. We need to show that $T^{-1}(aw_1 + bw_2) = aT^{-1}(w_1) + bT^{-1}(w_2)$.

Put $v_1 := T^{-1}(w_1)$ and $v_2 := T^{-1}(w_2)$.

$$T(v_1) = w_1, \qquad T(v_2) = w_2.$$

Since T is linear,

$$T^{-1}(aw_1 + bw_2) = T^{-1}(T(av_1 + bv_2)) = av_1 + bv_2 = aT^{-1}(w_1) + bT^{-1}(w_2).$$

$$\Rightarrow T^{-1}(aw_1 + bw_2) = av_1 + bv_2 = aT^{-1}(w_1) + bT^{-1}(w_2).$$

18 $\square$

Lemma (15.b.5). Let $T : V \to W$, $S : W \to U$ be linear maps. Then the composition $S \circ T : V \to U$ is also a linear.

Proof. Let $v_1, v_2 \in V$ and $a, b \in K$.

$$S \circ T(av_1 + bv_2) = S(aT(v_1) + bT(v_2)) = aS(T(v_1)) + bS(T(v_2)) = a(S \circ T)(v_1) + b(S \circ T)(v_2).$$

$\square$

Exercise — Show that "$\cong$" (being isomorphic) is an equivalence relation on the class of all vector spaces over a field K.

Definition. A linear map $T : V \to V$ is called an *endomorphism* of V. We denote the set of all endomorphisms $T : V \to V$ by

$$\mathrm{End}_K(V) \quad (= \mathrm{Hom}_K(V, V)).$$

Definition. A linear map $T : V \to V$ which is also an isomorphism is called an *automorphism* of V. The set of autos (automorphisms) of V is denoted by

$$\mathrm{Aut}(V).$$

Definition (More jargon). Let $T : V \to W$ be a linear map.

1. If T is injective. we also say that T is a *monomorphism*.
2. If T is surjective, we also say that T is a *epimorphism*.

Lemma (15.b.8)**.** Let $T : V \to W$ be a linear map. Let $v_1, \dots, v_n \in V$ be a list that spans V. Then

$$\operatorname{Im}(T) = \operatorname{Sp}(T(v_1), \dots, T(v_n)).$$

Proof. We'll first show $\operatorname{Im}(T) \subseteq \operatorname{Sp}(T(v_1), \dots, T(v_n))$. Let $a_1, \dots, a_n \in K$. Let $\sum_{i=1}^{n} a_i T v_i \in \operatorname{Sp}(T(v_1), \dots, T(v_n))$.

$$\sum_{i=1}^{n} a_i T v_i = T\left(\sum_{i=1}^{n} a_i v_i\right) \in \operatorname{Im}(T).$$

We'll show now that $\operatorname{Im}(T) \supseteq \operatorname{Sp}(Tv_1, \dots, T(v_n))$. Let $w \in \operatorname{Im}(T)$. By definition $\exists v \in V$ s.t. $T(v) = w$. Since $v_1, \dots, v_n$ span V, $\exists a_1, \dots, a_n \in K$ s.t.

$$v = \sum_{i=1}^{n} a_i v_i.$$

$$\Rightarrow w = T(v) = T\left(\sum_{i=1}^{n} a_i v_i\right) = \sum_{i=1}^{n} a_i T(v_i) \in \operatorname{Sp}(T(v_1), \dots, T(v_n)).$$

This shows $\operatorname{Im}(T) \supseteq \operatorname{Sp}(T(v_1), \dots, T(v_n))$. $\square$

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Theorem (15.b.9: The Rank Theorem)**.** Let $T : V \to W$ be a linear map, where V is finite dimensional. Then $\dim \ker(T) + \dim \operatorname{Im}(T) = \dim V$.

Proof. Write $n \coloneqq \dim V$. Let $n_1, \dots, n_k$ be a basis for $\ker(T)$ ($k = \dim \ker(T)$). (Note that $\ker(T)$ is finite dimensional, being a subspace of V which is finite dimensional). Extend this basis to a basis

$$u_1, \dots, u_k, v_1, \dots, v_{n-k} \text{ of } V.$$

By *Lemma 15.b.8*, $\operatorname{Im}(T) = \operatorname{Sp}(T(u_1), \dots, T(u_k), T(v_1), \dots, T(v_{n-k}))$.

$$\dots.$$

Well show that $T(v_1), \dots, T(v_{n-k})$ is linearly independent. Too see this, assume that

$$\sum_{i=1}^{n-k} a_i T(v_i) = 0_W (*).$$

Since T is linear,

$$(*) = T\left(\sum_{i=1}^{n-k} a_i v_i\right).$$

So,

$$T\left(\sum_{i=1}^{n-k} a_i v_i\right) = 0_W \Rightarrow \sum_{i=1}^{n-k} a_i v_i \in \ker(T).$$

Now, $u_1, \ldots, u_k$ form a basis for $\ker(T)$, $\Rightarrow \exists b_1, \ldots, b_k \in K$ s.t.

$$\sum_{i=1}^{n-k} a_i v_i = b_1u_1 + \ldots + b_ku_k \Rightarrow b_1u_1 + \ldots + b_ku_k + (-a_1)v_1 + \ldots + (-a_{n-k})v_{n-k} = 0_V.$$

18 Since $u_1, \ldots, u_k, v_1, \ldots, v_{n-k}$ form a basis for V, we have:

$$b_1 = \ldots = b_k = 0, \quad a_1 = \ldots = a_{n-k} = 0.$$

This shows that $T(v_1), \ldots, T(v_{n-k})$ are linearly independent. We have already seen that those span $\operatorname{Im}(T)$, so they actually form a basis of $\operatorname{Im}(T)$.

$$\Rightarrow \dim \operatorname{Im}(T) = n - k =$$
$$= \dim V - \dim \ker(T). \Rightarrow \dim \ker(T) + \dim \operatorname{Im}(T) = \dim V.$$

□

Lecture 19

21.11.2025

Corollary. Let $T : V \to W$ be a linear map, where V & W are finite dimensional. Then:

1. If $\dim W < \dim V$, then T is *not* injective.
2. If $\dim W > \dim V$, then T is *not* surjective.
3. If $\dim V = \dim W$, then T is bijective $\Leftrightarrow$ T is injective $\Leftrightarrow$ T is surjective.

Proof.

1. Since $\operatorname{Im}(T)$ is a subspace of W, we have:

   $$\dim \operatorname{Im}(T) \leq \dim W < \dim V.$$

   19 By the rank theorem (15.b.9)

   $$\dim \ker(T) = \dim V - \dim \operatorname{Im}(T) > 0 \Rightarrow \ker(T) \neq \{0_V\}.$$

   By proposition 15.b.2, T is *not* injective.
2. $\dim \operatorname{Im}(T) = \dim V - \dim \ker(T) \leq \dim V < \dim W$.

   $$\Rightarrow \operatorname{Im}(T) \subsetneq W \Rightarrow T \text{ is } \textit{not} \text{ surjective..}$$

3. T is injective $\Leftrightarrow \ker(T) = \{0_V\} \Leftrightarrow \dim \ker(T) = 0 \Leftrightarrow \dim \operatorname{Im}(T) = \dim V \Leftrightarrow$ $\dim \operatorname{Im}(T) = \dim W$ (b.c. $\dim V = \dim W$) $\Leftrightarrow \operatorname{Im}(T) = W \Leftrightarrow T$ is surjective.

$\square$

Corollary. Two finite dimensional vector spaces V and W are isomorphic iff $\dim V = \dim W$.

Proof. "$\Rightarrow$" : Suppose $T : V \to W$ is an isomorphism. By proposition 15.b.2, $\ker(T) = \{0_V\}$ and $\operatorname{Im}(T) = W$. By the rank theorem (15.b.9) we have:

$$\dim W = \dim \operatorname{Im}(T) = \dim V - \underbrace{\dim \ker(T)}_{=0} = \dim V..$$

"$\Leftarrow$" : Assume $\dim V = \dim W$. Denote by $n := \dim V = \dim W$. Let $v_1, \ldots, v_n$ be a basis of V. Let $w_1, \ldots, w_n$ be a basis of W. Define a linear map $T : V \to W$ by $T(v_i) := w_i \forall 1 \in (*)$ By Theorem 15.a.3, $\exists$ a linear map T with these (*) properties. By Lemma 15.b.8, $\operatorname{Im}(T) = \operatorname{Sp}(T(v_1), \ldots, T(v_n)) = \operatorname{Sp}(w_1, \ldots, w_n) = W$. So T is surjective. By corollary 15.b.10 we have that T is bijective. $\square$

Theorem. Let $T : V \to W$ be an isomorphism between two vector spaces V & W. Let S be a set of vectors from V. Write

$$T(S) := \{T(v) \mid v \in S\}.$$

Then:

1. S is linearly independent iff T(S) is linearly independent.
2. S spans V iff T(S) spans W.
19 3. S is a basis of V iff T(S) is a basis of W.

$$\begin{aligned} V &\cong W \\ \exists T &: V \to W \\ \& \exists R &: W \to V \\ \text{linear s.t. } R \circ T &= id_V \\ \text{and } R \circ T &= id_W \\ &. \end{aligned}$$

$$\boxed{V \cong K^n}$$
$$K[x]_3 \cong K^4.$$

Definition (15.b.13)**.** Let $T : V \to W$ be a linear map. Define the rank of T, to be $\operatorname{rank(T)} := \dim \operatorname{Im}(T)$.

Exercise — Let $T : V \to W, S : W \to U$ be linear maps, where U, V, W are finite dimensional. Show that:

1. $\operatorname{rank}(S \circ T) \leq \min\{\operatorname{rank}(T), \operatorname{rank}(S)\}$.
2. If S is injective, then $\operatorname{rank}(S \circ T) = \operatorname{rank}(T)$.
3. If T is surjective, then $\operatorname{rank}(S \circ T) = \operatorname{rank}(S)$.

Hint: Just use the definitions.

2.16.1 Writing linear maps using bases/coordinates

In this discussion all vector spaces are finite dimensional. We'll write a basis as an *ordered* tuple B= $(v_1, \ldots, v_n)$.

Definition (16.1). Let V be a vector space over K and $B = (v_1, \ldots, v_n)$ a basis for V. Let $v \in V$. Define the *coordinate vector* of v with respect to $mscriptB$ as

$$[v]_B \coloneqq \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in K^n,$$

where $a_1, \ldots, a_n \in K$ are the unique list of scalars in K s.t. $v = a_1v_1 + \cdots + a_nv_n$. Define a map $\Phi_B : V \to K^n$ by

$$\Phi_B(v) \coloneqq [v]_B \in K^n \quad \forall v \in V.$$

Proposition (16.2). Φ_B is linear and is an isomorphism.

Proof. Linearity of Φ_B. Let $v, w \in V, a, b \in K$. Write v & v' a linear combination of the elements of B.

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$$v = a_1v_1 + \cdots + a_nv_n$$
$$w = b_1v_1 + \cdots + b_nv_n$$

$$\Rightarrow [v]_B = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \quad [w]_B = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

$$\Rightarrow av + bw = (aa_1 + bb_1)v_1 + \cdots + (aa_n + bb_n)v_n.$$

$$\Rightarrow [av + bv']_B = \begin{pmatrix} aa_1 + bb_1 \\ \vdots \\ aa_n + bb_n \end{pmatrix} = a \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + b \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

$$\Phi_B(av+bv') = [av+bv']_B = a \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + b \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = a[v]_B + b[v']_B = a\Phi_B(v) + b\Phi_B(v').$$

This proves that Φ_B is linear. We'll show now that Φ_B is an isomorphism. We'll show that $Ker(\Phi_B) = \{0_V\}$. Indeed, if $v \in \ker(\Phi_B)$, then $[v]_B = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \in K^n$.

$\Rightarrow v = 0v_1 + \cdots + 0v_n = 0$. So $\ker(\Phi_B) = \{0\}$. Since $\dim V = \dim K^n, \Phi_B$ is an isomorphism. (Alternatively: show Φ_B is surjective). $\square$

Lecture 20

26.11.2025

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Remark. The inverse Φ_B^{-1} of Φ_B can be written as:

$$\Phi_B^{-1} : K^n \to V,$$
$$\Phi_B^{-1}\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1v_1 + \cdots + a_nv_n.$$

Example. Consider $V = \mathbb{R}^2$ and let $\mathcal{E} = \big((1,0),(0,1)\big)$ be the standard basis. Let $B = \big((1,1),(1,-1)\big)$ be another basis (check it's a basis). Let $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$.

$$\left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{E}} = \begin{pmatrix} x \\ y \end{pmatrix}.$$

Why? Because $x\begin{pmatrix} 1 \\ 0 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$. What is $\left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_B$? Consider first $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

$$\text{Write } \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right]_B = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}.$$

$$a_1\begin{pmatrix} 1 \\ 1 \end{pmatrix} + a_2\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

$$\begin{cases} a_1 + a_2 = 1, \\ a_1 - a_2 = 0, \end{cases} \Rightarrow a_1 = a_2 = \tfrac{1}{2}.$$

So

$$\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right]_B = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}.$$

$$\left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]_B = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix},$$

$$b_1\begin{pmatrix} 1 \\ 1 \end{pmatrix} + b_2\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

$$\begin{cases} b_1 + b_2 = 0 \\ b_1 - b_2 = 1 \end{cases} \Rightarrow \quad b_1 = \tfrac{1}{2},\ b_2 = -\tfrac{1}{2}.$$

$$\left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]_B = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}.$$

Using linearity of Φ_B,

$$\left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_B = x\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right]_B + y\left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]_B .$$

$$= \begin{pmatrix} \frac{x}{2} \\ \frac{x}{2} \end{pmatrix} + \begin{pmatrix} \frac{y}{2} \\ -\frac{y}{2} \end{pmatrix} = \begin{pmatrix} \frac{x+y}{2} \\ \frac{x-y}{2} \end{pmatrix}.$$

Let V, W be vector spaces (finite dimensional) $n := \dim V$, $m := \dim W$. Let T be a linear map $T : V \to W$.

$$T : V \longrightarrow W. \quad \text{Fix a basis } \mathcal{B} \text{ for } V, \text{ and } \mathcal{C} \text{ for } W.$$

$$T : V \longrightarrow W.$$

Fix a basis B for V, and a basis C for W.

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[dr, phantom, "\copyright"] & W \arrow[d, "\Phi_C"] \\ K^n \arrow[u, "\Phi_B^{-1}"] \arrow[r, "\Phi_C \circ T \circ \Phi_B^{-1}"'] & K^m \end{tikzcd} \qquad \text{diag. commutes}$$

Recall that composition of linear maps is linear, hence

$$\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} : K^n \to K^m$$

20 is linear.

By previous knowledge (15,a,4), $\exists!\, A \in M_{m\times n}(K)$ such that

$$\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} = T_A$$

(where $T_A : K^n \to K^m$ is $T_A x := Ax$).

The matrix A is called the representation of T with respect to the bases $\mathcal{B}$ and $\mathcal{C}$. It is denoted

$$[T]_{\mathcal{B}}^{\mathcal{C}} := A, \qquad [T]_{\mathcal{B}}^{\mathcal{C}} \in M_{m\times n}(K).$$

How to calculate $[T]_{\mathcal{C}}^{B}$?

Put $A := [T]_{\mathcal{C}}^{B}$.

$$\begin{tikzcd} v_i \arrow[rrr, bend left, ""] & \in V \arrow[r, "T"] \arrow[d, "\Phi_B^{-1}"'] \arrow[dr, phantom, "c"] & W \in \arrow[d, "\Phi_{\mathcal{C}}"] & Tv_i \arrow[d] \\ e_i \arrow[u] & \in K^u \arrow[r, "T_A"'] & K^w \in & [Tv_i]_{\mathcal{C}} \end{tikzcd}$$

$$\begin{aligned} T_A(e_i) &= \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}(e_i) \\ &= \Phi_{\mathcal{C}}\big(T(v_i)\big) \\ &= [Tv_i]_{\mathcal{C}} \end{aligned}$$

In other words

$$A = [T]_{\mathcal{C}}^{B} = \begin{array}{c} \begin{array}{ccccc} & & \text{col } \#j & & \end{array} \\ \begin{pmatrix} | & & | & & | \\ [Tv_1]_{\mathcal{C}} & \cdots & [Tv_j]_{\mathcal{C}} & \cdots & [Tv_n]_{\mathcal{C}} \\ | & & | & & | \end{pmatrix} \end{array} \in M_{m\times n}(K) \longrightarrow \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}$$

Another way to describe$[T]_{\mathcal{C}}^{B}$:
Write $[T]_{\mathcal{C}}^{B} = (a_{ij})_{\substack{1\le i\le m \\ 1\le j\le n}}$. The entries a_{ij} are uniquely defined by:

$$Tv_j = a_{1j}w_1 + \cdots + a_{mj}w_m = \sum_{i=1}^{m} a_{ij}w_i$$

$\forall j$ (Here $\mathcal{C} = (w_1, \ldots, w_m)$).

Proposition ((16.4)). Let $T : V \to W$ be a linear map between two finite dimensional vector spaces. Let $\mathcal{B}, \mathcal{C}$ be bases for V & W respectively. Put $A = [T]_{\mathcal{C}}^{\mathcal{B}}$. Then $\forall v \in V, [Tv]_{\mathcal{C}} = A[v]_{\mathcal{B}}$. Equivalently:

$$[Tv]_{\mathcal{C}} = [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}}.$$

Proof.

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$$\begin{tikzcd}
\omega \arrow[d] \arrow[rrrrr, bend left, "T"] & \in & V \arrow[r, "T"] \arrow[d, "\Phi_B"'] \arrow[dr, phantom, "\odot"] & W \arrow[d, "\Phi_e"] & \ni & T\omega \arrow[d] \\
{[\omega]_B} & \in & K^n \arrow[r, "T_A"'] & K^m & \ni & {[T\omega]_e}
\end{tikzcd}$$

So $[T\omega]_e = T_A\big([\omega]_B\big) = A\,[\omega]_B$. $\square$

$\square$

Example.

1. Let
$$V = K^n, \qquad W = K^m,$$
and let
$$T : K^n \to K^m \text{ be } T = T_Q,$$
where
$$Q \in M_{m\times n}(K).$$
So $Tv = Qv$. Take $\mathcal{B} = (e_1, \ldots, e_n)$, $\mathcal{E} = (e_1, \ldots, e_m)$. Then
$$[T]_{\mathcal{E}}^{\mathcal{B}} = Q.$$

2.
$$V = K[x]_{\le 3}, \qquad W = K[x]_{\le 2}.$$

$$D : V \to W, \qquad D(p(x)) := p'(x).$$

D is linear. Take $B = (1, x, x^2, x^3)$ and $\mathcal{C} = (1, x, x^2)$. Find $[D]_{\mathcal{C}}^{B} \in M_{3\times 4}(K)$.

$$\begin{array}{rcl|rcl} D(1) & = & 0 = 0\cdot 1 + 0\cdot x + 0\cdot x^2 & D(x^2) & = & 2x = 0\cdot 1 + 2\cdot x + 0\cdot x^2 \\ D(x) & = & 1 = 1\cdot 1 + 0\cdot x + 0\cdot x^2 & D(x^3) & = & 3x^2 = 0\cdot 1 + 0\cdot x + 3\cdot x^2 \end{array}$$

$$[D]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

2'. Another variation of the problem. View D as a map $D : K[x]_3 \to K[x]_3$. Take $\mathcal{C} = \mathcal{B}$. Then

$$[D]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

which is just as valid and even makes sense because you will never get as a function value from D a polynomial of degree 3.

Let $T : V \to W$ be linear maps. Let

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$$\mathcal{A} = (v_1, \ldots, v_n) \text{ be a basis for } V,$$
$$\mathcal{B} = (w_1, \ldots, w_m) \text{ be a basis for } W..$$

and $e = (u_1, \ldots, u_n)$ be a basis for U. Consider $S \circ T : V \to U$. *Question:* What is the relation between $[S \circ T]_e^{\mathcal{A}}$ and $[T]_{\mathcal{B}}^{\mathcal{A}}, [S]_e^{\mathcal{B}}$? Write $[T]_{\mathcal{B}}^{\mathcal{A}} = (a_{ij})_{\substack{1\le i\le m \\ 1\le j\le n}} \in M_{m\times n}$,

$$[S]_e^{\mathcal{B}} = (b_{ij})_{\substack{1\le i\le p \\ 1\le j\le m}} \in M_{p\times m}$$
$$[S \circ T]_e^{\mathcal{A}} = (c_{ij})_{\substack{1\le i\le p \\ 1\le j\le n}} \in M_{p\times n}.$$

We know that

$$Tv_j = a_{1j}w_1 + \ldots + a_{mj}w_m = \sum_{i=1}^{m} a_{ij}w_i \forall 1 \le j \le n$$

$$Sw_j = b_{1j}u_1 + \ldots + b_{pj}u_p = \sum_{i=1}^{p} b_{ij}u_i \forall 1 \le j \le m$$

$$(S \circ T)v_j = c_{1j}u_1 + \ldots + c_{pj}u_p = \sum_{i=1}^{p} c_{ij}u_i \forall 1 \le j \le n.$$

But $(S \circ T)v_j = S(Tv_j) = S\left(\sum_{i=1}^{m} a_{kj}w_k\right) = \sum_{i=1}^{m} a_{kj}S(w_k) = \sum_{i=1}^{m} a_{kj}\left(\sum_{l=1}^{p} b_{ik}u_l\right) =$

$= \sum_{k=1}^{m}\sum_{i=1}^{p} a_{kj}b_{ik}U_i = \sum_{i=1}^{p}\left(\sum_{k=1}^{m} b_{ik}a_{kj}\right)U_i.$

So $c_{ij} = \sum_{k=1}^{m} b_{ik} a_{kj}$.
Hence we find the matrix (is cols = M?????)

$$[S \circ T]^{\mathcal{A}}_{\mathcal{C}} = \begin{pmatrix} c_{11} & \dots & c_{1j} & \dots & c_{1m} \\ \vdots & & \vdots & & \vdots \\ c_{i1} & \dots & c_{ij} & \dots & c_{im} \\ \vdots & & \vdots & & \vdots \\ c_{p1} & \dots & c_{pj} & \dots & c_{pm} \end{pmatrix}.$$

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$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[rr, bend left, "S \circ T"] & W \arrow[r, "S"] & U \\
K^n \arrow[u, "\Phi_{\mathcal{A}}"] \arrow[r, "T_{[T]^{\mathcal{A}}_{\mathcal{B}}}"] \arrow[rr, bend right, "T_{[T]^{\mathcal{A}}_{\mathcal{B}} \cdot [S]^{\mathcal{B}}_{\mathcal{C}}}"'] & K^m \arrow[u, "\Phi_{\mathcal{B}}"] \arrow[r, "T_{[S]^{\mathcal{B}}_{\mathcal{C}}}"] & K^p \arrow[u, "\Phi_{\mathcal{C}}"']
\end{tikzcd}$$

We are looking for the unique matrix representing the linear map $T_{[T]^{\mathcal{A}}_{\mathcal{B}} \cdot [S]^{\mathcal{B}}_{\mathcal{C}}} := T_{[T]^{\mathcal{A}}_{\mathcal{B}}} \circ T_{[S]^{\mathcal{B}}_{\mathcal{C}}}$, where T_X just maps $v \to Xv$, via vector-matrix multiplication. We found them, they are given by

$$c_{ij} = \sum_{k=1}^{m} b_{ik} a_{kj},$$

Lecture 21

28.11.2025

So let's try to see what happens in total entry notation

$$\begin{pmatrix} \vdots & & \vdots & & \\ b_{i1} & \dots & b_{ik} & \dots & b_{im} \\ \vdots & & \vdots & & \end{pmatrix} \begin{pmatrix} a_{1j} \\ \vdots \\ a_{kj} \\ \vdots \\ a_{mj} \end{pmatrix} = \begin{pmatrix} & \vdots & \\ \dots & c_{ij} & \dots \\ & \vdots & \end{pmatrix}$$

21 Notice that we take the vector formed by the i-th row of the first matrix and multiply element-wise and sum with the j-th column of the second matrix, giving you a number which you put into the entry i, j of the resulting matrix. Let's look at this purely from the numerical-matrix-entry view.

Definition (Matrix multiplication (17.a.1)). Let $A \in \mathrm{M}_{m \times n}(K)$ and $B \in \mathrm{M}_{n \times p}(K)$. Their entries will be

$$A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}, \quad B = (b_{ij})_{1 \leq j \leq n, 1 \leq j \leq p}.$$

Define a new matrix $C \in \mathrm{M}_{n\times p}(K)$, with entries $(c_{ij})_{1\le i\le m, 1\le j\le p}$, where

$$c_{ij} := a_{i1}b_{1j} + \cdots + a_{ij}b_{kj} + \cdots + a_{in}b_{nj} = \sum_{\boldsymbol{k}=1}^{n} a_{i\boldsymbol{k}}b_{\boldsymbol{k}j},$$

where the index k is highlighted for mnemonic purposes. We denote this new matrix C by $A\cdot B$ and call it the *multiplication* of A and B or the product of A and B.

Remark. Finally, we can truly understand why it makes sense to define matrix multiplication exactly this way. After all, if you had only seen the index by index definition of the product, there would be no real information hidden in there. But because we motivated this definition by looking at the coordinate representation of linear operators between two *arbitrary* finite dimensional vector spaces, we can say that we know where this comes from.

Important — $A\cdot B$ is defined only when #columns A = #rows B.

Another way to describe $A\cdot B$:

$$(A)\begin{pmatrix} | & & | & & \\ z_1 & \dots & z_j & \dots & z_p \\ | & & | & & \end{pmatrix} = \begin{pmatrix} | & & | & & \\ Az_1 & \dots & Az_j & \dots & Az_p \\ | & & | & & \end{pmatrix}.$$

21 **Example.** $\begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 0 \end{pmatrix} =$

$$= \begin{pmatrix} 3\cdot 1 + 2\cdot 0 + 1\cdot 4 & 3\cdot 2 + 2\cdot 1 + 1\cdot 0 \\ 1\cdot 1 + 0\cdot 0 + 2\cdot 4 & 1\cdot 2 + 0\cdot 1 + 2\cdot 0 \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 9 & 2 \end{pmatrix} = A\cdot B.$$

Example. $\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} =$

$$= \begin{pmatrix} 1\cdot 3 + 2\cdot 1 & 1\cdot 2 + 2\cdot 0 & 1\cdot 1 + 2\cdot 2 \\ 0\cdot 3 + 1\cdot 1 & 0\cdot 2 + 1\cdot 0 & 0\cdot 1 + 1\cdot 2 \\ 4\cdot 3 + 0\cdot 1 & 4\cdot 2 + 0\cdot 0 & 4\cdot 1 + 0\cdot 2 \end{pmatrix} = \begin{pmatrix} 5 & 2 & 5 \\ 1 & 0 & 2 \\ 12 & 8 & 4 \end{pmatrix} = B\cdot A$$

Special Cases

1. Let $A \in M_{m\times n}, v \in K^n_{col}$. We defined (long time ago)

$$A\cdot v \in K^m_{col}.$$

We can view v as an $n\times 1$ matrix. Then

$$\begin{pmatrix} & A & \end{pmatrix} \cdot \begin{pmatrix} | \\ v \\ | \end{pmatrix} = A\cdot v.$$

2. Let $B \in M_{n\times p}, v \in K^n_{row}$. We can define $v \cdot B \in M_{1\times p} = K^p_{row}$.

$$\begin{pmatrix} — & v & — \end{pmatrix} \begin{pmatrix} & B & \end{pmatrix} = \begin{pmatrix} —— \end{pmatrix}.$$

So matrix multiplication can also be described like:

$$\text{If } \begin{pmatrix} — & v_1 & — \\ & \vdots & \\ — & v_n & — \end{pmatrix} \begin{pmatrix} & B & \end{pmatrix} = \begin{pmatrix} — & v_1 \cdot B & — \\ & \vdots & \\ — & v_n \cdot B & — \end{pmatrix}.$$

3. Let $v = (a_1, \ldots, a_n) \in K^n_{row} = M_{1\times n}$ and $w = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in K^n_{col} = M_{n\times 1}$. Then $v \cdot w \in M_{1\times 1} = K$:

$$\begin{pmatrix} a_1 & \ldots & a_n \end{pmatrix} \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = a_1 \cdot b_1 + \ldots + a_n \cdot b_n.\text{scalar product}$$

Conclusion — $[S \circ T]^{\mathcal{A}}_e = [S]^{\mathcal{B}}_e \cdot [T]^{\mathcal{A}}_{\mathcal{B}}$.

Notation — Let I be an index set. For all $i, j \in I$, define

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$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j \end{cases}$$

(Kronecker delta).

Definition (Identity matrix). Define the identity matrix $I_n \in M_{n\times n}(K)$ as

$$I_n := (\delta_{ij})_{\substack{1\le i\le n \\ 1\le j\le n}} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Remark. Let V be a finite dimensional vector space, and $n := \dim V$. Let $\mathcal{B}$ be any basis for V. Then $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}} = I_n$. ← check!

Proposition ((17.a.)).

1. $\forall A \in M_{m\times n}, B \in M_{n\times p}, C \in M_{p\times q}$ we have:

$$(A \cdot B) \cdot C = A \cdot (B \cdot C) \quad \text{(associativity)}.$$

Hence, ther is a meaning to write $A \cdot B \cdot C$ without parentheses.

2. $\forall A, B \in M_{m\times n}, C \in M_{n\times p}, C' \in M_{q\times m}$ we have:

$$\begin{aligned} (A + B) \cdot C &= A \cdot C + B \cdot C \quad \text{(left distributivity)} \\ C' \cdot (A + B) &= C' \cdot A + C' \cdot B \quad \text{(right distributivity)}. \end{aligned}$$

3. $\forall A \in M_{m\times n}$ we have:

$$I_m \cdot A = A, \quad I_n \cdot A = A \quad \text{(identity element).}$$

4. $\forall \alpha \in K, A \in M_{m\times n}(K), B \in M_{n\times p}(K)$ then
$\alpha\cdot(A\cdot B) = (\alpha\cdot A)\cdot B = A\cdot(\alpha\cdot B)$ (compatibility with scalar multiplication).

Proof. We need to cook!

1. Let $T_A : K^n \to K^m$, $T_B : K^p \to K^n$, $T_C : K^q \to K^p$ be the linear maps defined using the matrices A, B, C respectively. (i.e. $T_A(v) := A \cdot v$ etc.)
$\forall \ell$, denote by $\mathcal{E}_\ell$ the standard basis for K^ℓ.

$$[T_A]^{\mathcal{E}_m}_{\mathcal{E}_n} = A, [T_B]^{\mathcal{E}_n}_{\mathcal{E}_p} = B, [T_C]^{\mathcal{E}_p}_{\mathcal{E}_q} = C.$$

We have:

$$\underset{(K^p\to K^m)}{\underset{\uparrow}{(T_A \circ T_B)}} \circ \underset{(K^q\to K^p)}{\underset{\uparrow}{T_C}} \quad = \quad \underset{(K^n\to K^m)}{\underset{\uparrow}{T_A}} \circ \underset{(K^q\to K^n)}{\underset{\uparrow}{(T_B \circ T_C)}}$$

Composition is $K^q \to K^m$.

$$\Rightarrow [(T_A \circ T_B)\circ T_C]^{\mathcal{E}_m}_{\mathcal{E}_q} = [T_A \circ (T_B \circ T_C)]^{\mathcal{E}_m}_{\mathcal{E}_q}.$$

By what we have done earlier:

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$$[T_A \circ T_B]^{\mathcal{E}_m}_{\mathcal{E}_p} \cdot [T_C]^{\mathcal{E}_p}_{\mathcal{E}_q} = [T_A]^{\mathcal{E}_m}_{\mathcal{E}_n} \cdot [T_B \cdot T_C]^{\mathcal{E}_n}_{\mathcal{E}_q}$$
$$\left([T_A]^{\mathcal{E}_m}_{\mathcal{E}_n} \cdot [T_B]^{\mathcal{E}_n}_{\mathcal{E}_p}\right) \cdot [T_C]^{\mathcal{E}_p}_{\mathcal{E}_q} = [T_A]^{\mathcal{E}_m}_{\mathcal{E}_n} \cdot \left([T_B]^{\mathcal{E}_n}_{\mathcal{E}_p} \cdot [T_C]^{\mathcal{E}_p}_{\mathcal{E}_q}\right).$$

2. If $T, S : V \to W$ are linear, B is a basis for V, and $\mathcal{E}$ is a basis for W, then

$$[T+S]^B_{\mathcal{E}} = [T]^B_{\mathcal{C}} + [S]^B_{\mathcal{C}}.$$

$$\Rightarrow (A\cdot B)\cdot C = A \cdot (B \cdot C),$$

$$\begin{aligned} 2) \quad \underbrace{\left[(T_A + T_B)\circ T_C\right]^{\varepsilon_p}_{\varepsilon_M}}_{(A+B)C} &= \left[T_A \circ T_C + T_B \circ T_C\right]^{\varepsilon_p}_{\varepsilon_M} \\ &= \left[T_A \circ T_C\right]^{\varepsilon_p}_{\varepsilon_M} + \left[T_B \circ T_C\right]^{\varepsilon_p}_{\varepsilon_M} \\ &= A\cdot C + B \cdot C. \end{aligned}$$

The other identity in 2) – exc.

3.

$$I_m = [\mathrm{id}_{k^m}]^{\varepsilon_m}_{\varepsilon_m},$$
$$\mathrm{id}_{k^m} T_A = T_A \quad \Rightarrow \quad \underbrace{[\mathrm{id}]^{\varepsilon_m}_{\varepsilon_m}}_{=I_m} \cdot A = A.$$

□

Lecture 22

03.12.2025

Last time, we defined matrix multiplication as a way to compose linear maps in matrix form w.r.t. certain bases we picked. Then we defined the special identity matrix and started proving computation rules for matrices. Today we ask how we can invert the matrix, in the similar sense we can invert a map.

Definition (Invertible matrix)**.** An $n \times n$ matrix $A \in \mathrm{M}_{n\times n}$ is called invertible if there exists $B \in \mathrm{M}_{n\times n}$ such that

$$A \cdot B = B \cdot A = I_n.$$

Remark. If $A \in \mathrm{M}_{n\times n}$ is invertible, the inverse B is unique. Indeed, assume that $B, C \in \mathrm{M}_{n\times n}$ and

$$A \cdot B = B \cdot A = I_n, \quad A \cdot C = C \cdot A = I_n,$$

then

$$B = B \cdot I_n = B \cdot (A \cdot C) = (B \cdot A) \cdot C = I_n \cdot C = C,$$

hence $B = C$ and the inverse is unique and we denote it $A^{-1} \in \mathrm{M}_{n\times n}$.

Definition (General linear group)**.** We denote the very special set

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$$\mathrm{GL}_n(K) := \{A \in \mathrm{M}_{n\times n}(K) \mid A \text{ is invertible}\}.$$

Remark. Very clearly, $\mathrm{GL}_n(K) \neq \emptyset$, because $I_n \in \mathrm{GL}_n(K)$ and additionally, $0 \notin \mathrm{GL}_n(K)$, hence $\mathrm{GL}_n(K)$ is a proper subset of $\mathrm{M}_{n\times n}(K)$. It is also immediately *so far from a subspace that you shouldn't even think about it*, as zero is not in there. It is a group though, which will be studied later in other subjects, not really linear algebra.

Proposition (Invertibility rules)**.** Let $A, B \in \mathrm{GL}_n(K)$. Then, $A \cdot B \in \mathrm{GL}_n(K)$ and also $A^{-1} \in \mathrm{GL}_n(K)$ and moreover

$$(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}, \quad \left(A^{-1}\right)^{-1} = A.$$

Proof. We just need to verify this, which is pretty easy. Consider that

$$(B^{-1} \cdot A^{-1}) \cdot (A \cdot B) = B^{-1} \cdot (A^{-1} \cdot A) \cdot B = B^{-1} \cdot I_n \cdot B = B^{-1} \cdot B = I_n$$

and from the other side

$$(A \cdot B) \cdot (B^{-1} \cdot A^{-1}) = A \cdot (B \cdot B^{-1}) \cdot A^{-1} = A \cdot A^{-1} = I_n,$$

hence the product is in $\mathrm{GL}_n(K)$ and it is as claimed.

For the inverse,

$$A \cdot A^{-1} = I_n, \ A^{-1} \cdot A = I_n \Rightarrow \left(A^{-1}\right)^{-1} = A,$$

by definition of inverse applied to A^{-1}. $\square$

Let's see how we can apply this to solve LSEs.

Corollary. Let $A \in \mathrm{GL}_n(K)$. Then $\forall b \in K^n$, the linear system of equations $A \cdot x = b$ has a unique solution, which is $x = A^{-1}b$.

Proof. If $Ax = b$, then $A^{-1}Ax = A^{-1}b$, so $I_n x = x = A^{-1}b$. Vice versa, if $x = A^{-1}b$, then $Ax = b$. $\square$

Proposition (Criterion for invertibility)**.** Let $A \in \mathrm{M}_{n\times n}(K)$. A is invertible if and only if

$$T_A : K^n \to K^n : v \mapsto Av$$

is an isomorphism. Moreover

$$(T_A)^{-1} = T_{A^{-1}}$$

Proof. Assume first that A is invertible. We claim that

$$T_{A^{-1}} \circ T_A = \mathrm{id}_{K^n}$$

and the other way around

$$T_A \circ T_{A^{-1}} = \mathrm{id}_{K^n}.$$

22

This will show that T_A is an isomorphism and that $(T_A)^{-1} = T_{A^{-1}}$. Indeed,

$$\forall v \in K^n : (T_{A^{-1}} \circ T_A)(v) = T_{A^{-1}}(Av) = A^{-1} \cdot (Av) = (A^{-1} \cdot A)v = v,$$

similarly for the other direction.

Conversely, assume that $T_A : K^n \to K^n$ is an isomorphism. Denote

$$S := (T_A)^{-1}.$$

Now S is a linear map $K^n \to K^n$. Recall that for any such linear map $S : K^n \to K^n$, there exists a unique matrix $M \in \mathrm{M}_{n\times n}(K)$ such that

$$S(v) = Mv \Rightarrow S = T_M.$$

So, $\exists B \in \mathrm{M}_{n\times n}$ such that $S = T_B$.

Now, $\forall v \in K^n$, we have

$$v = (S \circ T_A)(v) = (T_B \circ T_A)(v) = BAv \Rightarrow (BA)v = v.$$

Apply this to the standard basis $v = e_1, v = e_2, \ldots, v = e_n$ and we deduce that $B \cdot A = I_n$. In a similar way, one shows $A \cdot B = I_n$. It follows that A is invertible. $\square$

Proposition. Let $A \in \mathrm{M}_{m\times n}$ and $B \in \mathrm{M}_{n\times p}$. Then,

$$(AB)^\top = B^\top A^\top.$$

Moreover, if $A \in \mathrm{GL}_n(K)$, then $A^\top \in \mathrm{GL}_n(K)$ and

$$\left(A^\top\right)^{-1} = \left(A^{-1}\right)^\top.$$

Proof. This is left as a very simple exercise. Use the component definition and verify the candidate for the inverse using the first rule. □

Definition (Triangular and diagonal matrices). A matrix $A \in \mathrm{M}_{n\times n}$ with $A = (a_{ij})_{1\leq i,j\leq n}$ is called *upper triangular* if

$$\forall i > j : a_{ij} = 0.$$

The matrix A is called *lower triangular* if

$$\forall i < j : a_{ij} = 0.$$

The matrix A is called *diagonal* if

$$\forall i \neq j : a_{ij} = 0.$$

Lemma. Let $A, B \in \mathrm{M}_{n\times n}$ be two upper triangular matrices (resp. lower triangular or diagonal) matrices, then the product $A \cdot B \in \mathrm{M}_{n\times n}$ is the same.

Proof. Done in Problem Sheet 11. □

This is allegedly a very important process in the application of applied math *(aka physics)* and other applications. What we want to study is: *How does $[T]^{\mathcal{B}}_{\mathcal{C}}$ depend on the choices*
22 *of $\mathcal{B}$ and $\mathcal{C}$?*

Corollary. Let V, W be finite dimensional vector spaces. Let $n = \dim V, m = \dim W$. Let $\mathcal{B}, \mathcal{B}'$ be two bases for V and $\mathcal{C}, \mathcal{C}'$ two bases for W. Let $T : V \to W$ be a linear map.Then

$$[T]^{\mathcal{B}'}_{\mathcal{C}'} = [\mathrm{id}_W]^{\mathcal{C}}_{\mathcal{C}'} \cdot [T]^{\mathcal{B}}_{\mathcal{C}} \cdot [\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}.$$

Furthermore,

$$[\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}} \in \mathrm{GL}_n(K), [\mathrm{id}_W]^{\mathcal{C}'}_{\mathcal{C}} \in \mathrm{GL}_m(K)$$

and

$$[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}'} = \left([\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}\right)^{-1} \text{ and } [\mathrm{id}_W]^{\mathcal{C}}_{\mathcal{C}'} = \left([\mathrm{id}_W]^{\mathcal{C}'}_{\mathcal{C}}\right)^{-1}$$

Proof. So, the map T is trivially

$$T = \mathrm{id}_W \circ T \circ \mathrm{id}_V = \mathrm{id}_W \circ (T \circ \mathrm{id}_V).$$

Hence, because of definitions,

$$[T]^{\mathcal{B}'}_{\mathcal{C}'} = [\mathrm{id}_W]^{\mathcal{C}}_{\mathcal{C}'} \cdot [T \circ \mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{C}} = [\mathrm{id}_W]^{\mathcal{C}}_{\mathcal{C}'} \cdot [T]^{\mathcal{B}}_{\mathcal{C}} \cdot [\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}.$$

22 For the second identity, consider that $\mathrm{id}_V = \mathrm{id}_V \circ \mathrm{id}_V$. Hence,

$$[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}} = [\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}} \cdot [\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}'},$$

the same for id_W. □

Lecture 23: Equivalence classes of matrices

Last time, at the end, we started inspecting what happens to the matrix of linear map when you change the bases. What we found was that there are two matrices generated by the identity map that are completely independent of the map T itself, that are only dependent on the bases themselves.

Definition (Transition matrix)**.** Let V be a finite dimensional vector space and $n = \dim V$. Let $\mathcal{B}, \mathcal{B}'$ be two bases for V. Then the $n \times n$ matrix

$$[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}'} \in \mathrm{GL}_n(K)$$

is called the change of basis matrix, or transition matrix, between $\mathcal{B}$ and $\mathcal{B}'$.

If we write $\mathcal{B} = (v_1, \ldots, v_n)$ and $\mathcal{B}' = (v_1', \ldots, v_n')$, then by definition,

$$[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}'} = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{B}'} & \ldots & [v_n]_{\mathcal{B}'} \\ | & & | \end{pmatrix}.$$

This makes it especially easy to find these matrices in for example $\mathbb{R}^n$.

23 **Proposition.** Let $T : V \to W$ be an isomorphism between two finite dimensional vector spaces and let $\mathcal{B}$ and $\mathcal{C}$ be bases for V and W, respectively (this forces $n := \dim V = \dim W$). Then

$$[T]^{\mathcal{B}}_{\mathcal{C}} \in \mathrm{GL}_n(K)$$

and moreover,

$$\left[T^{-1}\right]^{\mathcal{C}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{C}}\right)^{-1}.$$

Proof. We have very simply that

$$\left[T^{-1}\right]^{\mathcal{C}}_{\mathcal{B}} \cdot [T]^{\mathcal{B}}_{\mathcal{C}} = [T^{-1} \circ T]^{\mathcal{B}}_{\mathcal{B}} = [\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}} = I_n,$$

and similarly,

$$[T]^{\mathcal{B}}_{\mathcal{C}} \cdot \left[T^{-1}\right]^{\mathcal{C}}_{\mathcal{B}} = \left[T \circ T^{-1}\right]^{\mathcal{C}}_{\mathcal{C}} = [\mathrm{id}_V]^{\mathcal{C}}_{\mathcal{C}} = I_n.$$

□

Let us now consider some endomorphism $T : V \to V$, with $\mathcal{B}, \mathcal{B}'$ two bases for V. We want to ask what is the relation between $[T]^{\mathcal{B}'}_{\mathcal{B}'}$ and $[T]^{\mathcal{B}}_{\mathcal{B}}$. It turns out to be pretty easy.

Corollary. So let $T : V \to V$ be an endomorphism and $\mathcal{B}, \mathcal{B}'$ two bases for V. We have

$$[T]^{\mathcal{B}'}_{\mathcal{B}'} = \left([\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}\right)^{-1} \cdot [T]^{\mathcal{B}}_{\mathcal{B}} \cdot [\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}.$$

Proof. This is an immediate consequence of expanding $T = \mathrm{id}_V \circ T \circ \mathrm{id}_V$ and applying the last theorem. □

We want to be able to compare different matrices and hope to discover some sort of classification for them.

Definition (Equivalent and similar matrices)**.** Let $A, B \in \mathrm{M}_{m\times n}(K)$. We say that A and B are *equivalent* if $\exists P \in \mathrm{GL}_m(K)$ and $\exists Q \in \mathrm{GL}_n(K)$ such that

$$B = PAQ.$$

This time let $A, B \in \mathrm{M}_{n\times n}(K)$, then we say that A and B are *similar*, if $\exists P \in \mathrm{GL}_n(K)$ such that

$$B = P^{-1}AP.$$

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Remark. Equivalence between matrices as defined here is an equivalence relation on $\mathrm{M}_{m\times n}(K)$. Furthermore, $A, B \in \mathrm{M}_{m\times n}(K)$ are equivalent if and only if $\exists$ vector spaces V and W with $\dim V = n, \dim W = m$ and two bases $\mathcal{B}, \mathcal{B}'$ for V and two bases $\mathcal{C}, \mathcal{C}'$ for W and a linear map $T : V \to W$ such that

$$A = [T]^{\mathcal{B}}_{\mathcal{C}}, \quad B = [T]^{\mathcal{B}'}_{\mathcal{C}'}.$$

Similarity is also an equivalence relation on $\mathrm{M}_{n\times n}(K)$. Again, $M, N \in \mathrm{M}_{n\times n}(K)$ are equivalent if and only if $\exists$ n-dimensional vector space V and $T : V \to V$ and two bases $\mathcal{B}$ and $\mathcal{B}'$ for V such that

$$M = [T]^{\mathcal{B}}_{\mathcal{B}}, \quad N = [T]^{\mathcal{B}'}_{\mathcal{B}'}.$$

The proofs are left as an exercise, they should follow from the proofs we already did.

Proposition (Idenitity-like representation of a map)**.** Let V, W be finite dimensional vector spaces over K of dimension $n = \dim V$ and $m = \dim W$. Let $T : V \to W$ be a linear map such that $\mathrm{rank}(T) = \dim \mathrm{Im}(T) = r$. Then there exist a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V and a basis $\mathcal{C} = (w_1, \dots, w_m)$ of W such that

$$[T]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}.$$

Proof. So let $\ell = \dim \mathrm{Ker}(T)$. By the rank theorem $r = n - \ell$. Let $v_1, \dots, v_\ell$ be a basis of $\mathrm{Ker}(T)$. Extend this to a basis $v_1, \dots, v_\ell, v_{\ell+1}, \dots, v_n$ of V. We'll take $\mathcal{B}$ to be

$$\mathcal{B} = (v_{\ell+1}, \dots, v_n, v_1, \dots, v_\ell).$$

We have seen in the proof of the rank theorem (and in a problem sheet, probably) that

$$Tv_{\ell+1}, \dots, Tv_n$$

form a basis for $\mathrm{Im}(T)$. Either way, it's obvious. Then we define

$$w_i = Tv_{\ell+i} \quad 1 \leq i \leq r.$$

Note that $w_r = Tv_{\ell+r} = Tv_n$. Now extend $w_1, \dots, w_r, w_{r+1}, \dots, w_m$ to a basis of W,

$$\mathcal{C} = (w_1, \dots, w_r, w_{r+1}, \dots, w_m).$$

But now

$$[T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | & | & & | \\ [Tv_{\ell+1}]_{\mathcal{C}} & \dots & [Tv_n]_{\mathcal{C}} & [Tv_1]_{\mathcal{C}} & \dots & [Tv_\ell]_{\mathcal{C}} \\ | & & | & | & & | \end{pmatrix}$$

We know that $Tv_1 = \cdots = Tv_\ell = 0$ and $Tv_{\ell+i} = w_i$ for all $1 \leq i \leq r$. The matrix hence takes on the desired form. $\square$

Corollary (Equivalent to identity-like matrix)**.** Take a matrix $A \in \mathrm{M}_{m\times n}(K)$. Then $\exists P \in \mathrm{GL}_m(K)$ and $\exists Q \in \mathrm{GL}_n(K)$ such that

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$$PAQ = \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}$$

where $r = \text{col-rank}(A)$. In particular, $\text{col-rank}(A) \leq \min\{m, n\}$.

Proof. Consider the map $T_A : K^n \to K^m : v \to Av$. Then we have by definition of the column space that

$$\mathrm{ColS}(A) = \mathrm{Im}(T_A),$$

so $r = \mathrm{rank}(T_A)$. By the above proposition, there exist bases $\mathcal{B}$ and $\mathcal{C}$ for K^n and K^m respectively, such that

$$[T_A]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}.$$

But we also know that

$$[T_A]_{\mathcal{C}}^{\mathcal{B}} = [\mathrm{id}_{K^m}]_{\mathcal{C}}^{\mathcal{E}_m} \cdot [T_A]_{\mathcal{E}_m}^{\mathcal{E}_n} \cdot [\mathrm{id}_W]_{\mathcal{E}_n}^{\mathcal{B}},$$

so we arrived at the desired decomposition, as $[T_A]_{\mathcal{E}_m}^{\mathcal{E}_n} = A$, if we let $\mathcal{E}_m$ and $\mathcal{E}_n$ be the respective standard bases. The inequality holds because $\text{col-rank}(A) = \dim \mathrm{Im}(T_A) = n - \dim \mathrm{Ker}(T) \leq n$ and $\text{col-rank}(A) \leq \dim K^m = m$. $\square$

Corollary (Column rank equality is equivalence)**.** Let $A, B \in M_{m\times n}(K)$. Then $A \sim B$ (equivalent) if and only if $\text{col-rank}(A) = \text{col-rank}(B)$.

Proof. Assume $r := \text{col-rank}(A) = \text{col-rank}(B)$. Then

$$A \sim \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}.$$

but also

$$B \sim \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}.$$

Hence by transitivity, $A \sim B$.

Now assume that $A \sim B$. Then by definition,

$$\exists P \in \mathrm{GL}_m(K), Q \in \mathrm{GL}_n(K) \text{ s.t. } B = PAQ.$$

23 This however implies that

$$T_B = T_P \circ T_A \circ T_Q.$$

But therefore,

$$\mathrm{Im}(T_B) = T_B\left(K^n\right) = T_P\left(T_A\left(T_Q\left(K^n\right)\right)\right) = T_P\left(T_A\left(K^m\right)\right),$$

since Q is invertible, hence T_Q is an isomorphism, so we have

$$T_P\left(T_A\left(K^m\right)\right) = T_P\left(\mathrm{Im}(T_A)\right),$$

which has to same dimension as $\mathrm{Im}(T_A)$, as T_P is an isomorphism as well. So in total,

$$\dim \mathrm{Im}(T_B) = \dim \mathrm{Im}(T_A) \Rightarrow \text{col-rank}(B) = \text{col-rank}(A).$$

□

Lecture 24

10.12.2025

Preparations for the proof

Lemma (1 (1.7.c.2)). Let $T : V \to W$ be a lin. map, and let $S : W \xrightarrow{\cong} U$ be an iso, and $L : P \xrightarrow{\cong} V$ also an iso. Then:

1. $\mathrm{rank}(S \circ T) = \mathrm{rank}(T)$.
2. $\mathrm{rank}(T \circ L) = \mathrm{rank}(T)$.

24 *Remind.* — $T : V \to W$, $\mathrm{rank}(T) := \dim(\mathrm{Im}(T))$.

Proof.
1. $\mathrm{rank}(S \circ T) = \dim S(T(V))$. (*)
   Note that $S|_{T(V)} : T(V) \to S(T(V))$ is a linear map which is also an isomorphism. So (*) $= \dim T(V) = \mathrm{rank}(T)$.
2. $\mathrm{rank}(T \circ L) = \dim T(L(P)) = \dim T(V) = \mathrm{rank}(T)$. (because L(P) = V)

exercise — generalize to S is injective. L is surjective. □

Lemma (2 (17.c.3)). Let $A \in M_{n\times n}$. Then A is invertible iff $T_A : K^n \to K^n$ is an isomorphism. Moreover,

$$(T_A)^{-1} = T_{A^{-1}},$$

where

$$T_A(v) := Av.$$

Lemma (3 (17.c.4)). Let $A, B \in M_{m\times n}(K)$. Let P $\in GL_m(K)$, $Q \in GL_n(K)$. Then:

1. col-rank(A) = col-rank(PAQ)
2. row-rank(A) = row-rank(PAQ)

In other words, if $B \sim A$,

$$col - rank(A) = col - rank(B), \quad row - rank(A) = row - rank(B).$$

Proof. 1) Let $B := PAQ$. This means

$$T_B = T_P \circ T_A \circ T_Q.$$

But since T_P and T_Q are isomorphisms by Lemma 2, we know that

$$\text{col-rank}(B) = \text{rank}(T_B) = \text{rank}(T_A) = \text{col-rank}(A).$$

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2) We have $B^\top = Q^\top A^\top P^\top$. Since P and Q are invertible, so are $Q^\top$ and $P^\top$, hence by 1,

$$\text{row-rank}(B) = \text{col-rank}(B^\top) = \text{col-rank}(A^\top) = \text{row-rank}(A)$$

finishing the Lemma. □

Theorem. Let $A \in \mathrm{M}_{m\times n}(K)$. Then col-rank$(A)$ = row-rank(A). We define with this just the rank(A).

Proof. We learned last time that there exist $P \in GK_m(K), Q \in \mathrm{GL}_n(K)$ such that

$$PAQ = \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix} := B$$

for some $r \geq 0$ and denote by B this matrix.

$$\begin{array}{cc} & \qquad\quad \overset{r}{\longleftrightarrow} \\ r \updownarrow & \left(\begin{array}{cc|c} 1 & 0 & \\ & \ddots & 0 \\ 0 & 1 & \\ \hline \multicolumn{2}{c|}{0} & 0 \end{array}\right) =: B \end{array}$$

Clearly

$$row - rank(B) = col - rank(B) = .$$

But row-rank(A) = row-rank$(B) = r$ Lemma 17.c.4 col-rank(A) = col-rank$(B) = r$

$$\Rightarrow \text{row-rank}(A) = \text{col-rank}(A).$$

□

From now on: rank $(A) :=$ row-rank(A) = col-rank(A).

Remark. Suppose A' is a row-reduced echelon matrix which is row-equivalent to A. We know that RowS(A) = RowS(A'). So rank(A) = rank(A'). But rank(A') = # pivots in A.

2.16.2 Back to linear equations

Let $A \in M_{m\times n}(K)$, $b \in K^m$. Consider the system of linear equations $A \cdot x = b$.

$$\text{Denote by. } \operatorname{Sol}(A, b) := \{x \in K^n : A \cdot x = b\}.$$

set of solutions. We know that for $b = 0$, Sol$(A, 0)$ is a vector space (a subspace of K^n).

Claim. — $\dim \operatorname{Sol}(A, 0) = n - \operatorname{rank}(A)$. This is b.c.:

24

$$\dim \operatorname{Sol}(A, 0) = \dim \ker(T_A) = n - \dim \operatorname{Im}(T_A) = n - \operatorname{rank}(A) \text{ (by rank theorem).}$$

$A \in M_{m\times n}, T_A : K^n \to K^m$.
Im (T_A) = ColS(A).
rank(T_A) = rank(A).

Let A' be a row-reduced echelon matrix which is row-equivalent to A.

$$\underbrace{\begin{pmatrix} 0 & \dots & 0 & 1 & * & \dots & \dots & * & \\ & & & & 1 & * & \dots & * & \\ & & & & & 1 & \dots & * & \\ & 0 & & & & & \ddots & \vdots & \\ & & & & & & & 1 & * \cdots * \end{pmatrix}}_{A'} \begin{pmatrix} x_1 \\ \vdots \\ \vdots \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ \vdots \\ 0 \end{pmatrix}$$

$$\#\ \text{pivots}(A') = \operatorname{rank}(A') = \operatorname{rank}(A).$$

$$n - \operatorname{rank}(A') = \#\text{free variables} = \dim \operatorname{Sol}(A', 0) = \dim \operatorname{Sol}(A, 0).$$

$$\operatorname{Sol}(A, 0) = \operatorname{Sol}(A', 0).$$

Denote by $x_{i_1}, \dots, x_{i_\ell}$ the free variables $1 \le i_1 < \dots < i_\ell \le n, \ell = n - \operatorname{rank}(A)$. Denote by $x_{j_1}, \dots, x_{j_r}, 1 \le j_1 < \dots < j_r \le n, r = \operatorname{rank}(A)$ the pivot variables. For each choice of values for the free variables $x_{i_1}, \dots, x_{i_\ell}$, the pivot variables are uniquely determined.

In other words we have a map

$$\Phi : K^\ell \to \operatorname{Sol}(A, 0).$$

$\Phi(\lambda_1, \dots, \lambda_\ell) :=$ the unique solution of $A \cdot x = 0$ or $A' \cdot x = 0$ with

$$x_{i_1} = \lambda_1, \dots, x_{i_\ell} = \lambda_\ell.$$

Proposition (17.d.1)**.** The map Φ is linear, and moreover it is an isomorphism.

Proof. Linearity of Φ. Consider row #k of A' and of $A' \cdot x = 0$. If $1 \le k \le r$, it looks like

$$x_{j_k} + \big(\text{linear combin. of free variables that come after } j_k\big) = 0$$

$$\text{So,} \quad \underset{\substack{\uparrow \\ \text{pivot} \\ \text{var } \#k}}{x_{j_k}} = - \sum_{\{q \mid 1 \le q \le l,\ i_q > j_k\}} a'_{k,iq} \cdot \underset{\substack{\uparrow \\ \text{free} \\ \text{vars,}}}{x_{i_q}} \qquad \begin{pmatrix} \text{the } * \\ \text{coeffs} \\ \text{in row} \\ \#k \text{ of } A'. \end{pmatrix}$$

$\Rightarrow x_{j_1}, \dots, x_{j_r}$ (the pivots) depend linearly on the free variables $x_{i_1}, \dots, x_{i_\ell}$. This
24 shows Φ is linear. We will show that Φ is an isomorphism. We will 1st show $\ker(\Phi) = \{0\}$.
If $\Phi(\lambda_1, \dots, \lambda_\ell) = 0$, then by definition $\lambda_1 = \dots = \lambda_\ell = 0$.
But $\Phi : K^\ell \to \operatorname{Sol}(A, 0)$, and $\dim \operatorname{Sol}(A, 0) = n - \operatorname{rank}(A) = n - \#\text{pivots}(A')\#\text{freevariables} = \ell$. Since Φ is injective, Φ is an isomorphism.

$$\Phi^{-1} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_{i_1} \\ \vdots \\ x_{i_\ell} \end{pmatrix}.$$

Sol(A,0) = the values of the free variables. □

Corollary (17.d.2)**.** Let $b \in K^m$, $A \in M_{m \times n}$. Then:

1. Sol(A,b)$\neq \emptyset$ iff $b \in \operatorname{ColS}(A)$.
2. If Sol(A,b) $\neq \emptyset$, and $y \in \operatorname{Sol}(A, b)$, then

$$\operatorname{Sol}(A, b) = y + \operatorname{Sol}(A, 0) := \{y + x : x \in \operatorname{Sol}(A, 0)\}.$$

Proof. 1. If $A = \begin{pmatrix} | & \dots & | \\ v_1 & \dots & v_n \\ | & \dots & | \end{pmatrix}$, $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$, then $A \cdot x = x_1 \cdot \begin{pmatrix} | \\ v_1 \\ | \end{pmatrix} + \dots + x_n \cdot \begin{pmatrix} | \\ v_n \\ | \end{pmatrix}$.

2. If $x \in \mathrm{Sol}(A, 0)$, then

$$A(y + x) = Ay + Ax = b.$$

$$\text{If } z \in \mathrm{Sol}(A, b), \text{ then } x := z - y \in \mathrm{Sol}(A, 0).$$

□

Proposition ((17.d.3)). Let $A \in M_{m\times n}$, $b \in k^m$. Then the following statements are equiv.:

1. $\mathrm{rank}(A \mid b) = \mathrm{rank}(A)$
2. The syst. of eq. $Ax = b$ has a solut.

Proof. Write $A = \begin{pmatrix} v_1 & \dots & v_n \end{pmatrix}$.

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$$\begin{aligned}(1) &\Leftrightarrow \text{col-rank}(A \mid b) = \text{col-rank}(A)\\ &\Leftrightarrow \mathrm{Sp}(v_1, \dots, v_n, b) = \mathrm{Sp}(v_1, \dots, v_n)\\ &\Leftrightarrow b \in \mathrm{Sp}(v_1, \dots, v_n) \Leftrightarrow b \in \mathrm{colS}(A)\\ &\Leftrightarrow \exists \text{ solut. to } Ax = b.\end{aligned}$$

□

Proposition (17.d.4). Let $A \in M_{m\times n}$ and $b \in K^m$. Then the following are equivalent:

1. $\mathrm{rank}(A) = \mathrm{rank}(A \mid b) = n$.
2. The system $Ax = b$ has a unique solution.

Proof.

Outline of proof.

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}.$$

$$\mathrm{rank}(A) = n \iff \dim \mathrm{Sp}(v_1, \dots, v_n) = n \iff v_1, \dots, v_n \text{ form a basis for } \mathrm{Cols}(A).$$

□

Lecture 25

12.12.2025

Elementary row operations

Main point. These operatioins can be written by matrix multiplication. *Notation.* Let $n \in \mathbb{Z}_{\geq 1}$, $1 \leq i \leq n$, $1 \leq j \leq n$. Denote by $E_{i,j} \in M_{n\times n}(K)$ the matrix that has all entries 0, except (i,j)-where the entry is 1.

$$E_{i,j} = \left(\begin{array}{ccc|c|ccc} 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \hline 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ \hline 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array}\right),$$

We will define now 3 other matrices: **Type I.** Let $1 \leq i \leq n$, $1 \leq j \leq n$, $i \neq j$, and let $\alpha \in K$. Define $Q_{i,j}(\alpha)$ to be the matrix obtained from I_n after applying $R_i + \alpha R_j \longrightarrow R_1$.

We have $Q_{i,j}(\alpha) = I_n + \alpha E_{i,j}$..

25

$$Q_{i,j}(\alpha) := I_n + \alpha E_{ij} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} + \alpha \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \alpha & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

Type II. Let $1 \leq i \leq n$, $1 \leq j \leq n$. Define $P_{i,j}$ to be the matrix obtained from I_n by exchanging row #i and row #j. ($R_i \longleftrightarrow R_j$)

$$P_{i,j} = \left(\begin{array}{ccc|c|c|ccc} \ddots & & & \vdots & \vdots & & & \\ & \ddots & & \vdots & \vdots & & & \\ & & \ddots & \vdots & \vdots & & & \\ \hline \cdots & \cdots & \cdots & 0 & 1 & \cdots & \cdots & \cdots \\ \hline & & & \vdots & \vdots & \ddots & & \\ \hline \cdots & \cdots & \cdots & 1 & 0 & \cdots & \cdots & \cdots \\ \hline & & & \vdots & \vdots & & \ddots & \\ & & & \vdots & \vdots & & & \ddots \end{array}\right), \qquad (i \neq j).$$

Type III. Let $1 \leq i \leq n$, $0 \neq \alpha \in K$. Denote by $S_i(\alpha)$ the matrix obtained from I_n by applying $\alpha R_i \longrightarrow R_i$.

$$S_i(\alpha) = \begin{array}{c} \hspace{3em}\text{col } i \\ \left(\begin{array}{ccc|c|ccc} 1 & 0 & \cdots & 0 & \cdots & 0 & 0 \\ 0 & \ddots & & \vdots & & & \vdots \\ \vdots & & 1 & 0 & & & 0 \\ \hline 0 & \cdots & 0 & \alpha & 0 & \cdots & 0 \\ \hline 0 & & & 0 & 1 & & \vdots \\ \vdots & & & \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 & 1 \end{array}\right) \end{array} \begin{array}{l} \\ \\ \\ \\ \xleftarrow{\text{row } i} \\ \\ \\ \\ \end{array}$$

Example. - In $M_{4\times 4}(K)$ $Q_{1,3}(\alpha) = \begin{pmatrix} 1 & 0 & \alpha & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ $Q_{3,1}(\beta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \beta & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$P_{1,2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ $P_{2,4} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$ $S_3(\alpha) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \alpha & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

25 **Lemma** (17.e.1)**.** Let $A \in M_{n\times p}(K)$.

1. Multiplying A from the left by $Q_{i,j}(\alpha)$ results in applying the row operation $R_i + \alpha R_j \longrightarrow R_i$ to A.
2. Multiplying A from the left by $P_{i,j}$ gives the application of $R_i \longleftrightarrow R_j$ on A.
3. Multiplying A from the left by $S_i(\alpha)$ results in applying $\alpha R_i \longrightarrow R_i$ to A.

Proof. Can be done by explicit straightforward calculation.

$$q_{ij}(\alpha) = \begin{pmatrix} \ddots & & & \\ & 1 & \alpha & \\ & 0 & 1 & \\ & & & \ddots \end{pmatrix}$$

$$Q_{ij}(\alpha) = \underline{I_n} + \alpha E_{ij}$$

$$Q_{ij}(\alpha) \cdot A = \left(\underline{I_n} + \alpha E_{ij}\right) \cdot A$$

$$= A + \alpha\, E_{ij} \cdot A$$

$$E_{ij}A = \begin{pmatrix} 0 \\ \vdots \\ \text{row } j \text{ of } A \\ \vdots \\ 0 \end{pmatrix} \quad \text{(this appears in row } i\text{)}$$

$$\Rightarrow \quad Q_{ij}(\alpha)A \text{ is } A \text{ but with row } i \leftarrow \text{row } i + \alpha \cdot (\text{row } j \text{ of } A).$$

$$|\,|\,| \qquad \text{check details} \ldots$$

□

Elementary column operations.

1. ' Multiplying A from the right by $Q_{i,j}(\alpha)$ results the column operation $A \xleftarrow{C_j+\alpha C_i \longrightarrow C_j} A \cdot Q_{i,j}(\alpha)$.

2. ' Multiplying A from the right by $P_{i,j}$ results in $C_i \longleftrightarrow C_j$.

$$A \xleftarrow{C_i \longleftrightarrow C_j} A \cdot P_{i,j}.$$

3. ' Multiplying A from the right by $S_i(\alpha)$ results in $\alpha C_i \longrightarrow C_i$.

25

$$A \xleftarrow{\alpha C_i \longrightarrow C_i} A \cdot S_i(\alpha).$$

To prove that 1'-3' just use transposed matrices.

$$\text{e.g.} \quad (A \cdot Q_{i,j}(\alpha))^\top = (*) Q_{i,j}(\alpha)^\top \cdot A^\top.$$

$$\text{, but } Q_{i,j}(\alpha)^\top = Q_{j,i}(\alpha).$$

$$\text{So } (*) = Q_{j,i}(\alpha) \cdot A^\top$$

$$= \text{the result of applying } R_j + \alpha R_i \longrightarrow R_j \text{ to } A^\top.$$

$$\Rightarrow A \cdot Q_{i,j}(\alpha) = (*)^\top \text{ is the result of applying } C_j + \alpha C_i \longrightarrow C_j.$$

ect. (exc. complete the proof).

Lemma (17.e.2)**.** Every elementary matrix is invertible and the inverse of each of them is also an elementary matrix. In fact:

$$(Q_{i,j}(\alpha))^{-1} = Q_{i,j}(-\alpha), \quad (P_{i,j})^{-1} = P_{i,j}, \quad (S_i(\alpha))^{-1} = S_i(\alpha^{-1}).$$

Proof. $Q_{i,j}(-\alpha) \cdot Q_{i,j}(-\alpha)$ = applying $R_i + (-\alpha)R_j$ to the matrix $Q_{i,j}(\alpha)$ <− (apply $R_i + \alpha R_j \longrightarrow R_i$ to I_n) = Applying ist $(R_i + \alpha R_j \longrightarrow R_i)$ to I_n and then $R_i + (-\alpha)R_j \longrightarrow R_i$ to the previous result = do nothing to $I_n = I_n$. Similary one shows $Q_{i,j}(\alpha) \cdot Q_{i,j}(-\alpha) = I_n$. *exercise* — complete the proof (i.e. check for $P_{i,j}$ and $S_i(\alpha)$). □

Theorem (17.e.3). $\forall A \in M_{m\times n}(K)$, $\exists k \geq 1$, elementary matrices $T_1, \dots, T_k$ such that

$$T_k \cdots T_{k-1} \cdot \dots \cdot T_1 \cdot A = I_n.$$

Furthermore we have $A = T_1^{-1} \cdots T_k^{-1}$, and $A^{-1} = T_k \cdots T_1$.

Proof. We proved that $\forall A \in M_{m\times n}(K)$ can be brought to row-reduced echelon matrix A' by applying a sequence of elementary row operations. Since A is invertible, rank(A) = n. Now rank(A') = rank $(A) = n$.

$$\begin{aligned} &\Rightarrow A' \text{has n pivots.} \\ &\Rightarrow A' = I_n.. \end{aligned}$$

But A' is obtained from A by applying a sequence of elementary row operations, hence $\exists$ elementary matrices $T_1, \dots, T_k$ such that

$$T_k \left(\cdots \left(T_2 \left(T_1 A\right)\right) \cdots\right) = A' = I_n.$$

The rest follows by straightforward calculation. □

Theorem (17.e.4). Let $A \in M_{n\times n}(K)$. Place A together with I_n on the right to A $(A \mid I_n)$. Then A is invertible iff the matrix $(A \mid I_n)$ can be brought by a sequence of elementary row operations to the form $(I_n \mid B)$ for some $B \in M_{n\times n}(K)$. In this case, $B = A^{-1}$.

Lemma (17.e.5). Let B, C, D be matrices $n \times n$. Then

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$$B \cdot (C \mid D) = (B \cdot C \mid B \cdot D).$$

Proof — Exercise (by direct calculation.)

Proof. Assume A is invertible. By Theorem 17.e.3, $\exists$ elementary matrices $T_1, \dots, T_k$ such that

$$T_k \cdots T_{k-1} \cdot \dots \cdot T_1 \cdot A = I_n.$$

By Lemma 17.e.5,

$$T_k \cdots T_{k-1} \cdot \dots \cdot T_1 \cdot (A \mid I_n) = (I_n \mid \underbrace{T_k \cdots T_1}_{\text{denote by } B}).$$

By Theorem 17.e.3, $A^{-1} = B$. But $T_k \cdots T_1(A \mid I_n)$ is precisely the matrix obtained from $(A \mid I_n)$ after applying a sequence of k elementary row operations. Conversely, assume $(A \mid I_n)$ can be brought to $(I_n \mid B)$ by a sequence of elementary row operations. $\Rightarrow$A itself can be brought to I_n by the same sequence of elementary row operations.

$$\begin{aligned} &\Rightarrow \text{A is row-equivalent to } I_n \\ &\Rightarrow \text{A is invertible} \end{aligned}$$

.

Exercise — show that $B = A^{-1}$. □

Example. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in M_{2\times 2}(\mathbb{Q})$.

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right) \xrightarrow{R_2 - 3R_1 \to R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right).$$

25 $$\xrightarrow{-\frac{1}{2}R_2 \to R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right) \xrightarrow{R_1 - 2R_2 \to R_1} \left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right).$$

So $A^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$.

V, W vector spaces $\Rightarrow \operatorname{Hom}(V,W)$ is also a vector space. If $V \& W$ are finite-dimensional, then $\operatorname{Hom}(V,W)$ is also finite-dimensional and $\dim \operatorname{Hom}(V,W) = \dim V \cdot \dim W$.

Lecture 26

17.12.2025

We want to learn about new vector spaces that naturally arise from other vector spaces we have studied this far. We start with $\operatorname{Hom}(V,W)$ for vector spaces V and W over a common field K.

Proposition (Hom is a vector space)**.** Given V, W vector spaces over K, $\operatorname{Hom}(V,W)$ has the structure of a vector space over K as follows:

- $T_1, T_2 \in \operatorname{Hom}(V,W)$, we define $(T_1 + T_2) : V \to W$ as $(T_1 + T_2)(v) = T_1(v) + T_2(v) \forall v \in V$.
- $\forall T \in \operatorname{Hom}(V,W), \forall \alpha \in K$, we define $(\alpha T)(v) = \alpha(T(v)) \forall v \in V$.

Proof. We 1st claim that $T_1 + T_2$ is a linear map. Indeed, $\forall a \in K, v \in V$ we have

26 $$\begin{aligned}(T_1 + T_2)(av) &= T_1(av) + T_2(av) \\ &= aT_1(v) + aT_2(v) \\ &= a(T_1(v) + T_2(v)) \\ &= a(T_1 + T_2)(v)..\end{aligned}$$

Let $v_1, v_2 \in V$. We have $(T_1 + T_2)(v_1 + v_2) = T_1(v_1 + v_2) + T_2(v_1 + v_2) = T_1(v_1) + T_1(v_2) + T_2(v_1) + T_2(v_2) = (T_1 + T_2)(v_1) + (T_1 + T_2)(v_2)$. This shows $T_1 + T_2) : V \to W$ is linear. Next one shows that αT is also a linear map (exc.) Now one needs to check that the operations

$$(T_1, T_2) \longmapsto T_1 + T_2, \qquad (\alpha, T) \longmapsto \alpha \cdot T$$

satisfy all the axioms of a vector space.

The neutral element, $0 \in \operatorname{Hom}(V,W)$, is the zero map

$$0 : V \to W, \qquad v \mapsto 0.$$

We denote it by 0.

One needs to verify that

$$0 + T = T, \qquad T + 0 = T,$$

etc.

This completes the proof. $\square$

Theorem. Let V, W be *finite* dimensional vector spaces and $\mathcal{B} = (v_1, \dots, v_n)$ a basis for V and $\mathcal{C} = (w_1, \dots, w_m)$ a basis for W. Define

$$\Psi_{\mathcal{C}}^{\mathcal{B}} : \mathrm{Hom}(V, W) \to \mathrm{M}_{m\times n}(K)$$

by the value

$$\Psi_{\mathcal{C}}^{\mathcal{B}}(T) := [T]_{\mathcal{C}}^{\mathcal{B}} \in \mathrm{M}_{m\times n}(K).$$

$$\begin{pmatrix} | & | & & | \\ T(v_1)_{\mathcal{C}} & T(v_2)_{\mathcal{C}} & \cdots & T(v_n)_{\mathcal{C}} \\ | & | & & | \end{pmatrix}.$$

Then, the map $\Phi_{\mathcal{C}}^{\mathcal{B}}$ is linear and an isomorphism.

Proof. Let $T \in \mathrm{Hom}(V, W)$, and

$$M := [T]_{\mathcal{C}}^{\mathcal{B}} \in M_{m\times n}(K).$$

26

Recall: we have

$$T_M = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}.$$

$$\begin{tikzcd} V \arrow[r, "T"] & W \arrow[d, "\Phi_{\mathcal{C}}"] \\ K^n \arrow[u, "\Phi_{\mathcal{B}}^{-1}"] \arrow[r, "T_M"'] & K^m \end{tikzcd}$$

$$\text{where } \Phi_{\mathcal{B}}(v) := [v]_{\mathcal{B}} \quad \forall v \in V,$$
$$\Phi_{\mathcal{C}}(w) := [w]_{\mathcal{C}} \quad \forall w \in W.$$

$$\Phi_{\mathcal{B}}^{-1}\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1 v_1 + \cdots + a_n v_n,$$

$$\Phi_{\mathcal{C}}^{-1}\begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} = b_1 w_1 + \cdots + b_m w_m.$$

26

We first claim that $\Psi_{\mathcal{C}}^{\mathcal{B}}$ is bijective. To do this, define

$$\Theta : M_{m\times n}(K) \longrightarrow \mathrm{Hom}(V,W).$$

For all $A \in M_{m\times n}(K)$,

$$\Theta(A) := \Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}.$$

$$\begin{tikzcd} V \arrow[r, dashed, "\Theta(A)"] \arrow[d, "\Phi_{\mathcal{B}}"'] & W \\ K^n \arrow[r, "T_A"'] & K^m \arrow[u, "\Phi_{\mathcal{C}}^{-1}"'] \end{tikzcd}$$

Consider $\Psi_{\mathcal{C}}^{\mathcal{B}} \circ \Theta : M_{m\times n}(K) \to M_{m\times n}(K)$. Let $A \in M_{m\times n}(K)$. Then

$$\begin{aligned}(\Psi_{\mathcal{C}}^{\mathcal{B}} \circ \Theta)(A) &= \Psi_{\mathcal{C}}^{\mathcal{B}}\big(\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}\big) \\ &= \begin{pmatrix} | & & | \\ \big[(\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}})(v_1)\big]_{\mathcal{C}} & \cdots & \big[(\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}})(v_n)\big]_{\mathcal{C}} \\ | & & | \end{pmatrix}.\end{aligned}$$

□

Proof. Let $A \in M_{m\times n}(K)$. Then

$$\begin{aligned}\begin{pmatrix} | & & | \\ T_A \circ \Phi_{\mathcal{B}}(v_1) & \cdots & T_A \circ \Phi_{\mathcal{B}}(v_n) \\ | & & | \end{pmatrix} &= \begin{pmatrix} | & & | \\ T_A e_1 & \cdots & T_A e_n \\ | & & | \end{pmatrix} \qquad (\text{since } \Phi_{\mathcal{B}}(v_j) = e_j,\ j = 1,\dots,n) \\ &= A.\end{aligned}$$

Therefore,

$$\Rightarrow\ \Psi_{\mathcal{C}}^{\mathcal{B}} \circ \Theta = \mathrm{id}_{M_{m\times n}(K)}\,.$$

Now let $T \in \mathrm{Hom}(V,W)$ and set $M := \Psi_{\mathcal{C}}^{\mathcal{B}}(T)$. Consider $\Theta \circ \Psi_{\mathcal{C}}^{\mathcal{B}}(T) = \Theta(M)$:

$$\Theta(M) = \Phi_{\mathcal{C}}^{-1} \circ T_M \circ \Phi_{\mathcal{B}} = T \qquad \text{(by the 1st diagram we had earlier).}$$

Hence,

$$\Theta \circ \Psi_{\mathcal{C}}^{\mathcal{B}} = \mathrm{id}_{\mathrm{Hom}(V,W)}\,.$$

It remains to show that $\Psi_{\mathcal{C}}^{\mathcal{B}}$ is linear. Indeed, let $\alpha \in K$ and $T \in \operatorname{Hom}(V, W)$. Then

$$\begin{aligned}
\Psi_{\mathcal{C}}^{\mathcal{B}}(\alpha T) &= \begin{pmatrix} | & & | \\ [(\alpha T)(v_1)]_{\mathcal{C}} & \cdots & [(\alpha T)(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \\
&= \begin{pmatrix} | & & | \\ \alpha[T(v_1)]_{\mathcal{C}} & \cdots & \alpha[T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \\
&= \alpha \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \\
&= \alpha \Psi_{\mathcal{C}}^{\mathcal{B}}(T).
\end{aligned}$$

Similarly, $\Psi_{\mathcal{C}}^{\mathcal{B}}(T_1 + T_2) = \Psi_{\mathcal{C}}^{\mathcal{B}}(T_1) + \Psi_{\mathcal{C}}^{\mathcal{B}}(T_2)$. (exc.) $\square$

Corollary (Dimension of $\operatorname{Hom}(V, W)$)**.** If V, W are finite dimensional then so is $\operatorname{Hom}(V, W)$ and

$$\dim \operatorname{Hom}(V, W) = \dim V \cdot \dim W.$$

Proof. Let $n = \dim V, m = \dim W$. By the previous theorem,

$$\operatorname{Hom}(V, W) \cong \mathrm{M}_{m \times n}(K).$$

Since $\mathrm{M}_{m\times n}(K)$ is finite dimensional, so it $\operatorname{Hom}(V, W)$ and $\dim \operatorname{Hom}(V, W) =$
26 $\mathrm{M}_{m\times n}(K) = m \cdot n$. $\square$

Definition (Ring)**.** A ring $(R, +, \cdot)$ is defined similarly to a field, but we do *not* require that $a \cdot b = b \cdot a$, and we do not require that $\forall a \in R \setminus \{0\}\ \exists a^{-1}$. The element 1 is called the unit or the unity of R.

Example.

1. Every field K is a commutative ring.
2. $R = \mathbb{Z}$ is a commutative ring.
3. $M_{n\times n}(K)$ is a ring. Multiplication: $A, B \in M_{n\times n}(K)$, then $A \cdot B$ is the multiplication of A & B as matrices. I_n is the unity of this ring. We get a non-commutative ring.

$$\begin{aligned}
A \cdot (B + C) &= A \cdot B + A \cdot C, \\
A \cdot (B + C) &= (A \cdot B) \cdot C..
\end{aligned}$$

Definition (Homomorphism of rings)**.** Let R, S be rings. A *homomorphism of rings* is a map $\Psi : R \to S$ such that $\forall r_1, r_2 \in R$,

$$\begin{aligned}\Psi(r_1 + r_2) &= \Psi(r_1) + \Psi(r_2),\\ \Psi(r_1 \cdot r_2) &= \Psi(r_1) \cdot \Psi(r_2),\\ \Psi(1) &= 1,\\ \Psi(0) &= 0.\end{aligned}$$

Corollary (18.a.4). Let V be a vector space over a field K. Then $\mathrm{End}(V) := \mathrm{Hom}(V, V)$ is a ring (with a unit), in general non-commutative, with the following operations:

+ *operation:* — the same as before for $\mathrm{Hom}(V, V)$.
· *operation:* $T_1 \cdot T_2 := T_1 \circ T_2$ $\forall T_1, T_2 \in \mathrm{End}(V)$. The unity is id_V. Moreover, if V is finite-dimensional, and $B = (v_1, \dots, v_n)$ is a basis of V, then $\Phi_B^B : \mathrm{End}(V) \to M_{n\times n}(K)$ is an isomorphism of rings.

Proof. $\forall T_1, T_2, T_3 \in \mathrm{Hom}(V, V)$,

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$$T_1 \circ (T_2 \circ T_3) = (T_1 \circ T_2) \circ T_3 \quad \Rightarrow \text{(multiplication in End}(V) \text{ is associative).}$$

$$T \circ id_V = id_V \circ T = T \forall T \in \mathrm{End}(V) \quad \Rightarrow id_V \text{ is the unity.}$$

Also $\forall T, T_1, T_2$

$$T \cdot (T_1 + T_2) = T \circ (T_1 + T_2) = T \circ T_1 + T \circ T_2 = T \cdot T_1 + T \cdot T_2.$$

etc... (check all the other axioms of a ring)

For the second part, assume that V is finite dimensional. Let $T_1, T_2 \in$. We already know that

$$\Psi_{\mathcal{B}}^{\mathcal{B}}(T_1 + T_2) = \Psi_{\mathcal{B}}^{\mathcal{B}}(T_1) + \Psi_{\mathcal{B}}^{\mathcal{B}}(T_2)$$

and that $\Psi_{\mathcal{B}}^{\mathcal{B}}$ is bijective. For the second requirement, notice that

$$\Psi_{\mathcal{B}}^{\mathcal{B}}(T_1 \circ T_2) = [T_1 \circ T_2]_{\mathcal{B}}^{\mathcal{B}} = [T_1]_{\mathcal{B}}^{\mathcal{B}} \cdot [T_2]_{\mathcal{B}}^{\mathcal{B}} = \Psi_{\mathcal{B}}^{\mathcal{B}}(T_1) \cdot \Psi_{\mathcal{B}}^{\mathcal{B}}(T_2),$$

hence this is a ring isomorphism. □

Lecture 27: Last lecture of Linear Algebra 1

19.12.2025

2.17 Dual Space

Special case of $\mathrm{Hom}(V, W)$ when $W = K$.

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Definition. Let Let V be a vector space over the field K. The *dual space* of V is defined

$$V^* := \mathrm{Hom}(V, K).$$

The elements of V^* are called *linear maps* $\ell : V \to K$ and are sometimes called *functionals* on V.

Corollary. Let V be a finite-dimensional vector space K. Then V^* is also finite-dimensional and $\dim V^* = \dim V$.

Proof. $\dim V^* = \dim \operatorname{Hom}(V, K) = \dim V \cdot \dim K = \dim V$ □

Example. Take $V = K^n_{col}$. We will identify V^* with K^n_{row}. If $\ell \in K^n_{row}$, then ℓ defines a functional $f_\ell : K^n_{col} \to K$ by

$$f_\ell(v) = \ell \cdot v \in M_{1\times 1}(K) \cong K.$$

The map

$$D : K^n_{row} \to (K^n_{col})^*, \qquad D(\ell) := f_\ell.$$

Exercise — D is linear and it is an isomorphism.

Let V be a finite-dimensional vector space over K, and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis of V. Define n functionals $v_1^*, \dots, v_n^* \in V^*$ as follows: $v_i^* : V \to K$ is the unique linear map such that

$$v_i^*(v_j) = \delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

In other words, $v_i^*(v_i) = 1$ and $v_i^*(v_j) = 0 \forall j \neq i$. Since B is a basis for V, this determines a functional $v_i^* : V \to K$ uniquely.

27 **Lemma** (18.a.7). The elements $v_1^*, \dots, v_n^* \in V^*$ form a basis for V^*. We denote this basis by $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$. It is called the *dual basis* of $\mathcal{B}$. Moreover, $\forall \ell \in V^*$,

$$\ell = \sum_{i=1}^n \ell(v_i) v_i^*.$$

$$\text{i.e. } \forall v \in V, \ell(v) = \sum_{i=1}^n \ell(v_i) v_i^*(v).$$

Proof. Let $\ell \in V^*$. We claim that

$$\ell = \sum_{i=1}^n \ell(v_i)\, v_i^*, \tag{1}$$

id est

$$\forall v \in V, \qquad \ell(v) = \sum_{i=1}^n \ell(v_i)\, v_i^*(v). \tag{2}$$

LHS of *(1)* is a linear map $V \to K$, and so is also RHS of *(1)*. So to show LHS = RHS it is enough to check *(2)* on the elements of the basis of V, exempli gratia with the basis $\mathcal{B} = (v_1, \dots, v_n)$. So we will check *(2)* for $v = v_1, \dots, v = v_n$.

Let $1 \leq k \leq n$, and substitute $v = v_k$ in *(2)*:

$$\text{RHS} = \sum_{i=1}^{n} \ell(v_i) \underbrace{v_i^*(v_k)}_{=\delta_{ik}} = \ell(v_k) \cdot \underbrace{\delta_{kk}}_{=1} = \ell(v_k).$$

So *(2)* holds for $v = v_k$.

The formula we have just proved, shows also that $v_1^*, \dots, v_n^*$ span V^*. But $\dim V^* = n$, hence $v_1^*, \dots, v_n^*$ form a basis of V^*. □

Question — Let V be a finite-dimensional vector space over K. Let $\mathcal{B}, \mathcal{C}$ be two bases of V. Then we also get two bases $\mathcal{B}^*, \mathcal{C}^*$ for V^*. What is the relationship between $[\mathrm{id}_V^*]_{\mathcal{C}^*}^{\mathcal{B}^*}$ and $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{C}}$?

Proposition (18.a.8)**.**

$$\begin{aligned} [\mathrm{id}_V^*]_{\mathcal{B}^*}^{\mathcal{C}^*} &= \left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}}\right)^{\top}, \\ [\mathrm{id}_V^*]_{\mathcal{C}^*}^{\mathcal{B}^*} &= \left(\left([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{C}}\right)^{\top}\right)^{-1}. \end{aligned}$$

Proof. Write $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_n)$. Then

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$$A := [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} | & & | & & | \\ [v_1]_{\mathcal{C}} & \cdots & [v_i]_{\mathcal{C}} & \cdots & [v_n]_{\mathcal{C}} \\ | & & | & & | \end{pmatrix}.$$

Write also $A = (a_{ij})$. We have

$$v_j = \sum_{k=1}^{n} a_{kj} w_k. \tag{$*$}$$

Let $\mathcal{C}^* = (w_1^*, \dots, w_n^*)$ be the dual basis of $\mathcal{C}$. By the previous lemma (18.a.7), if we take $\ell = w_j^*$, then

$$w_j^* = \sum_{i=1}^{n} w_j^*(v_i) v_i^* \underbrace{=}_{\text{by } (*)} \sum_{i=1}^{n} w_j^* \left(\sum_{k=1}^{n} a_{ki} w_k \right) v_i^* \underbrace{=}_{\substack{\text{the only term that is } \neq 0 \\ \text{in } w_j^*(\sum a_{ki} w_k) \text{ is when } k=j}} \sum_{i=1}^{n} a_{ji} v_i^*.$$

$$\Rightarrow [\mathrm{id}_V^*]_{\mathcal{B}^*}^{\mathcal{C}^*} = (a_{ji}) = A^{\top}.$$

□

2.17.1 The dual map

Definition (Lemma and Definition 18.a.9)**.** Let V, W be vector spaces over a field K, and let $T : V \to W$ be a linear map. Define a new map $T^* : W^* \to V^*$ by

$$T^*(\ell) = \ell \circ T, \qquad \forall \ell \in W^*.$$

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[dr, dashed, "\ell \circ T"'] & W \arrow[d, "\ell"] \\ & K \end{tikzcd}$$

The map $T^* : W^* \to V^*$ is called the *dual map* to T. Note that $\ell \circ T$ is indeed in V^*, since $\ell \circ T$ is linear (because T and ℓ are linear). Thus $T^* : W^* \to V^*$ is a linear map.

Proof. Let $\ell \in W^*$ and $\alpha \in K$. We have that $T^*(\alpha\ell)$ is the map $(\alpha\ell) \circ T$. So, for all $v \in V$,

$$\begin{aligned} T^*(\alpha\ell)(v) &= (\alpha\ell)(T(v)) \\ &= \alpha \cdot \ell(T(v)) \\ &= \alpha(\ell \circ T)(v) \\ &= \alpha T^*(\ell)(v). \end{aligned}$$

$$\Rightarrow T^*(\alpha\ell) = \alpha T^*(\ell).$$

Let $\ell_1, \ell_2 \in W^*$, and let $v \in V$.

$$\begin{aligned} T^*(\ell_1 + \ell_2)(v) &= (\ell_1 + \ell_2)(T(v)) \\ &= \ell_1(T(v)) + \ell_2(T(v)) \\ &= T^*(\ell_1)(v) + T^*(\ell_2)(v) \\ &= (T^*(\ell_1) + T^*(\ell_2))(v). \end{aligned}$$

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So

$$T^*(\ell_1 + \ell_2) = T^*(\ell_1) + T^*(\ell_2).$$

□

Proposition. Let $T : V \to W$, $S : W \to U$ be linear maps. Consider $V^* \leftarrow W^* : T^*$, $W^* \leftarrow U^* : S^*$. Then

$$(S \circ T)^* = T^* \circ S^*.$$

Proof. The proof is left as an exercise for the reader. □

Exercise — Let V, W be finite-dimensional vector spaces.

1. $(O : V \to W)^* = (O : W^* \to V^*)$

2. $(\mathrm{id}_V : V \to V)^* = (\mathrm{id}_{V^*} : V^* \to V^*)$

Lemma. Let $S : V \to W$ be a linear map between finite-dimensional vector spaces. Let $\mathcal{B}$ be abasis of V and $\mathcal{C}$ a basis of W. Consider $S^* : W^* \to V^*$, and the dual bases $\mathcal{B}^*, \mathcal{C}^*$. Then:

$$[S^*]^{\mathcal{C}^*}_{\mathcal{B}^*} = ([S]^{\mathcal{B}}_{\mathcal{C}})^\top.$$

Proof. Write $\mathcal{B} = (v_1, \dots, v_n)$, $\mathcal{C} = (w_1, \dots, w_m)$. $A = [S]_{\mathcal{C}}^{\mathcal{B}}, A = (a_{ij})$. $Sv_j = \sum_{i=1}^{n} a_{ij} w_i$. Now, for $w_j^* \in C^*$, we have

$$S^*(w_j^*) = w_j^* \circ S \underbrace{=}_{\text{by lemma 18.a.7}} \sum_{i=1}^{n} (w_j^* \circ S)(v_i) v_i^*.$$

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$$= \sum_{i=1}^{n} w_j^* \left(\sum_{k=1}^{m} a_{ki} w_k \right) v_i^* = \sum_{i=1}^{n} a_{ji} v_i^*.$$

$\Rightarrow$ The coordinates of $S^*(w_j^*)$ in the basis $\mathcal{B}^*$ aregiven $\begin{pmatrix} a_{j_1} \\ \vdots \\ a_{j_n} \end{pmatrix} = row\#j$ of A, made into a column. $\square$

Linear Algebra II

Semester 2

Lecture 28: Semester 2 Kickoff

18 Feb 2026

2.1 Semester 2 Setup

Recap

This recap reviews key facts from Semester 1 that we use throughout this lecture.

Dual space and functionals. Let V be a vector space over K. We define its dual space by

$$V^* := \mathrm{Hom}(V, K).$$

An element $\ell \in V^*$ is a linear map, also called a functional, of the form

$$\ell : V \to K.$$

Dual basis and coordinate expansion. To relate coordinates in V and V^*, let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V, and let $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ be the dual basis for V^*. Each map $v_i^* : V \to K$, for $i = 1, \dots, n$, is characterized by

$$v_i^*(v_j) = \delta_{ij} \qquad \boxed{\mathcal{B}^* \text{ is useful!}}$$

Therefore, for every $\ell \in V^*$, we can write

$$\ell = \sum_{i=1}^{n} \ell(v_i) \cdot v_i^*.$$

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Dual map. We now recall the dual-map construction. Let V, W be vector spaces over K, and let $T : V \to W$ be a linear map. Then there exists a linear map $T^* : W^* \to V^*$, defined by

$$T^*(\ell) := \ell \circ T \qquad \text{for every } \ell \in W^*.$$

This construction is summarized by the following diagram:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[dr, "\ell\circ T"'] & W \arrow[d, "\ell"] \\ & K \end{tikzcd}$$

2.2 Reflexiivity

Let V be a vector space. Define $V^{**} := (V^*)^*$. V^{**} is sometimes called the *bidual* of V.

Theorem. Let V be a finite-dimensional vector space. Then $\exists$ a canonical isomorphism $\tau : V \to V^{**}$ given by:

$$\forall v \in V, \qquad \tau(v) : V^* \to K, \qquad \tau(v)(\ell) := \ell(v).$$

In other words:

$$V \ni v \stackrel{\tau}{\mapsto} \big(V^* \ni \ell \mapsto \ell(v)\big).$$

Proof. *Linearity.* We claim that τ is linear. Indeed, if $v \in V$ and $\alpha \in K$, we have

$$\begin{aligned}\tau(\alpha v) &= \big(V^* \ni \ell \mapsto \ell(\alpha v)\big)\\ &= \big(V^* \ni \ell \mapsto \alpha\ell(v) \in K\big)\\ &= \alpha\big(V^* \ni \ell \mapsto \ell(v) \in K\big)\\ &= \alpha\tau(v).\end{aligned}$$

Exercise — Show that $\tau(v' + v'') = \tau(v') + \tau(v'')\ \forall v', v'' \in V$.

Injectivity. We now claim τ is injective. We will sow that $\ker\tau = \{0\}$. Indeed, assume $\tau(v) = 0$. $\Rightarrow \ell(v) = 0 \forall \ell \in V^*$. (*) choose a basis $\mathcal{B} = (v_1, \dots, v_n)$ for V. Write $v = \sum_{i=1}^n c_i v_i$ with $c_i \in K$. Apply (*) to $\ell = v_i^*$. We get

$$\begin{aligned}v_k^*(c_1v_1 + \cdots + c_nv_n) = c_k &= 0 \quad (\text{kernel: } v_k^*(v) = \delta_{ki})\\ &= 0 \quad \text{this holds } \forall k\\ \Rightarrow c_k = 0 \quad \forall k \Rightarrow v = 0.&\end{aligned}$$

This proves $\ker\tau = \{0\} \Rightarrow \tau$ is injective. Recall now $\dim V = \dim V^* = \dim V^{**}$ since $\tau : V \to V^{**}$ is injective, it must also be an isomorphism. $\square$

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2.2.1 naturality

The map $\tau : V \to V^{**}$ can be defined for every vect. sp. V. To emphas. dependence on V, we'll write here τ_V instead of τ.

Let V, W be vect. sp. over K. Then $\forall$ lin. map $T : V \to W$ we have the following commut. diagram:

$$\boxed{\text{Naturality}} \longrightarrow \begin{tikzcd} V \arrow[r, "\tau_V"] \arrow[d, "T"'] & V^{**} \arrow[d, "(T^*)^*"] \\ W \arrow[r, "\tau_W"'] & W^{**} \end{tikzcd}$$

Exc. : prove this.

Exercise:

Let V be a *f.dim*. Vect. sp. over K.
Let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V.
Let $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ be the dual basis,
and $\mathcal{B}^{**} = (v_1^{**}, \dots, v_n^{**})$ be the dual basis of $\mathcal{B}^*$.

2.3 Direct Sums

Proposition (18.b.1)**.** Let V, W be vector spaces over K. Then the cartesian product $V \times W$ becomes a vector space over K if we endow it with the following operations:

- $(v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2) \quad \forall v_1, v_2 \in V,\ w_1, w_2 \in W.$
- $\alpha(v, w) := (\alpha v, \alpha w) \quad \forall \alpha \in K,\ v, w \in V \times W.$
- $0_{V\times W} := (0_V, 0_W).$

Proof. Just go over the axioms of vector spaces one by one. $\boxed{\textit{Exercise!}}$ □

$V \times W$ is denoted by $V \oplus W$ and is called the *direct sum* of V and W. Its elements are denoted by v, w (or sometimes (ugly!) $(v \oplus w)$).

Remark. $V \oplus W$ connects with the following cannonical maps:

$$\left.\begin{array}{ll} \textit{1)}\ i_V : V \longrightarrow V \oplus W, & i_V(v) := (v, 0), \\ \textit{2)}\ i_W : W \longrightarrow V \oplus W, & i_W(w) := (0, w) \end{array}\right\} \quad \text{embeddings, or inclusions.}$$

$$\left.\begin{array}{ll} \textit{3)}\ p_V : V \oplus W \longrightarrow V, & p_V(v, w) := v, \\ \textit{4)}\ p_W : V \oplus W \longrightarrow W, & p_W(v, w) := w \end{array}\right\} \quad \text{projections.}$$

Using i_V & i_W we can view V as a subspace of $V \oplus W$. $V \subseteq V \oplus W$ and also $W \subseteq V \oplus W$.

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Proposition (18.b.2)**.** Let V be a vector space over K and $U \subseteq V$ be a subspace. Let $U' \subseteq V$ be another subspace which is a *complement* of U (recall: this means $U + U' = V$ and $U \cap U' = \{0\}$). Then the map:

$$\varphi : U \oplus U' \to V, \quad \text{defined by } \varphi(u, u') := u + u'$$

is linear and is an isomorphism.

Proof. Linearity of φ: exercise.

Injectivity of φ: Suppose $\varphi(u, u') = 0$. Then

$$u + u' = 0 \implies u = -u'.$$

Now $u \in U_b$ and $-u' \in U_b'$, hence $u \in U_b \cap U_b'$. Since U_b' is a complement of U_b, we have $U_b \cap U_b' = \{0\}$. Therefore $u = 0$, and then $u' = 0$. So $(u, u') = (0, 0)$, i.e. $\ker(\varphi) = \{0\}$.

Surjectivity of φ: By assumption, $U_b + U_b' = V$. Let $v \in V$. Then there exist $u \in U_b$ and $u' \in U_b'$ such that $v = u + u'$. Hence

$$\varphi(u, u') = u + u' = v.$$

Thus φ is surjective. □

Generalization: to arbitrarily many spaces.

Let $U_1, \dots, U_n$ be finitely many vector spaces over K. We can define

$$U_1 \oplus \cdots \oplus U_n \coloneqq U_1 \times \cdots \times U_n$$

similarly to what we have done before.
Let $\{U_i\}_{i \in I}$ be a family of vector spaces over K, parameterized by $i \in I$.

$$\bigtimes_{i \in I} U_i = \prod_{i \in I} U_i \coloneqq \left\{ t : I \to \bigcup_{i \in I} U_i \mid t(i) \in U_i, \forall i \in I \right\}$$

28 We can endow this with the structure of a vector space over K. Define also $\bigoplus\limits_{i \in I} U_i$:

$$\bigoplus_{i \in I} U_i \coloneqq \left\{ t \in \prod_{i \in I} U_i \mid t(i) \neq 0 \text{ only for finitely many } i \in I \right\}$$

If I is finite, then $\prod_{i \in I} U_i = \bigoplus_{i \in I} U_i$. But consider the example $I = \mathbb{Z}_{\geq 0}$:

$$\begin{aligned}
U_i &= K \quad \forall i \in I \\
\prod_{i \in I} U_i &= \{(a_1, a_2, \dots, a_n, \dots) \mid a_i \in K, \forall i\} \\
\bigoplus_{i \in I} U_i &= \{(a_1, a_2, \dots, a_n, \dots) \mid a_i \in K, \forall i, \text{ and } a_j = 0 \text{ for all large enough } j\}
\end{aligned}$$

Lecture 29: Quiotient spaces

23 Feb 2026

Recap

Dual space. For a vector space V, its dual space is V^*.

Bidual. The bidual is $V^{**} = (V^*)^*$.

Canonical map. There exists a canonical map $V \to V^{**}$. If V is finite-dimensional, then this map is an isomorphism.

Direct sum. The direct sum of vector spaces is denoted $U \oplus V$.

29

2.4 Quotient spaces

Let V be a vector space over a field K and let $U \subseteq V$ be a subspace. We will define a new vector space V/U together with a linear map $\pi : V \to V/U$ such that π is surjective and $\ker(\pi) = U$.

We begin by defining an equivalence relation on the set V. Let $v_1, v_2 \in V$.We declare $v_1 \sim v_2$ if $v_1 - v_2 \in U$.

Claim. $\sim$ is an equivalence relation.

Proof. *Reflexivity.* For every $v \in V$, we have $v - v = 0 \in U$, hence $v \sim v$.

Symmetry. If $v_1 \sim v_2$, then $v_1 - v_2 \in U$. Therefore

$$v_2 - v_1 = -(v_1 - v_2) \in U,$$

so $v_2 \sim v_1$.

Transitivity. If $v_1 \sim v_2$ and $v_2 \sim v_3$, then

$$v_1 - v_3 = (v_1 - v_2) + (v_2 - v_3) \in U,$$

so $v_1 \sim v_3$ (because $U \subseteq V$ is a subspace). □

Denote by $[v]$ the equivalence class of $v \in V$. One can also think of $[v]$ as

$$[v] = v + U = \{v + u \mid u \in U\}..$$

29

Definition (18.c.1). Let $V \& U$ be as above. We define

$$V/U \coloneqq V/\sim \coloneqq \text{ the set of equivalence classes of } \sim.$$

Elements of V/U are $[v]$, where $v \in V$. V/U is called the V quotient U'' or V modulo U. We will turn the set V/U into vertor space over K. Addition: Let $x_1, x_2 \in V/U$. Pick resentatives $v_1, v_2 \in V$ of x_1, x_2 respectively, id est $x_1 = [v_1]$ and $x_2 = [v_2]$. Define

$$x_1 + x_2 \coloneqq [v_1] + [v_2] \in V/U.$$

Multiplication by scalars: Let $\alpha \in K$ and $x \in V/U$. Pick $v \in V$, with $[v] = x$. Define

$$\alpha x \coloneqq [\alpha v] \in V/U.$$

Zero verctor: Define $0_{V/U} \coloneqq [0_V] \in V/U$.

Proposition (18.c.2). The operations above are well-defined. Moreover, they give V/U the structure of a vector space over K.

Proof. *Well definedness.*

Multiplication by a scalar. Suppose $v, v' \in V$ represent the same class $x \in V/U$, i.e. $v \sim v'$. Let $\alpha \in K$. Since $v - v' \in U$, we get

$$\alpha v - \alpha v' = \alpha(v - v') \in U.$$

Hence $\alpha v \sim \alpha v'$, so $[\alpha v] = [\alpha v']$. Therefore αx is well-defined.

Addition. Let $v_1, v_1', v_2, v_2' \in V$ such that $[v_1] = [v_1'] = x$ and $[v_2] = [v_2'] = x'$. We need to show $[v_1 + v_2] = [v_1' + v_2']$. Since $v_1 \sim v_1'$ and $v_2 \sim v_2'$,

$$(v_1 + v_2) - (v_1' + v_2') = (v_1 - v_1') + (v_2 - v_2') \in U.$$

So $v_1 + v_2 \sim v_1' + v_2'$, hence $[v_1 + v_2] = [v_1' + v_2']$. Therefore $x + x'$ is well-defined. $\square$

Define now a map $\pi : V \to V/U$ by $\pi(v) = [v] \forall v \in V$.

Proposition (18.c.3). π is a linear map. Moreover, $\mathrm{Im}(\pi) = V/U$ and $\ker(\pi) = U$.

Proof. *Linearity.* Let $v_1, v_2 \in V$. Then

$$\pi(v_1 + v_2) = [v_1 + v_2] = [v_1] + [v_2] = \pi(v_1) + \pi(v_2).$$

Let $\alpha \in K$ and $v \in V$. Then

$$\pi(\alpha v) = [\alpha v] = \alpha[v] = \alpha\pi(v).$$

Surjectivity. Let $x \in V/U$. By definition, there exists $v \in V$ such that $x = [v]$. Hence $x = \pi(v)$, so $\mathrm{Im}(\pi) = V/U$.

Kernel. If $v \in \ker(\pi)$, then $\pi(v) = 0_{V/U}$, so $[v] = [0_V]$. Thus $v \sim 0_V$, i.e. $v - 0_V = v \in U$. Therefore $\ker(\pi) \subseteq U$.

Conversely, let $u \in U$. Then $u \sim 0_V$, so $[u] = [0_V]$, hence $\pi(u) = 0_{V/U}$. Therefore $u \in \ker(\pi)$, and $U \subseteq \ker(\pi)$.

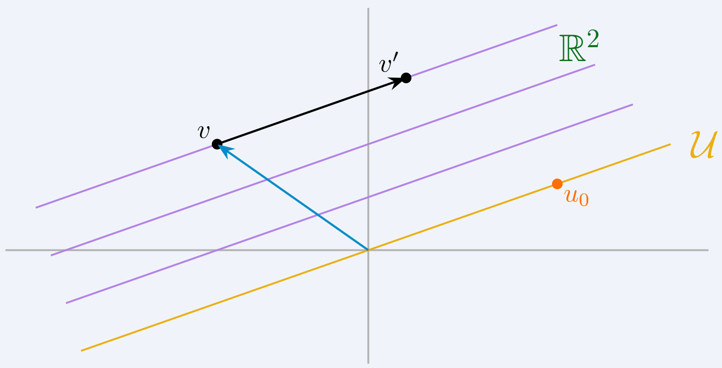
29 So $\ker(\pi) = U$. $\square$

$$V/U, \qquad V/\{0\} = V, \qquad V/V = \{0\}.$$

Example. A geometric way to think about V/U is the following. Let $K = \mathbb{R}$ and $V = \mathbb{R}^2$. Choose $0 \neq u_0 \in \mathbb{R}^2$, and define

$$U := \mathrm{span}(u_0),$$

which is the line through 0 and u_0. The elements of $\mathbb{R}^2/U$ are in one-to-one



correspondence with lines $\ell \subseteq \mathbb{R}^2$ that are parallel to U:

$$[v] \in \mathbb{R}^2/U \longleftrightarrow v + U.$$

Proposition (18.c.4). Suppose V is finite-dimensional. Then V/U is also finite-dimensional, and

$$\dim(V/U) = \dim V - \dim U.$$

Proof. Since $\pi : V \to V/U$ is surjective, V/U is finite-dimensional. Indeed, if $T : W' \to W''$ is a surjective linear map and W' is finite-dimensional, then W'' is finite-dimensional: if $w_1', \dots, w_m'$ is a basis of W', then $T(w_1'), \dots, T(w_m')$ spans $\mathrm{Im}(T) = W''$.

We now apply the rank theorem to π:

$$\dim V = \dim \ker(\pi) + \dim \mathrm{Im}(\pi) = \dim U + \dim(V/U).$$

□

Theorem (The isomorphism theorem 18.c.5). Let V, W be vector spaces over K, and let $T : V \to W$ be a linear map. Define a new map

$$\tilde{T} : V/\ker(T) \to \mathrm{Im}(T)$$

29 as follows. Let $x \in V/\ker(T)$. Choose $v \in V$ with $[v] = x$, and define

$$\tilde{T}(x) := T(v).$$

Then:

1. $\tilde{T}$ is well-defined and linear.
2. $\tilde{T} : V/\ker(T) \to \mathrm{Im}(T)$ is an isomorphism.
3. The following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\pi"'] \arrow[dr, phantom, "\copyright"] & W \\ V/\ker(T) \arrow[r, "\tilde{T}"'] & \mathrm{Im}(T) \arrow[u, "\iota \text{ (inclusion)}"'] \end{tikzcd}$$

Proof. *Well-definedness.* Suppose $x = [v] = [v']$ in $V/\ker(T)$. We must show that $T(v) = T(v')$. Since $[v] = [v']$, we have $v - v' \in \ker(T)$. Therefore

$$T(v - v') = 0,$$

which implies $T(v) = T(v')$. Hence $\tilde{T}$ is well-defined.

Linearity. Let $\alpha \in K$ and $v \in V$. Then

$$\tilde{T}(\alpha[v]) = \tilde{T}([\alpha v]) = T(\alpha v) = \alpha T(v) = \alpha \tilde{T}([v]).$$

Let $v_1, v_2 \in V$. Then

$$\tilde{T}([v_1] + [v_2]) = \tilde{T}([v_1 + v_2]) = T(v_1 + v_2) = T(v_1) + T(v_2) = \tilde{T}([v_1]) + \tilde{T}([v_2]).$$

So $\tilde{T}$ is linear.

Injectivity. Let $x \in \ker(\tilde{T})$, and write $x = [v]$ for some $v \in V$. Then $\tilde{T}(x) = T(v) = 0$, so $v \in \ker(T)$. Hence $x = [v] = [0]$, and therefore $\tilde{T}$ is injective.

Surjectivity onto $\mathrm{Im}(T)$. Let $w \in \mathrm{Im}(T)$. Then there exists $v \in V$ with $T(v) = w$. Thus

$$\tilde{T}([v]) = T(v) = w,$$

so $w \in \mathrm{Im}(\tilde{T})$. Therefore $\mathrm{Im}(T) \subseteq \mathrm{Im}(\tilde{T})$. On the other hand, by definition of $\tilde{T}$, we have $\mathrm{Im}(\tilde{T}) \subseteq \mathrm{Im}(T)$. Hence $\mathrm{Im}(\tilde{T}) = \mathrm{Im}(T)$, and $\tilde{T}$ is surjective onto $\mathrm{Im}(T)$.

Commutativity of the diagram. For every $v \in V$,

$$(\iota \circ \tilde{T} \circ \pi)(v) = \iota(\tilde{T}([v])) = \iota(T(v)) = T(v).$$

29 So the diagram commutes. □

Theorem (18.c.6). Let V be a vector space over K and let $U \subseteq V$ be a subspace. The quotient V/U has the following universal property: $\exists$ a space W and every linear map $T : V \to W$ with $T(U) = 0$ (i.e. $\ker(T) \supseteq U$) $\exists!$ a linear map $\tilde{T} : V/U \to W$ such that the following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\pi"'] \arrow[dr, phantom, "©"] & W \\ V/U \arrow[ur, "T'\ (\exists!)"'] & \end{tikzcd}$$

commutes. Moreover

$$\ker(T') = \ker(T)/U.$$

Jargon: Every lin. map $T \colon V \to W$ that sends U to 0 factors through V/U, i.e., $T = T' \circ \pi$

Lecture 30: Quotient spaces & complements

25 Feb 2026

Recap

Last time: quotient spaces. If $\mathcal{U} \subseteq V$ is a subspace, then $V/\mathcal{U}$ is a quotient space.

30 *Last time: induced isomorphism.* If $T : V \to W$ is linear, then it induces an isomorphism

$$\overline{T} : V/\ker(T) \xrightarrow{\cong} \mathrm{Im}(T), \qquad [v] \mapsto T(v).$$

Theorem (18.c.6)**.** Let V be a vector space over K and let $U \subseteq V$ be a subspace. The quotient V/U has the following universal property: $\exists$ a space W and every linear map $T : V \to W$ with $T(U) = 0$ (i.e. $\ker(T) \supseteq U$) $\exists!$ a linear map $\tilde{T} : V/U \to W$ such that the following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\pi"'] \arrow[dr, phantom, "©"] & W \\ V/U \arrow[ur, "T' \ (\exists!)"'] & \end{tikzcd}$$

commutes. Moreover

$$\ker(T') = \ker(T)/U.$$

Jargon: Every lin. map $T\colon V \to W$ that sends U to 0 factors through V/U, i.e., $T = T' \circ \pi$ T descendes to $T' : V/U \to W$.

Proof. Define $T' : V/U \to W$ in the following way. Let $x \in V/U$. Choose $v \in V$ with $[v] = x$, and define

$$T'(x) := T(v).$$

Exercise 1.: Show that T' is well-defined, i.e., if $[v] = [v']$, then $T(v) = T(v')$.

Exercise 2.: Show that T' is linear.

30 We claim that the diagram commutes. Indeed, $\forall v \in V$,

$$T'(\pi(v)) = T'([v]) = T(v).$$

Uniqueness of T': Suppose that $T = T' \circ \pi = T'' \circ \pi$. We will show that $T' = T''$. Indeed, let $x \in V/U$. Choose $v \in V$ with $[v] = x$. Then

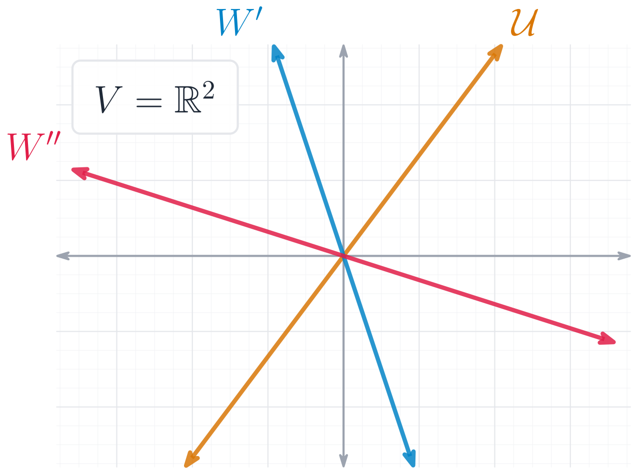
$$\begin{aligned} &\Rightarrow T'(v) = T'(\pi(v)) \text{ and } T''(v) = T''(\pi(v)) \\ &\Rightarrow T'(x) = T''(x). \end{aligned}$$

□

2.5 Quotient spaces & complements

Let V be a vector space and let $U \subseteq V$ be a subspace. Let $W \subseteq V$ be a complement of U, i.e., $W \subseteq V$ is a subspace and $W \cap U_0 = \{0\}$ and $W + U = V$.

Proposition (18.c.7)**.** $\exists$ a *canonical* isomorphism. $Q : W \xrightarrow{\cong} V/U$, defined by $Q(w) := [w] \in V/U \forall w \in W$.



Proof. Let's consider the inclusion $i : W \to V$, given by $i(w) = w\ \forall w \in W$, and the projection map $\pi : V \to V/U$. Recall that $\pi(v) = [v]$, so $Q = \pi \circ i$. Hence Q is linear.

We claim that $\ker Q = 0$. Indeed, if $w \in \ker Q$, then $[w] = 0 \Rightarrow w \in U$. But $w \in W$ and $U \cap W = \{0\} \Rightarrow w = 0$.

We claim now that Q is surjective. Let $x \in V/U$. Choose $v \in V$ with $[v] = x$. Since $V = W + U$, $\exists w \in W$ and $u \in U$ such that $v = w + u$.

30

$$\begin{aligned} &\Rightarrow [v] = [w] \in V/U \\ &\Rightarrow \underbrace{[w]}_{=Q(w)} = x. \end{aligned}$$

Thus Q is surjective. $\square$

2.6 Relations to the dual space

Proposition (18.c.8)**.** Let V be a vector space and let $U \subseteq V$ be a subspace. Define $U^\perp := \{\ell \in V^* \mid_U \equiv 0\}$. $U^\perp =$ all the functionals $\ell : V \to K$ that vanish on U.

1. $U^\perp \subseteq V^*$ is a linear subspace.
2. $\exists$ a canonical isomorphism.

$$(V/U)^* \cong U^\perp.$$

If we assume that V is finite-dimensional, then $\exists$ also a canonical isomorphism $V^*/U^\perp \cong U^*$.

Proof. *1.* Clearly $0 \in U^\perp$. If $\ell_1, \ell_2 \in U^\perp$ and $\alpha, \beta \in K$, then $(\alpha\ell_1 + \beta\ell_2) \in U^\perp$, because $\forall u \in U$, we have:

$$(\alpha\ell_1 + \beta\ell_2)(u) = \alpha\ell_1 \underbrace{(u)}_{=0} + \beta\ell_2 \underbrace{(u)}_{=0} = 0.$$

2. Define a map $L : (V/U)^* \to U^\perp$ as follows: let $s \in (V/U)^*$, i.e. $s : V/U \to K$ is linear. Define $L(s) \in V^*$ to be $L(s) \coloneqq s \circ \pi : V \to K$. So, L is: $s \mapsto s \circ \pi$.

$$\begin{tikzcd}
V \arrow[r, dashed, "s\circ\pi"] \arrow[d, "\pi"'] & K \\
V/Ub \arrow[ur, "s"'] &
\end{tikzcd}$$

Exercise. 1) Show that $s \circ \pi$ is indeed in $U^\perp$ (i.e. $s \circ \pi \mid_U \equiv 0$). 2) Show that L is linear. We claim that L is an isomorphism. To see this, we define another map

$$P : U^\perp \to (V/U)^*.$$

Let $t \in U^\perp$, i.e. $t : V \to K$ such that $t \mid_U \equiv 0$. By the universal property of quotient spaces, $\exists! t' : V/U \to K$ such that $t = t' \circ \pi$.

$$\begin{tikzcd}
V \arrow[r, "t"] \arrow[d, "\pi"] & K \\
V/W \arrow[ur, "t'"'] &
\end{tikzcd}$$

30

Define $P(t) \coloneqq t'$. We claim that

$$L \circ P = \mathrm{id}_{U^\perp} \quad \text{and} \quad P \circ L = \mathrm{id}_{(V/U)^*}.$$

Indeed, if $s : V/U \to K$, then

$$(P \circ L)(s) = P(s \circ \pi) = s,$$

so $P \circ L = \mathrm{id}_{(V/U)^*}$. Also, for every $t \in U^\perp$, we have

$$(L \circ P)(t) = L(t') = t' \circ \pi = t,$$

so $L \circ P = \mathrm{id}_{U^\perp}$. Since L is also a linear map and both of $L\&P$ are isomorphisms, assume now that V is finite-dimensional. Consider the map $R : V^* \to U^*$, defined by

$$R(\varphi) \coloneqq \varphi \mid_U \in U^*, \qquad \forall \varphi \in V^*.$$

(Restriction map). (BTW check that if $i : U \to V$ is the inclusion, then $R = i^* : V^* \to U^*$). Exercise: Check that R is linear. We claim that R is surjective. Indeed, let $\psi \in U^*$. We need to show that $\exists \varphi \in V^*$ such that $\varphi \mid_U = \psi$. Pick a basis $\{u_1, \dots, u_k\}$ for U, and extend it to a basis $\{u_1, \dots, u_k, v_{k+1}, \dots, v_n\}$ of V. Define $\varphi : V \to K$ by

$$\varphi(u_i) = \psi(u_i) \ \forall 1 \le i \le k, \qquad \varphi(v_j) = 0 \ \forall k+1 \le j \le n.$$

Since $\{u_1, \ldots, u_k, v_{k+1}, \ldots, v_n\}$ is a basis for V, φ is well-defined and clearly $\varphi\mid_U = \psi$. This shows $\mathrm{Im}(R) = U^*$. Also, $\ker(R) = U^\perp$ (by def.). By theorem 18.c.5 (the isomorphism theorem), we get that

$$V^*/\ker(R) \cong \mathrm{Im}(R).$$

30 So $V^*/U^\perp \cong U^*$. □

Exc. Let $T : V \to W$ be a linear map. Then

$$(\mathrm{Im}(T))^\perp = \ker(T^* : W^* \to V^*).$$

Chapter 3

Determinants

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2\times 2}(K) \rightsquigarrow \det(A) = ad - bc.$$

Lecture 31: Introduction to determinants

02 Mar 2026

3.1 Introduction

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(K)$.

Proposition (20.a.1)**.** A is inverible iff $ad - bc \neq 0$. Moreover if A is invertible

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The scalar $ad - bc$ is called the determinant of A and is denoted by $\det(A)$ or sometines $|A|$.

31 **Proof.** This is in the homework, but here is the outline A is not invertiblke iff the rank of A is strictly less than 2, $\Leftrightarrow$ the columns of A are linearly dependent $\Leftrightarrow$ either $\begin{pmatrix} a \\ c \end{pmatrix} = \lambda \begin{pmatrix} b \\ d \end{pmatrix}$ for some $\lambda \in K$ or $\begin{pmatrix} b \\ d \end{pmatrix} = \tau \begin{pmatrix} a \\ c \end{pmatrix}$ for some $\tau \in K$. In the first case

$$A = \begin{pmatrix} \lambda b & b \\ \lambda d & d \end{pmatrix}, \&a = \lambda b, c = \lambda d$$

and in the second case

$$A = \begin{pmatrix} a & \tau a \\ c & \tau c \end{pmatrix}, \&b = \tau a, d = \tau c.$$

Exc. Complete the proof. $\square$

Definition (20.a.2)**.** Let $D : M_2(K) \to K$ be a function. We say that D is $n-$linear if $\forall 1 \leq i \leq n$, D is a linear function of the i-th row, when the other $n-1$ rows are fixed. We can think of $M_n(K)$ as $M_{n\times n}(K)$ as $K^n_{\text{row}} \times \cdots \times K^n_{\text{row}}$ and if

$$A = \begin{pmatrix} \rule{2cm}{0.4pt}\ \alpha_1\ \rule{2cm}{0.4pt} \\ \vdots \\ \rule{2cm}{0.4pt}\ \alpha_n\ \rule{2cm}{0.4pt} \end{pmatrix}$$

we write $D(A) = D(\alpha_1, \ldots, \alpha_n)$ saying that D is $n-$ linear, means: $\forall 1 \leq i \leq n$,

1. $D(\alpha_1, \ldots, \alpha_i + \alpha_i', \ldots, \alpha_n) = D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + D(\alpha_1, \ldots, \alpha_i', \ldots, \alpha_n)$
2. $D(\alpha_1, \ldots, c \cdot \alpha_i, \ldots, \alpha_n) = c \cdot D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) \forall c \in K$

Example. For a matrix A, write $A(i,j)$ for its (i,j)-entry. One also writes a_{ij} or A_{ij}.

Fix integers $k_1, \ldots, k_n$ with $1 \leq k_i \leq n$ for all i. Fix also $a \in K$. Define $D : M_{n\times n}(K) \to K$ by

$$D(A) := a \cdot A(1, k_1) \cdots A(n, k_n).$$

Claim. The map D is n-linear.

Proof. Proof. Consider

31

$$\begin{pmatrix} — & \alpha_1 & — \\ & \vdots & \\ — & \alpha_i & — \\ & \vdots & \\ — & \alpha_n & — \end{pmatrix} = \begin{pmatrix} A(1,1) & \cdots & A(1,n) \\ \vdots & \ddots & \vdots \\ A(i,1) & \cdots & A(i,n) \\ \vdots & \ddots & \vdots \\ A(n,1) & \cdots & A(n,n) \end{pmatrix}.$$

We have $D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) = b \cdot A(i, k_i)$, where $b \in K$ does <u>NOT</u> depend on row i. (b depends only on rows $1, \ldots, i-1, i+1, \ldots, n$).

Let $\alpha_i' = (\alpha_{i,1}', \ldots, \alpha_{i,n}')$. Then

$$\begin{aligned} D(\alpha_1, \ldots, \alpha_i + \alpha_i', \ldots, \alpha_n) &= b \cdot (A(i, k_i) + \alpha_{i,k_i}') \\ &= D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + D(\alpha_1, \ldots, \alpha_i', \ldots, \alpha_n). \end{aligned}$$

Let $c \in K$. Then

$$\begin{aligned} D(\alpha_1, \ldots, c \cdot \alpha_i, \ldots, \alpha_n) &= b \cdot cA(i, k_i) \\ &= c \cdot D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n). \end{aligned}$$

$\square$

special case

$D(A) \coloneqq A(1,1) \cdot A(2,2) \cdots A(n,n)$ (the product of the elements along the diagonal) is n-linear.

Let's find all 2-linear functions $D : M_2(K) \to K$.

Write

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} — & \varepsilon_1 & — \\ — & \varepsilon_2 & — \end{pmatrix}$$

If D is 2-linear, then

$$\begin{aligned}
&D\left(\begin{pmatrix} A(1,1) & A(1,2) \\ A(2,1) & A(2,2) \end{pmatrix}\right) = \\
&= D\Big((A(1,1), A(1,2)), (A(2,1), A(2,2))\Big) \\
&= D\Big(A(1,1)\varepsilon_1 + A(1,2)\varepsilon_2,\ A(2,1)\varepsilon_1 + A(2,2)\varepsilon_2\Big) \\
&= A(1,1)D\Big(\varepsilon_1, A(2,1)\varepsilon_1 + A(2,2)\varepsilon_2\Big) + A(1,2)D\Big(\varepsilon_2, A(2,1)\varepsilon_1 + A(2,2)\varepsilon_2\Big) \\
&= A(1,1)A(2,1)D(\varepsilon_1, \varepsilon_1) + A(1,1)A(2,2)D(\varepsilon_1, \varepsilon_2) \\
&\quad + A(1,2)A(2,1)D(\varepsilon_2, \varepsilon_1) + A(1,2)A(2,2)D(\varepsilon_2, \varepsilon_2).
\end{aligned}$$

D is completely determined by the values of $D(\varepsilon_1, \varepsilon_1), D(\varepsilon_1, \varepsilon_2), D(\varepsilon_2, \varepsilon_1)$, and $D(\varepsilon_2, \varepsilon_2)$.

31 **Lemma (20.a.3).** Any linear combination of $n-$linear function is again an $n-$linear function.

Proof. Enough to prove for a linear combination of two n-linear functions. (WHY?)

Let $D, E : M_n(K) \to K$ be two n-linear functions, and let $a, b \in K$. Consider $aD + bE : M_n(K) \to K$, where $(aD + bE)(A) \coloneqq aD(A) + bE(A)$.

Let $1 \leq i \leq n$. For additivity in the i-th row, we have:

$$\begin{aligned}
&(aD + bE)(\alpha_1, \ldots, \alpha_i + \alpha_i', \ldots, \alpha_n) = \\
&\qquad = aD(\alpha_1, \ldots, \alpha_i + \alpha_i', \ldots, \alpha_n) + bE(\alpha_1, \ldots, \alpha_i + \alpha_i', \ldots, \alpha_n) \\
&\qquad = a \cdot \Big(D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + D(\alpha_1, \ldots, \alpha_i', \ldots, \alpha_n)\Big) \\
&\qquad\quad + b\Big(E(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + E(\alpha_1, \ldots, \alpha_i', \ldots, \alpha_n)\Big) \\
&\qquad = (a \cdot D + bE)(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + (aD + bE)(\alpha_1, \ldots, \alpha_i', \ldots, \alpha_n).
\end{aligned}$$

For scalar multiplication by $\lambda \in K$ in the i-th row, we have:

$$\begin{aligned}(aD+bE)(\alpha_1,\ldots,\lambda\cdot\alpha_i,\ldots,\alpha_n) &= \\ &= aD(\alpha_1,\ldots,\lambda\alpha_i,\ldots,\alpha_n)+bE(\alpha_1,\ldots,\lambda\alpha_i,\ldots,\alpha_n) \\ &= \lambda aD(\alpha_1,\ldots,\alpha_i,\ldots,\alpha_n)+\lambda bE(\alpha_1,\ldots,\alpha_i,\ldots,\alpha_n) \\ &= \lambda\cdot(aD+bE)(\alpha_1,\ldots,\alpha_i,\ldots,\alpha_n).\end{aligned}$$

□

Example. Consider $D : M_{2\times 2}(K) \to K$ $D(A) = A_{11}A_{22} - A_{12}A_{21}$. $(\det A)$ is 2-linear.

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}.$$

Proof. $D = D_1 - D_2$ where $D_1(A) = A_{11}A_{22}$ and $D_2(A) = -A_{12}A_{21}$. By a previeos exxample, both D_1 and D_2 are 2-linear. $\Rightarrow$ D is also 2-linear. (b.c. its a linear combination of $D_1 \& D_2$). □

Remark. $D = \det A$ as a function $M_{2\times 2}(K) \to K$ has two additional properties:

1. $D(I) = D\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1.$
2. If A' is obtained from A by interchanging two rows, then $\det A' = -\det A$.

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Indeed, if $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$, $A' = \begin{pmatrix} A_{21} & A_{22} \\ A_{11} & A_{12} \end{pmatrix}$, then $\det A' = A_{21}A_{12} - A_{22}A_{11} = -(A_{11}A_{22} - A_{12}A_{21}) = -\det A$.

Definition (20.a.4). Let $D : M_{n\times n}(K) \to K$ be an n-linear function. We say that D is alternating if for every matrix A, in which some two rows are equal we have $D(A) = 0$.

Proposition (20.a.5). Let D be an n-linear function. Assume that D has the propery that whenever $A \in M_{n\times n}(K)$ has two equal adjecent rows then $D(A) = 0$. Then:

1. D is alternating.
2. $\forall$ matrix B, if B' is obtained from B by interchanging *two* of its rows then $D(B') = -D(B)$.

Proof. We begin with proving (emph2). Assume 1st, that B' is obtained from B

by interchanging two adjacent rows.

$$B = \begin{pmatrix} — & \beta_1 & — \\ — & \beta_k & — \\ — & \beta_{k+1} & — \\ — & \beta_n & — \end{pmatrix}, \quad B' = \begin{pmatrix} — & \beta_1 & — \\ — & \beta_{k+1} & — \\ — & \beta_k & — \\ — & \beta_n & — \end{pmatrix}.$$

Consider $D\left(\beta_1, \dots, \beta_{k-1}, \underbrace{\beta_k + \beta_{k+1}}_{\text{entry } \#k}, \underbrace{\beta_k + \beta_{k+1}}_{\text{entry } \#(k+1)}, \beta_{k+2}, \dots, \beta_n\right)$.

31 By assumption $(*) = 0$, By n-linearity,

$$\begin{aligned} (*) =& D(\beta_1, \dots, \beta_k, \beta_k + \beta_{k+1}, \dots, \beta_n) + \\ &+ D(\beta_1, \dots, \beta_{k+1}, \beta_k + \beta_{k+1}, \dots, \beta_n) \\ &= D(\beta_1, -, \underbrace{\beta_k, \beta_k}_{=0}, -, \beta_n) + \\ &\quad + D(\beta_1, -, \beta_k, \beta_{k+1}, -, \beta_n) + \qquad 1 \\ &\quad + D(\beta_1, -, \beta_{k+1}, \beta_k, -, \beta_n) + \qquad 2 \\ &\quad + D(\beta_1, -, \underbrace{\beta_{k+1}, \beta_{k+1}}_{0=}, -, \beta_n) \end{aligned}$$

□

Lecture 32: Determinants (continued)

04 Mar 2026

Recap

Last time: Proposition 20.a.5. Let $D : M_{n\times n}(K) \to K$ be an n-linear map. Assume that $D(A) = 0$ whenever two adjacent rows of A are equal. Then:

1. D is alternating.

2. If B' is obtained from B by interchanging two rows, then $D(B') = -D(B)$.

32 *Proof idea for (2), adjacent-row case.* We first prove the statement when B' is obtained from B by swapping rows k and $k+1$. Write

$$B = \begin{pmatrix} \text{———} \beta_1 \text{———} \\ \vdots \\ \text{———} \beta_k \text{———} \\ \text{———} \beta_{k+1} \text{———} \\ \vdots \\ \text{———} \beta_n \text{———} \end{pmatrix}, \quad B' = \begin{pmatrix} \text{———} \beta_1 \text{———} \\ \vdots \\ \text{———} \beta_{k+1} \text{———} \\ \text{———} \beta_k \text{———} \\ \vdots \\ \text{———} \beta_n \text{———} \end{pmatrix}.$$

Define

$$X := D(\beta_1, \dots, \beta_k + \beta_{k+1}, \beta_k + \beta_{k+1}, \dots, \beta_n).$$

The two middle rows in this input are equal, so $X = 0$ by assumption. Expand X by

n-linearity in the k-th and $(k+1)$-st rows:

$$X = \underbrace{D(\beta_1, \ldots, \beta_k, \beta_{k+1}, \ldots, \beta_n)}_{(1)} + \underbrace{D(\beta_1, \ldots, \beta_{k+1}, \beta_k, \ldots, \beta_n)}_{(2)}.$$

Therefore $(1) + (2) = 0$, and hence $D(B') = -D(B)$ for one adjacent swap.

Suppose now that B' is obtained from B by interchanging rows $\#k$ and $\#\ell$, where $k < \ell$ (not necessarily $\ell = k + 1$).

We can obtain B' from B by doing a sequence of interchanges of adjacent rows. Indeed, begin by exchanging rows k and $k + 1$, and continue in this way until we get

$$\beta_1, \ldots, \beta_{k-1}, \beta_{k+1}, \ldots, \beta_\ell, \beta_k, \beta_{\ell+1}, \ldots, \beta_n.$$

This requires $r \coloneqq \ell - k$ interchanges of adjacent rows.

Next, move β_ℓ back by a sequence of interchanges of adjacent rows until it arrives at position $\#k$. This requires $(\ell - 1) - k = r - 1$ interchanges of adjacent rows.

So in total, we can obtain B' from B by performing $r + (r - 1) = 2r - 1$ interchanges of adjacent rows.

So $D(B') = (-1)^{2r-1} D(B) = -D(B)$. This concludes the proof of (2).

32 *Proof of (1).* Let A be an $n \times n$ matrix in which row $\#i$ = row $\#j$, where $i < j$. If $j = i + 1$, then by assumption $D(A) = 0$. So assume $j > i + 1$.

We interchange now rows $\#i + 1$ and $\#j$. We get a new matrix A', in which rows $\#i$ and $\#i + 1$ are equal.

$$A = \begin{array}{cc} \begin{pmatrix} \text{———} & \alpha_1 & \text{———} \\ & \vdots & \\ \text{———} & \alpha_i & \text{———} \\ \text{———} & \alpha_{i+1} & \text{———} \\ & \vdots & \\ \text{———} & \alpha_j & \text{———} \\ & \vdots & \\ \text{———} & \alpha_n & \text{———} \end{pmatrix} & \begin{array}{c} \\ \\ \leftarrow \\ \\ \text{equal} \\ \leftarrow \\ \\ \\ \end{array} \end{array}$$

By (2): $D(A) = -D(A')$. By assumption, $D(A') = 0$. So $D(A) = 0$. This concludes the proof.

Definition (20.a.6)**.** A function $D : M_{n\times n}(K) \to K$ is called a *determinant function* if it is n-linear, alternating, and $D(I_n) = 1$. Warmup: Let's find *all* determinant functions $D : M_{2\times 2}(K) \to K$. Write $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \text{—} & \varepsilon_1 & \text{—} \\ \text{—} & \varepsilon_2 & \text{—} \end{pmatrix}$. $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$.

We show that every 2-linear function D satisfies:

$$\begin{aligned} D(A) &= A_{11}A_{21}D(\varepsilon_1,\varepsilon_1) + A_{11}A_{22}D(\varepsilon_1,\varepsilon_2) \\ &\quad + A_{12}A_{21}D(\varepsilon_2,\varepsilon_1) + A_{12}A_{22}D(\varepsilon_2,\varepsilon_2). \quad (*) \end{aligned}$$

Since D is alternating, $D(\varepsilon_1,\varepsilon_1) = D(\varepsilon_2,\varepsilon_2) = 0$, and $D(\varepsilon_1,\varepsilon_2) = -D(\varepsilon_2,\varepsilon_1)$. So $D(A) = A_{11}A_{22}D\underbrace{(\varepsilon_1,\varepsilon_2)}_{=I} - A_{12}A_{21}D\underbrace{(\varepsilon_1,\varepsilon_2)}_{=I}$. Since $D(I) = 1$, we get

$$D(A) = A_{11}A_{22} - A_{12}A_{21} = \det\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}.$$

Definition (20.a.7)**.** Let $A \in M_{n\times n}(K)$, $n \geq 2$, and $1 \leq i \leq n$, $1 \leq j \leq n$. Denote by $A(i|j) \in M_{(n-1)\times(n-1)}(K)$ the $(n-1)\times(n-1)$-matrix obtained from A after deleting row $\#i$ and column $\#j$. If D is an $(n-1)$-linear function, denote

$$D_{ij} := D(A(i|j)).$$

Theorem (20.a.8)**.** Let $n \geq 2$, and D an $(n-1)$-linear function, which is alternating. Let $1 \leq j \leq n$. Define a function $E_j : M_{n\times n}(K) \to K$ by

$$E_j(A) := \sum_{i=1}^{n} (-1)^{i+j} A_{ij} \cdot D_{ij}(A).$$

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Then E_j is an n-linear function and it is alternating. If D is a determinant function (i.e. $D(I_{n-1}) = 1$), then so is E_j.

Before the proof a quick

Corollary (20.a.9)**.** $\forall n \geq 1, \exists$ at least one determinant function $M_{n\times n}(K) \to K$.

Proof. Induction on n. For $n = 1$, $M_{1\times 1}(K) \to K, a \mapsto a$ is a determinant function. For $n = 2$, we have already seen that $\det : M_{2\times 2}(K) \to K$ is a determinant function. Continue by induction using Theorem 20.a.8.: if $\exists$ a determinant function for $n-1 \times n-1$ matrices, then $\exists$ a determinant function also for $n \times n$ matrices. □

Proof of Theorem 20.a.8.. Let $A \in M_{n\times n}(K)$. $D_{ij}(A)$ is independent of row $\#i$ of A. Since D is $(n-1)$-linear, $A \mapsto D_{ij}(A)$ is linear when viewed as a function of each row of A, except row $\#i$. This implies that the function $M_{n\times n}(K) \ni A \mapsto A_{ij} \cdot D_{ij}(A)$ is n-linear. Since linear combinations of n-linear functions are n-linear, it follows that

$$E_j(A) = \sum_{i=1}^{n} (-1)^{i+j} A_{ij} \cdot D_{ij}(A)$$

is n-linear. □

Claim. E_j is alternating.

Proof. Indeed, by Proposition 20.a.5, it is enough to show that $E_j(A) = 0$ whenever A has two equal adjacent rows. So let

$$A = \begin{pmatrix} \text{———} & \alpha_1 & \text{———} \\ & \vdots & \\ \text{———} & \alpha_n & \text{———} \end{pmatrix}$$

be such a matrix, where $\alpha_k = \alpha_{k+1}$ for some $1 \leq k \leq n-1$. Let $1 \leq i \leq n$, $i \neq k, k+1$. The matrix $A(i \mid j)$ has two equal adjacent rows (because $i \neq k, k+1$), hence $D_{ij}(A) = 0$ (b.c. $D_{ij}(A) = D(A(i \mid j))$ and D is assumed to be alternating).

$$\Rightarrow E_j(A) = (-1)^{k+j} A_{kj} \cdot D_{kj}(A) + (-1)^{(k+1)+j} A_{(k+1)j} \cdot D_{(k+1)j}(A).$$

hence $D_{ij}(A) = 0$ (b.c. $D_{ij}(A) = D(A(i|j))$ and D is assumed to be alternating).
① = ②

$$\Rightarrow E_j(A) = (-1)^{k+j} \overbrace{(A_{kj} D_{kj}(A))}^{①} + (-1)^{k+1+j} \overbrace{(A_{k+1,j} \cdot D_{k+1,j}(A))}^{②}$$

But $\alpha_k = \alpha_{k+1} \Rightarrow A_{kj} = A_{k+1,j}$ and $A(k|j) = A(k+1|j)$

But $\alpha_k = \alpha_{k+1} \Rightarrow A_{kj} = A_{k+1,j}$ and $A(k|j) = A(k+1|j)$

So $E_j(A) = (-1)^{k+j}① + (-1)^{k+1+j}①$

32

$$= (-1)^{k+j} \cdot (① - ①) = 0.$$

This proves E_j is alternating.

Finally, assume $D(I_{n-1}) = 1$.

$$\begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} = I_n$$

$I_n(j|j) = I_{n-1}, (I_n)_{ij} = \delta_{ij}$

So, $E_j(I_n) = \sum_{i=1}^{n} (-1)^{i+j} \delta_{ij} D(I_n(i|j))$

$$= (-1)^{2j} \cdot 1 \cdot D(I_{n-1}) = 1.$$

□

Example. For a 2×2 mat. $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$

we have that $B \longmapsto B_{11}B_{22} - B_{12}B_{21}$

is a determin funct. $\det(B)$ or $|B|$

Consider $A = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} \in M_{3\times 3}(K)$.

32 $$E_1(A) = A_{11}\begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{21}\begin{vmatrix} A_{12} & A_{13} \\ A_{32} & A_{33} \end{vmatrix} + A_{31}\begin{vmatrix} A_{12} & A_{13} \\ A_{22} & A_{23} \end{vmatrix}$$

$$E_2(A) = -A_{12}\begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} + A_{22}\begin{vmatrix} A_{11} & A_{13} \\ A_{31} & A_{33} \end{vmatrix} - A_{32}\begin{vmatrix} A_{11} & A_{13} \\ A_{21} & A_{23} \end{vmatrix}$$

Lecture 33: Uniqueness of the determinant function

09 Mar 2026

Recap

Last time. Determinant functions $M_{n\times n}(K) \to K$ do exist.

In fact. If D is an $(n-1)$-linear alternating function and if $1 \le j \le n$ is fixed, then the map $E_j : M_{n\times n}(K) \to K$ is defined by

$$E_j(A) = \sum_{i=1}^{n} (-1)^{i+j} A_{ij}\, D\begin{pmatrix} \text{row } i \quad \text{column } j \\ A_{ij} \end{pmatrix},$$

where we delete row i and column j. Then E_j is n-linear and alternating. If D is a determinant function, then $E_j = \det$.

33

3.2 Uniqueness of the determinant function

Let $D\colon M_n(K) \to K$ be an n-linear and alternating function.

$$A = \begin{pmatrix} — & \alpha_1 & — \\ — & \alpha_2 & — \\ — & \vdots & — \\ — & \alpha_n & — \end{pmatrix} \in M_n(K).$$

$$I := \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} — & \varepsilon_1 & — \\ & \vdots & \\ — & \varepsilon_n & — \end{pmatrix} \text{ this is the standard basis of } K^n_{row}.$$

We have $\alpha_i = \displaystyle\sum_{j=1}^{n} A_{ij}\varepsilon_j$

$$D(A) = D(\alpha_1, \dots, \alpha_n) = D\left(\sum_{j=1}^{n} A_{1j}\varepsilon_j, \alpha_2, \dots, \alpha_n\right) = \sum_{j=1}^{n} A_{1j} D(\varepsilon_j, \alpha_2, \dots, \alpha_n).$$

$$\sum_{j=1}^{n} A_{1j} D\left(\varepsilon_j, \sum_{k=1}^{n} A_{2k}\varepsilon_k, \ldots, \alpha_n\right) = \sum_{1 \leq j,k \leq n} A_{1,j} \cdot A_{2,k} \cdot D(\varepsilon_j, \varepsilon_k, \ldots, \alpha_n).$$

$$\sum_{1 \leq k_1, \ldots k_n \leq n} A_{1,k_1} \cdot A_{2,k_2} \cdots A_{n,k_n} \cdot D(\varepsilon_{k_1}, \varepsilon_{k_2}, \ldots, \varepsilon_{k_n}) \quad . \tag{*}$$

Since D is alternating, $D(\varepsilon_{k_1}, \ldots, \varepsilon_{k_n}) = 0$ whenever two of the indices $k_1, \ldots, k_n$ are equal. So in the previous sum, it's enough to sum up over $(k_1, \ldots, k_n$ with the property that $k_i \neq k_j, \forall i \neq j)$ such a sequence is called a permutation of degree n. (or n-permutation) We can think of permutations as functions

$$\sigma : \{1, \ldots, n\} \to \{1, \ldots, n\}$$

which are bijective. $(k_1, \ldots, k_n) = (\sigma(1), \ldots, \sigma(n))$. So (*) $= \sum_\sigma A_{1,\sigma(1)} \cdot A_{2,\sigma(2)} \cdots A_{n,\sigma(n)} \cdot D(\varepsilon_{\sigma(1)}, \ldots, \varepsilon_{\sigma(n)})$. Where the sum (*) runs over all permutations σon $\{1, \ldots, n\}$.

Number of permutations on n objects $= n!$. $(n \cdot (n-1) \cdot \cdots \cdot \ldots \cdot 2 \cdot 1)$.

Every peromutation $\sigma : \{1, \ldots, n\} \to \{1, \ldots, n\}$ can be written as a compostion $\sigma = \theta_1 \circ \ldots \circ \theta_m$, of transpositions θ_i. A transposition θ is a permutation such that $\exists i \neq j \in \{1, \ldots, n\} i \neq j$ such that $\theta(i) = j, \theta j = i$ and $\theta k = k \forall k \neq i, j$.
Why? Start with $1, \ldots, n$. If $\sigma(1) \neq 1$ then interchange $1 \& \sigma(1)$. Now we have $\sigma 1, \ldots, 1, \ldots, n$. If $\sigma(2) \neq 2$ then interchange them. Now we have to 3, etc.

33 **Example** (3,2,1). $\sigma(1) = 3, \sigma(2) = 2, \sigma(3) = 1$.

$$(1,2,3) \to (3,2,1) \text{ We are done.}$$

$$(1,2,3) \longrightarrow (2,1,3) \longrightarrow (2,3,1) \longrightarrow (3,2,1).$$

$$\sigma = \theta_1 = \theta'_1 \circ \theta'_2 \circ \theta'_3.$$

Every σ can be written as $\sigma = \theta_1 \circ \ldots \circ \theta_m$ but not in a unique way, i.e. $\sigma = \theta'_1 \circ \ldots \circ \theta'_{m'}$ with θ'_j transpositions.

Lemma (20.b.1). If $\sigma = \theta_1 \circ \ldots \circ \theta_m = \theta'_1 \circ \ldots \circ \theta'_{m'}$ with θ_i and θ'_j transpositions, then m and m' have the same parity, namely m is odd iff m' is odd $\Leftrightarrow (-1)^m = (-1)^{m'}$. Define $\operatorname{sgn}(\sigma) := (-1)^m$.

Proof. We know that a determinant function $M_{n \times n}(K) \to K$ exists for all $n \geq 1$. Fix one such function E. Consider $E(\varepsilon_{\sigma(1)}, \ldots, \varepsilon_{\sigma(n)})$. If $\sigma(1), \ldots, \sigma(n)$ can be obtained from $1, \ldots, n$ by a sequence of m transpositions, then the matrix

$$\begin{pmatrix} - & \varepsilon_{\sigma(1)} & - \\ & \vdots & \\ - & \varepsilon_{\sigma(n)} & - \end{pmatrix}$$

can be obtained from

$$I = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{pmatrix}$$

by a sequence of m interchanges of pairs of rows. Since E is alternating, each interchange changes the sign, hence

$$E(\varepsilon_{\sigma(1)}, \ldots, \varepsilon_{\sigma(n)}) = (-1)^m E(I) = (-1)^m,$$

because $E(I) = 1$.

By the same argument, using the decomposition $\sigma = \theta'_1 \circ \ldots \circ \theta'_{m'}$, we get

$$E(\varepsilon_{\sigma(1)}, \ldots, \varepsilon_{\sigma(n)}) = (-1)^{m'}.$$

Therefore

$$(-1)^m = (-1)^{m'},$$

so m and m' have the same parity. □

Back to $D(A)$,

$$D(A) = \sum_\sigma A_{1,\sigma(1)} \cdots A_{n,\sigma(n)}\, D(\varepsilon_{\sigma(1)}, \ldots, \varepsilon_{\sigma(n)})$$

33 and by Lemma 20.b.1,

$$D(\varepsilon_{\sigma(1)}, \ldots, \varepsilon_{\sigma(n)}) = \underbrace{\operatorname{sgn}(\sigma) \cdot D(I)}_{\text{depends only on } \sigma \text{ and } D(I)}.$$

Theorem (20.b.2)**.** For every n-linear and alternating function $D\colon M_n(K) \to K$, we have

$$D(A) = \sum_\sigma \operatorname{sgn}(\sigma)\, A_{1,\sigma(1)} \cdots A_{n,\sigma(n)}\, D(I).$$

In particular, if D is a determinant function, then

$$D(A) = \sum_\sigma \operatorname{sgn}(\sigma)\, A_{1,\sigma(1)} \cdots A_{n,\sigma(n)}.$$

So D is unique.

Proof. Substitute the previous formula for $D(\varepsilon_{\sigma(1)}, \ldots, \varepsilon_{\sigma(n)})$ into the expression for $D(A)$. □

Corollary (20.b.3)**.** $\forall$ n-linear and alternating function $D\colon M_{n\times n}(K) \to K$, we have $D(A) = D(I) \cdot \det(A)$.

Denote by S_n the set of all permutations on n elements. S_n is called a group. We can compose permutations, $\sigma, \tau : \{1, \ldots, n\} \to \{1, \ldots, n\}$. We have $\sigma \circ \tau$. We can also invert every $\sigma \in S_n$.

$$\sigma^{-1} : \{1, \dots, n\} \to \{1, \dots, n\}.$$

1 identity/unit : the trivial permutation.

Lemma. (20.b.4) $\forall \sigma, \tau \in S_n$, we have $\operatorname{sgn}(\sigma \circ \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau)$.

Proof. Suppose $\sigma = \theta_1 \circ \dots \circ \theta_m$ and $\tau = \lambda_1 \circ \dots \circ \lambda_\ell$, where θ_i and λ_j are transpositions. $\Rightarrow \sigma \circ \tau = \theta_1 \circ \dots \circ \theta_m \circ \lambda_1 \circ \dots \circ \lambda_\ell \Rightarrow \operatorname{sgn}(\sigma \circ \tau) = (-1)^{m+\ell} = (-1)^m \cdot (-1)^\ell$. $\square$

Theorem (20.b.5). $\forall A, B \in M_{n\times n}(K)$, we have $\det(A \cdot B) = \det(A) \cdot \det(B)$.

Proof. Fix $B \in M_{n\times n}(K)$, and consider the function $D\colon M_{n\times n}(K) \to K$ defined by $D(A) = \det(A \cdot B)$.

Claim. D is n-linear and alternating.

Proof. Write $A = \begin{pmatrix} — & \alpha_1 & — \\ & \vdots & \\ — & \alpha_n & — \end{pmatrix}$. Then

33

$$A \cdot B = \begin{pmatrix} — & \alpha_1 \cdot B & — \\ & \vdots & \\ — & \alpha_i \cdot B & — \\ & \vdots & \\ — & \alpha_n \cdot B & — \end{pmatrix},$$

so $D(A) = \det(\alpha_1 \cdot B, \dots, \alpha_n \cdot B)$.

Fix $1 \le i \le n$. If α_i and α_i' are two row vectors, then $(\alpha_i + \alpha_i') \cdot B = \alpha_i \cdot B + \alpha_i' \cdot B$, so

$$D(\alpha_1, \dots, \alpha_i + \alpha_i', \dots, \alpha_n) = D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + D(\alpha_1, \dots, \alpha_i', \dots, \alpha_n),$$

because det is n-linear.

Similarly, for all $c \in K$, $(c \cdot \alpha_i) \cdot B = c \cdot (\alpha_i \cdot B)$, hence

$$D(\alpha_1, \dots, c \cdot \alpha_i, \dots, \alpha_n) = c \cdot D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n).$$

If $\alpha_i = \alpha_j$, then $\alpha_i \cdot B = \alpha_j \cdot B$, hence $D(A) = \det(\alpha_1 \cdot B, \dots, \alpha_n \cdot B) = 0$. So D is alternating. $\square$

By a previous corollary (20.b.3), we have $D(A) = \det(A) \cdot D(I)$. But $D(I) = \det(B)$. Therefore,

$$D(A) = \det(A \cdot B) = \det(A) \cdot \det(B).$$

$\square$

Corollary (20.b.6)**.** If A is invertible, then $\det(A) \neq 0$. Moreover $\det(A^{-1}) = \frac{1}{\det(A)}$.

33 **Proof.** If A is invertible, then $A \cdot A^{-1} = I$. $\Rightarrow \det(A \cdot A^{-1}) = \det(I) = 1$. But $\det(A \cdot A^{-1}) = \det(A) \cdot \det(A^{-1})$. ☐

Lecture 34: Notes pending

11 Mar 2026

3.3 Lecture 34 Placeholder

Recall: $\forall M \in M_{m\times n}(K)$ we have the transposed matrix $M^\top \in M_{n\times m}(K)$ with $(M^\top)_{ij} = M_{ji}$.

Theorem (20.C.1)**.** $\forall A \in M \in M_{n\times n}(K)$ we have $\det(A) = \det(A^\top)$.

Proof. If σ is a permutation of $\{1, \dots, n\}$ then $(A^\top)_{i\sigma(i)} = A_{\sigma(i)i}$. Hence, $\det(A^\top) = \sum_\sigma \operatorname{sgn}(\sigma) A_{\sigma(1),1} \cdots A_{\sigma(n),n}$. If $\sigma_i = j$ then $A_{\sigma(i),i} = A_{j,\sigma^{-1}j}$.

Claim. $A_{\sigma(1),1} \cdot \dots \cdot A_{\sigma(n),n} = A_{1,\sigma^{-1}1} \cdot \dots \cdot A_{n,\sigma^{-1}n}$.

Proof. Every factor on LHS of the equation appears will appear and exactly once on the RHS of the equation. Ao the product on the LHS is the same as the product on the RHS. This proves the claim. ☐

34 $(\operatorname{sgn}(\sigma)) \cdot \operatorname{sgn}(\sigma \circ \sigma^{-1}) = 1$ so $\operatorname{sgn}(\sigma) = \operatorname{sgn}(\sigma^{-1})$ (recall: $\operatorname{sgn} \in \{-1, 1\}$). $\Rightarrow \det(A^\top) = \sum_{\sigma\in S_n} (\operatorname{sgn}\sigma) A_{\sigma(1),1} \cdots A_{\sigma(n),n} = \displaystyle\sum_{\sigma\in S_n} (\operatorname{sgn}\sigma^{-1}) A_{1,\sigma^{-1}1} \cdots A_{n,\sigma^{-1}n} =$ $displaystyle \displaystyle\sum_{\tau\in S_n} (\operatorname{sgn}\tau) A_{1,\tau 1} \cdots A_{n,\tau n} = \det(A)$. When σ runs over S_n, $\tau = \sigma^{-1}$ also runs over S_n and vice versa. ☐

Corollary (20.c.2)**.** If $D : M_{n\times n}(K) \to K$ is an alternating n-linear function (of the rows of the matrix) then D is also alternating $+n$-linear when viewed as a function of the columns of the matrix. In fact, $D(A\top) = D(A) \forall A \in M_{n\times n}(K)$.

Proof. $\forall A, D(A) = \det(A) \cdot D(I)$. $\Rightarrow D(A^\top) = \det(A^\top) \cdot D(I) = \det(A) \cdot D(I) = D(A)$. ☐

Theorem (20.C.3)**.**
1. If B is obtained from A by adding a multiple of one row of A to another row

$$A \xrightarrow{R_i + c\cdot R_j \longrightarrow R_i} (i \neq j(!))B,$$

then $\det(B) = \det(A)$.

2. If $A \xrightarrow{R_i \leftrightarrow R_j (i \neq j (!))} B$ then $\det(B) = -\det(A)$.
3. If $A \xrightarrow{c \cdot R_i \longrightarrow R_i (c \in K)} B$, then $\det(B) = c \cdot \det(A)$.

Proof. 2 Follows from det being alternating.

3 Follows from n-linearity of det.

1 Assume $j > i$ first. Write $A = \begin{pmatrix} — & \alpha_1 & — \\ & \vdots & \\ — & \alpha_n & — \end{pmatrix}$.

$\det B = \det(\alpha_1, \dots, \alpha_i + c \cdot \alpha_j, \dots, \alpha_j, \dots, \alpha_n) = \det(\alpha_1, \dots, \alpha_i, \dots, \alpha_j, \dots, \alpha_n) + c \cdot \det \underbrace{(\alpha_1, \dots, \alpha_j, \dots, \alpha_j, \dots, \alpha_n)}_{=0} = \det(\alpha_1, \dots, \alpha_n) = \det(A)$.

□

Determinants of block matrices.

Example of a block matrix:

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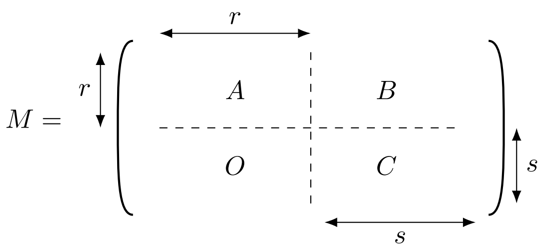

Consider M like here, where

$$A \in M_{r \times r}(K), \qquad C \in M_{s \times s}(K), \qquad B \in M_{r \times s}(K).$$

Proposition (20.c.4).

$$\det(M) = \det(A) \cdot \det(C).$$

Proof. Define

$$D : M_{r \times r}(K) \times M_{r \times s}(K) \times M_{s \times s}(K) \to K,$$

by

$$D(A, B, C) := \det \left(\begin{array}{c|c} A & B \\ \hline O & C \end{array} \right).$$

If we fix A, B, then the function

$$M_{s \times s}(K) \to K, \qquad C \mapsto D(A, B, C)$$

is s-linear and alternating. By a previous theorem (Corollary 20.b.3), we have

$$D(A,B,C) = \det(C) \cdot D(A,B,I_s).(*)$$

$$D(A,B,I_s) = \det\left(\begin{array}{c|c} A & B \\ \hline O & I_s \end{array}\right)$$

By applying row operations of the type $R_i + c \cdot R_j \longrightarrow R_i$ where $1 \leq i \leq r, r+1 \leq j \leq r+s$ we can get to the matrix

$$\begin{array}{c|c} A & O \\ \hline O & I_s \end{array}, \text{ and } \det\left(\begin{array}{c|c} A & B \\ \hline O & I_s \end{array}\right) = \det\left(\begin{array}{c|c} A & O \\ \hline O & I_s \end{array}\right).(**)$$

The function $M_{r\times r}(K) \ni A \mapsto \det\left(\begin{array}{c|c} A & O \\ \hline O & I_s \end{array}\right) \in K$ is r-linear and alternating.

$$\Rightarrow \det\left(\begin{array}{c|c} A & O \\ \hline O & I_s \end{array}\right) = \det(A) \cdot \det\left(\underbrace{\begin{array}{c|c} I_r & O \\ \hline O & I_s \end{array}}_{I_{r+s}}\right) = \det(A).$$

Back to $(*)$, and using $(**)$: $\det\left(\begin{array}{c|c} A & B \\ \hline O & C \end{array}\right) = \det(C) \cdot \det(A)$ □

34 **Example.**

$$\underline{\text{Exp;}} \qquad A = \begin{pmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 1 & 1 & -1 \\ 1 & 2 & 3 & 0 \end{pmatrix} \in M_{4\times 4}(\mathbb{R})$$

$$A \xrightarrow[\substack{R_2 - 2R_1 \to R_2 \\ R_3 - 4R_1 \to R_3 \\ R_4 - R_1 \to R_4}]{} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix} \xrightarrow[\substack{R_3 - \frac{5}{4}R_2 \to R_3 \\ R_4 - \frac{3}{4}R_2 \to R_4}]{} \left(\begin{array}{cc|cc} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ \hline 0 & 0 & -4 & -8 \\ 0 & 0 & 4 & 0 \end{array}\right)$$

$$\det(A) = 1 \cdot 4 \cdot (-8) \cdot 4 = 4 \cdot 32 = 128.$$

Let D be the determinant funct. on $M_{(n-1)\times(n-1)}(K)$. Recall that $\forall 1 \leq j \leq n$, $E_j : M_{n\times n}(K) \to K$, and

$$E_j(A) = \sum_{i=1}^{n} (-1)^{i+j} A_{ij}\, D\big(A(i \mid j)\big).$$

We know now that $\forall 1 \leq j \leq n$, $E_j(A) = \det(A)$.

The scalars $C_{ij} = (-1)^{i+j} \cdot \det A(i \mid j)$ are called the i, j cofactors of A. We have

$$\det(A) = \sum_{i=1}^{n} A_{ij}\, C_{ij},$$

$(*)$

Claim. If we replace A_{ij} in $(*)$ with A_{ik} (k is fixed), where $k \neq j$, then $(*)$ becomes 0.

Namely,

$$\sum_{i=1}^{n} A_{ik} C_{ij} = \begin{cases} \det(A) & k = j, \\ 0 & k \neq j. \end{cases}$$

Proof. Let $B \in M_{n\times n}(K)$ be the matrix obtained from A col $\#j$ of A by col $\#k$ of A. (Assume $k \neq j$) Clearly $\det(B) = 0$. (b.c. B has two equal columns).

$$\Rightarrow\ 0 = \det(B) = \sum_{i=1}^{n} (-1)^{i+j} B_{ij}\, \det\bigl(B(i \mid j)\bigr)$$

$$= \sum_{i=1}^{n} (-1)^{i+j} A_{ik}\, \det\bigl(A(i \mid j)\bigr) = \sum_{i=1}^{n} A_{ik} C_{ij}$$

34 $$\uparrow \qquad\qquad \uparrow$$
$$A_{ik} = B_{ij} \quad A(i \mid j) = B(i \mid j)$$

$\square$

Summary: $\forall A \in M_{n\times n}(K)$, $\forall j, k$ we have $\sum_{i=1}^{n} A_{ik} C_{ij} = \delta_{jk} \cdot \det(A)$. The matrix $(C_{ij})^\top$ is called the classical adjoint of A and it is denoted by $\operatorname{adj}(A) \in M_{n\times n}(K)$.

$$(\operatorname{adj}(A))_{ij} = C_{ji} = (-1)^{j+i} \cdot \det A(j \mid i)..$$

The formula (the one with the delta) says $\boxed{(\operatorname{adj}(A)) \cdot A = (\det(A)) \cdot I.}$

Claim. $A \cdot (\operatorname{adj} A) = (\det A) \cdot I$.

Proof. $A^T(i \mid j) = \bigl(A(j \mid i)\bigr)^T\ \forall\, i, j$.

$$\Rightarrow\ (\operatorname{adj} A^T)_{ij} = (-1)^{i+j} \det A^T(j \mid i) = (-1)^{i+j} \det\bigl(A(i \mid j)\bigr)^T = (-1)^{i+j} \det A(i \mid j) =$$
$$= (\operatorname{adj} A)_{j,i} = (\operatorname{adj} A)^T_{i,j}.$$

$$\Rightarrow \operatorname{adj}(A^T) = (\operatorname{adj} A)^T.$$

$\square$

Lecture 35: Notes pending

16 Mar 2026

Recap

Last time.

$$A \in M_{n\times n}(K), \qquad \text{adj}\, A \in M_{n\times n}(K).$$

Adjugate matrix. For $A \in M_{n\times n}(K)$, the adjugate matrix $\text{adj}\, A$ is defined by

$$(\text{adj}\, A)_{i,j} = (-1)^{i+j} \det A(j \mid i).$$

We have seen.
We have

$$(\text{adj}\, A) \cdot A = (\det A) \cdot I.$$

Also.
Transpose compatibility. Also,

$$\text{adj}(A^\top) = (\text{adj}\, A)^\top.$$

Claim.
Goal for today. We will prove

$$A \cdot (\text{adj}\, A) = (\det A) \cdot I.$$

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Theorem (20.c.6)**.** Let $A \in M_{n\times n}(K)$ A is invertible iff $\det A \neq 0$. When A is inveritible, we have,

$$A^{-1} = \frac{1}{\det A} \text{adj}\, A..$$

3.3.1 Cramer's rule

Let $A \in M_{n\times n}(K)$ and $b \in K^n$. Consider the system of equations, $A \cdot x = b$. where $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$. $Ax = b \Rightarrow (\text{ adj}\, A) \cdot A \cdot x = (\text{adj}\, A) \cdot b \Rightarrow (\det A) \cdot x = (\text{adj}\, A) \cdot b$ (*). If det $A \neq 0$, then:

$$\Rightarrow x = \frac{1}{\det A} (\text{adj}\, A) \cdot b.$$

$\Rightarrow \forall 1 \leq j \leq n$, we have:

$$(*) : (\det A) \cdot x_j = \sum_{i=1}^{n} (\text{adj}\, A)_{j,i} \cdot b_i = \sum_{i=1}^{n} (-1)^{i+j} b_i \cdot \det A(i \mid j) (**).$$

Consider the matrix

$$A(j,b) := \left(\begin{array}{c|c|c} A & b & A \end{array} \right) \qquad \leftarrow \text{col } \# \, j$$

the matrix obtained from A by replacing col # j with the vector b. Entry (i,j) of this matrix is exactly b_i.

So, by previous formulas (formula for E_j, but applied to $A(j,b)$), we have:

$$(**) = \det A(j,b).$$

Cramer's formula:

$$x_j = \frac{\det A(j,b)}{\det A}.$$

3.4 Determinants and Endomorphisms

Warm up. Let $A \in M_{n\times n}(K)$, $P \in GL(n,K)$ invertible.

$$\Rightarrow \det(P^{-1}AP) = \det(P^{-1}) \cdot \det(A) \cdot \det(P) = \det(A)$$

35 So, similar matrices

$$\left(\frac{1}{\det P}\right)$$

have the same det. Let V be a f. dim. vect. sp. over K.

Put $n := \dim V$. Let $T : V \to V$ be a lin. map. Pick a basis B for V.

$\rightsquigarrow$ We get a matrix $[T]^B_B$.

Consider $\det[T]^B_B \in K$. Q. Does $\det[T]^B_B$ depend on B?

A. Let B' be another basis for V.

Recap

Recall.
Change of basis. If B' is another basis of V, then

$$[T]^{B'}_{B'} = [\mathrm{id}_V]^{B'}_B \, [T]^B_B \, [\mathrm{id}_V]^B_{B'} = P^{-1} \, [T]^B_B \, P, \qquad P = [\mathrm{id}_V]^B_{B'}.$$

Recall that

$$[\mathrm{id}_V]^{B'}_B = \left([\mathrm{id}_V]^B_{B'}\right)^{-1},$$

so

$$\det[T]^{B'}_{B'} = \det[T]^B_B,$$

so $\det(T)$ is independent of the choice of basis for V.

Composition and inverses. Recall also that for linear maps $S, T : V \to V$, we have

$$[S \circ T]^B_B = [S]^B_B \cdot [T]^B_B, \qquad [T^{-1}]^B_B = ([T]^B_B)^{-1}$$

whenever T is an isomorphism. Applying det, we get

$$\det(S \circ T) = \det(S) \cdot \det(T), \qquad \det(T^{-1}) = \frac{1}{\det(T)}.$$

Special cases. So

$$\det(\mathrm{id}_V) = 1, \qquad \det(O_{V \longrightarrow V}) = 0.$$

Important distinction. For a general linear map $T : V \to W$, the quantity $\det[T]_C^B$ depends on the choices of bases B and C. Consequently, we cannot define $\det(T)$ for arbitrary maps $T : V \to W$. It is only well-defined for endomorphisms $T : V \to V$.

T is an iso if and only if, in some basis or in every basis—doesn't matter—which we choose here and that we choose here, the determinant of T, sorry, the matrix of T, is an invertible matrix.

So, once we know that T can be represented in some basis here and some basis here by an invertible matrix, you know this is equivalent to requiring that T is an iso.

Now, forget about T. We know how to decide whether a matrix is invertible or not. We just take the determinant of this matrix and see if the number is zero or not. The value of this determinant, assuming it's not zero, is going to depend on the choices. However, the property of this number not being zero is not going to matter.

As someone else in here phrased it, and as my dad phrased it, if in some basis here and
35 here the matrix has determinant zero, then this will continue to be the case in all other
bases. But if the determinant is not zero in some basis here and some basis here, it's going to continue not to be zero in every other choice of basis; however, the value which is not zero may depend on your choice.

Okay? So, let's get to the next question. I don't know; I have, now it's just confusing.

Right, so we begin a new topic—an extremely important, central topic when we are providing those sorts of genetics. This is quite a strange word game.

So, I am now using English, but the name in English is Eigen Vectors.

Chapter 4

Eigen Vectors

Definition (21.a.1). Let V be a vector space over K and $T\colon V \to V$ a linear map. We say that $\lambda \in K$ is an eigenvalue of T if $\exists\, 0 \neq v \in V$ s.t. $Tv = \lambda v$. Any vector $v \in V$ with this property (i.e. $Tv = \lambda v$) is called an eignevector of T for the eigenvalue λ.

35 **Definition (21.a.2).** Let V be a finite dimensional vector space over K. Let $T : V \to V$ be a linear map. We say that T is diagonalizable if $\exists$ a basis B of V such that

$$[T]_B^B = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \; (*)$$

Lemma (21.a.3). T is diagonalizable *iff* $\exists$ a basis $B = \{v_1, \ldots, v_n\}$ consisting of eigenvectors of T. In this case, if $Tv_i = \lambda_i v_i$ for some $\lambda_i \in K$. Moreover, if for some basis $B = \{v_1, \ldots, v_n\}$ the matrix $[T]_B^B$ has the shape (*), then $\forall k$, λ_i is an eigenvalue of T and v_i is an eigenvector of T.

Definition. A matrix $A \in M_{n\times n}(K)$ is called diagonalizable if the $T_A : K^n \to K^n$ is diagonalizable.

Example. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, K = \mathbb{R}$.

$$T_A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = A \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda = 3 \text{ is an eigenvalue of } T_A,$$

and

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ is an eigenvect. of } \lambda_1 = 3.$$

$$T_A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

so

$$v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ is an eigenvect. of the eigenvalue } \lambda_2 = 1.$$

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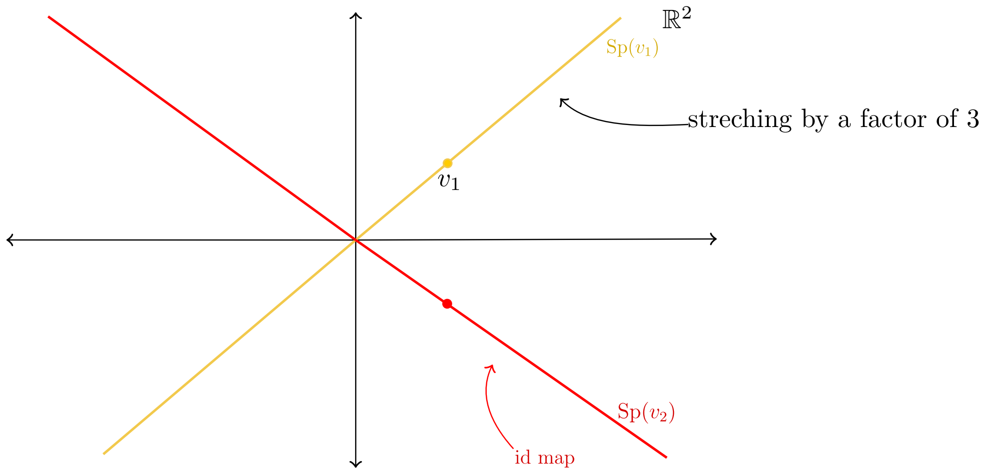

Figure 4.1: FigEigen1

4.1 Quick Sidequest

Goal: prove the following statement.

Theorem (Completeness)**.** Let $\mathscr{L}$ be an arbitrary (uncountable) signature, and let T be a consistent set of $\mathscr{L}$-sentences. Then T has a model.

Subgoal: construct an $\mathscr{L}$-structure M^* with domain A^*.

Lemma (Compactness)**.** Let T be an $\mathscr{L}$-theory such that every finite subset $\Phi \subseteq T$ has an $\mathscr{L}$-structure M_Φ satisfying $M_\Phi \vDash \Phi$. Then T has a model.

Definition. Let S be a set. An ultrafilter $\mathcal{U} \subseteq \mathcal{P}(S)$ over S is a filter on S such that

1. $\mathcal{U}$ is a filter on S.

2. For every $X \in \mathcal{P}(S)$, we have $X \in \mathcal{U}$ or $S \setminus X \in \mathcal{U}$.

Theorem. Every filter can be extended to an ultrafilter.

Definition. Let $\mathscr{L}$ be an arbitrary fixed signature, let $I \neq \varnothing$ be a set, and for each $i \in I$ let

$$M_i$$

be an $\mathscr{L}$-structure with domain A_i. Put

$$A := \bigtimes_{i \in I} A_i,$$

and identify elements of A with functions

$$g : I \to \bigcup_{i \in I} A_i$$

such that $g(i) \in A_i$ for each $i \in I$.

Definition. Let $\mathcal{U} \subseteq \mathcal{P}(I)$ be an ultrafilter over I. Define a relation on A by

$$f \sim g \Leftrightarrow \{\, i \in I \mid f(i) = g(i) \,\} \in \mathcal{U}.$$

Let $A^* := A/{\sim}$.

Definition. For each constant symbol $c \in \mathscr{L}$, define $f_c \in A$ by

$$f_c(i) := c^{M_i} \qquad (i \in I),$$

and define

$$c^{M^*} := [f_c].$$

Definition. For every n-ary function symbol $F \in \mathscr{L}$, define

$$F^{M^*} : (A^*)^n \to A^*$$

by

$$F^{M^*}([f_0], \ldots, [f_{n-1}]) = [f] \Leftrightarrow \{\, u \in I \mid F^{M_u}(f_0(u), \ldots, f_{n-1}(u)) = f(u) \,\} \in \mathcal{U}.$$

Definition. For every n-ary relation symbol $R \in \mathscr{L}$, define $R^{M^*} \subseteq (A^*)^n$ by

$$\langle [f_0], \ldots, [f_{n-1}] \rangle \in R^{M^*} \Leftrightarrow \{\, u \in I \mid \langle f_0(u), \ldots, f_{n-1}(u) \rangle \in R^{M_u} \,\} \in \mathcal{U}.$$

Definition. The $\mathscr{L}$-structure M^* with domain A^* is then an ultraproduct of the $\mathscr{L}$-structures M_i $(i \in I)$ with respect to $\mathcal{U}$.

Definition. If $M_i = M$ for every $i \in I$, then M^* is called an ultrapower of M (with respect to $\mathcal{U}$).

Theorem (Łoś). Let σ be an $\mathscr{L}$-sentence. Then

$$M^* \models \sigma \Leftrightarrow \{ i \in I \mid M_i \models \sigma \} \in \mathcal{U}.$$

Proof of compactness. Let

$$I := \{ \Phi \subseteq T \mid \Phi \text{ finite} \}.$$

For every $\Phi \in I$, choose an $\mathscr{L}$-structure M_Φ with domain A_Φ such that

$$M_\Phi \models \Phi.$$

For every $\Phi \in I$, define

$$\Delta(\Phi) := \{ \Phi' \in I \mid \Phi \subseteq \Phi' \}.$$

Then $\Phi \in \Delta(\Phi)$, so $\Delta(\Phi) \neq \varnothing$. Moreover, for all $\Phi_1, \Phi_2 \in I$,

$$\Delta(\Phi_1) \cap \Delta(\Phi_2) = \Delta(\Phi_1 \cup \Phi_2),$$

and $\Phi_1 \cup \Phi_2 \in I$. Hence the family

$$\mathcal{F} := \{ X \subseteq I \mid \exists \Phi \in I : \ \Delta(\Phi) \subseteq X \}$$

is a filter on I. By the previous theorem, $\mathcal{F}$ extends to an ultrafilter $\mathcal{U}$ on I.

Let M^* be the ultraproduct of the structures M_Φ $(\Phi \in I)$ with respect to $\mathcal{U}$.

Now fix an arbitrary sentence $\sigma_0 \in T$. Then $\{\sigma_0\} \in I$ and

$$M_{\{\sigma_0\}} \models \sigma_0.$$

For every $\Phi \in \Delta(\{\sigma_0\})$, we have $\sigma_0 \in \Phi$, and since $M_\Phi \models \Phi$, it follows that

$$M_\Phi \models \sigma_0.$$

Thus

$$\Delta(\{\sigma_0\}) = \{ \Phi \in I \mid \sigma_0 \in \Phi \} \subseteq \{ \Phi \in I \mid M_\Phi \models \sigma_0 \}.$$

Observe that $\Delta(\{\sigma_0\}) \in \mathcal{F} \subseteq \mathcal{U}$, so

$$\{ \Phi \in I \mid M_\Phi \models \sigma_0 \} \in \mathcal{U}.$$

By Łoś's theorem, it follows that

$$M^* \models \sigma_0.$$

Because $\sigma_0 \in T$ was arbitrary, $M^* \models T$. Thus T has a model. $\square$

Recap of the strategy. To prove completeness for arbitrary signatures, it is enough to show that every finite subset $\Phi \subseteq T$ has a model. For such a finite Φ, only finitely many non-logical symbols occur, so we pass to the finite subsignature $\mathscr{L}'_\Phi$, construct a model there, extend it back to an $\mathscr{L}$-structure, and then apply compactness.

Proof of completeness. Assume that T is consistent. Let $\Phi \subseteq T$ be finite. Then Φ is consistent as well.

Let $\mathscr{L}'_\Phi \subseteq \mathscr{L}$ be the finite subsignature consisting of all non-logical symbols appearing in sentences of Φ. Then Φ is a consistent $\mathscr{L}'_\Phi$-theory. Since $\mathscr{L}'_\Phi$ is finite, the finite-signature model-existence theorem applies, so there exists an $\mathscr{L}'_\Phi$-structure

$$M'_\Phi$$

with domain A_Φ such that

$$M'_\Phi \models \Phi.$$

Now extend M'_Φ to an $\mathscr{L}$-structure M_Φ with the same domain A_Φ by choosing arbitrary interpretations for symbols in $\mathscr{L} \setminus \mathscr{L}'_\Phi$. Since the sentences in Φ only use symbols from $\mathscr{L}'_\Phi$, we still have

$$M_\Phi \models \Phi.$$

Thus for every finite subset $\Phi \subseteq T$, there exists an $\mathscr{L}$-structure M_Φ such that

$$M_\Phi \models \Phi.$$

By Lemma 1, T has a model. □

Lecture 36: Notes pending

18 Mar 2026

4.2 Lecture 36 Placeholder

Proposition (21.a.4)**.** Let V be a vector space over a field K and let $T : V \to V$ be a linear map. Suppose $\lambda_1, \dots, \lambda_k$ are eigenvalues of T, s.t. they are pairwise distinct (i.e. $\lambda_i \neq \lambda_j \forall i \neq j$). Let $v_1, \dots, v_n$ be eigenvectors for $\lambda_1, \dots, \lambda_n$ respectively. Then $v_1, \dots, v_n$ are linearly independent. (put differently, eigenvectors corresponding to distinct eigenvalues are linearly independent).

36 **Proof.** Induction on n. For $n = 1$, v_1 is and eigenvector of λ_1. By definition of eigenvector, $v_1 \neq 0$. So v_1 is linearly independent. Let $n > 1$ and suppose the proposition holds for every list of n eigenvalues and eigenvectors. We will show the preposition holds also for any list of $n + 1$ eigenvalues and eigenvectors. Indeed, let $\lambda_1, \dots, \lambda_{n+1}$ be pairwise distinct eigenvalues of T and let $v_1, \dots, v_n, v_{n+1}$ be eigenvectors corresponding to $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$. Assume

$$a_1 v_1 + \ldots + a_n v_n + a_{n+1} v_{n+1} = 0 (*)$$

We will show that $a_1 = \ldots = a_n = a_{n+1} = 0$. Applying T to (*) we get

$$a_1 T v_1 + \ldots + a_n T v_n + a_{n+1} T v_{n+1} = 0$$

But $Tv_i = \lambda_i v_i$, so we get

$$a_1 \lambda_1 v_1 + \ldots + a_n \lambda_n v_n + a_{n+1} \lambda_{n+1} v_{n+1} = 0$$

Consider $(**) - \lambda_{n+1} \cdot (*)$:

$$a_1(\lambda_1 - \lambda_{n+1})v_1 + \ldots + a_n(\lambda_n - \lambda_{n+1})v_n = 0.$$

By the induction hypothesis, $v_1, \ldots, v_n$ are linearly independent, hence,

$$a_1(\lambda_1 - \lambda_{n+1}) = \ldots = a_n(\lambda_n - \lambda_{n+1}) = 0.$$

By assumption $\lambda_{n+1} \neq \lambda_i \forall 1 \leq i \leq n$, so $\lambda_i - \lambda_{n+1} \neq 0 \forall 1 \leq i \leq n$. Hence, $a_1 = \ldots = a_n = 0$. Substituting this back in (*):

$$a_{n+1} v_{n+1} = 0$$

. Since $v_{n+1} \neq 0$ (it is an eigenvector!), we deduce that $a_{n+1} \cdot v_{n+1} = 0$. This completes the induction step and the proof. □

An eigenvector for a given eigenvalue is not unique. If v is an eigenvector for λ and $c \in K^\times$, then cv is again an eigenvector for λ. More generally, all eigenvectors for λ, together with 0, form the eigenspace $\mathrm{Eig}_T(\lambda)$.

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4.3 The characteristic polynomial

Proposition (21.a.5). Let V be a vector space over a field K and let $T : V \to V$ be a linear map. Then $\lambda \in K$ is an eigenvalue of T iff the map $T - \lambda \cdot \mathrm{id}_V : V \to V$ is NOT injective.

Proof. λ is an eigenvalue of $T \Leftrightarrow \exists O \neq v \in V$ s.t. $Tv = \lambda v \Leftrightarrow \exists 0 \neq v \in V$ s.t. $(T - \lambda \cdot \mathrm{id}_V)(v) = 0 \Leftrightarrow \ker(T - \lambda \cdot \mathrm{id}_V) \neq \{0\} \Leftrightarrow T - \lambda \cdot \mathrm{id}_V$ is NOT injective. □

Definition (21.a.6). Let V be a vector space over a field K andlet $T : V \to V$ be a linear map. Define $P_T(x) := \det(T - x \cdot \mathrm{id}_V)$. We call $P_T(x)$ the *characteristic polynomial* of T. If $A \in M_{n\times n}(K)$, we define the characteristic polynomial of A, as $P_A(x) := \det(A - x \cdot I_n)$.

Definition (21.a.7). Let V be a vector space over a field K and let $T : V \to V$ be a linear map. Let $\lambda \in K$ be an eigenvalue of T. The eignespace of T corresponding to λ is $\mathrm{Eig}_T(\lambda) := \{v \in V : Tv = \lambda v\} = \ker(T - \lambda \cdot \mathrm{id}_V) = = \{v \in V \mid v \text{is an eigenvector of } T \text{ corresponding to } \lambda\} \cup \{0\}$. Clearly $\mathrm{Eig}_T(\lambda) \subseteq V$ is a linear subspace (b.c. it is a kernel of a linear map).

Lemma (21.a.8)**.** Let $A \in M_{n\times n}(K)$. Then:

1. $P_A(x)$ is a polynomial of degree n, with leading coefficient $(-1)^n$, namely,
$$P_A(x) = c_n x^n + c_{n-1}x^{n-1} + \ldots + c_1 x + c_0$$
with $c_i \in K$ for all i, and $c_n = (-1)^n$.
2. $P_{A^T}(x) = P_A(x)$.
3. If
$$A = \begin{pmatrix} \lambda_1 & \cdots & * \\ 0 & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}$$
has upper triangular shape, then
$$P_A(x) = (\lambda_1 - x)\cdots(\lambda_n - x).$$
4. If
$$A = \begin{array}{cc} \left(\begin{array}{c|c} B_1 & * \\ \hline 0 & B_2 \end{array}\right) & \begin{array}{l} \updownarrow r \\ \updownarrow s \end{array} \end{array}$$
is a block matrix, where
$$B_1 \in M_{r\times r}(K), \qquad B_2 \in M_{s\times s}(K),$$
36 then
$$P_A(x) = P_{B_1}(x) \cdot P_{B_2}(x).$$
5. If A and B are similar matrices, $B = P^{-1}AP$ for some invertible P, then
$$P_B(x) = P_A(x).$$

Proof. 1)
$$P_A(x) = \det(A - xI) = \det \begin{pmatrix} A_{11} - x & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} - x & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & \cdots & \cdots & A_{nn} - x \end{pmatrix}$$
Denote this matrix by B. Then
$$P_A(x) = \sum_{\sigma \in S_n} (\operatorname{sgn}\sigma)\, B_{1,\sigma(1)} \cdots B_{n,\sigma(n)}.$$
Hence
$$P_A(x) = (A_{11} - x)(A_{22} - x)\cdots(A_{nn} - x) + \sum_{\sigma \neq \mathrm{id}} (\operatorname{sgn}\sigma)\, B_{1,\sigma(1)} \cdots B_{n,\sigma(n)}.$$

Here the first term corresponds to $\sigma = \text{id}$, and

$$\text{sgn}(\text{id}) = 1.$$

If $\sigma \neq \text{id}$, then at least one of the factors

$$B_{1,\sigma(1)}, \ldots, B_{n,\sigma(n)}$$

will not have an x in it. Hence

$$P_A(x) = (-1)^n x^n + \text{ polynomial of deg } \leq n-1.$$

This proves 1.

2)
$$P_{A^T}(x) = \det(A^T - xI) = \det\big((A - xI)^T\big) = \det(A - xI) = P_A(x).$$

(3+4) Exc.

5)
$$\begin{aligned} P_B(x) &= \det(B - xI) = \det(P^{-1}AP - xI) \\ &= \det\big(P^{-1}(A - xI)P\big) = \det(A - xI) = P_A(x). \end{aligned}$$

□

As already suggested, the roots of the characteristic polynomial are the candidates for
36 eigenvalues. However, this depends on the field. For example, the polynomial x^2+1 has no roots over $\mathbb{R}$, but it has roots $\pm i$ over $\mathbb{C}$. Thus a real matrix may have no real eigenvalues, even though it has complex eigenvalues after extending scalars to $\mathbb{C}$.

Corollary. If V is an n-dimensional vector space over a field K and $T : V \to V$ is a linear map, then $P_T(x)$ is a polynomial in x of degree n with leading coefficient $(-1)^n$.

Proof. Let $\mathcal{B}$ be a basis of V. Then $[T - x\text{id}_V]^{\mathcal{B}}_{\mathcal{B}} = [T]^{\mathcal{B}}_{\mathcal{B}} - xI_n$. Hence

$$P_T(x) = \det(T - x\text{id}_V) = \det([T]^{\mathcal{B}}_{\mathcal{B}} - x \cdot I_n) = P_A(x),$$

where $A = [T]^{\mathcal{B}}_{\mathcal{B}}$. By Corollary follows from (1) of Lemma 21.a.8. □

Definition (21.a.9). Let $A \in M_{n\times n}(K)$. Define the *trace* of A as $\text{Tr}(A) := \sum_{i=1}^{n} A_{ii}$. If V is finite-dimensional and $T : V \to V$ is a linear map, define $\text{Tr}(T) := \big(\text{Tr}([T]^{\mathcal{B}}_{\mathcal{B}})\big)$, where $\mathcal{B}$ is any basis of V.

Lemma (21.a.10).
1. $\forall A, B \in M_{n\times n}(K)$, $\text{Tr}(A \cdot B) = \text{Tr}(B \cdot A)$.
2. If A and B are similar, then $\text{Tr}(A) = \text{Tr}(B)$. In particular, if $\mathcal{B}$ and $\mathcal{B}'$ are

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two bases for V, then $\mathrm{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \mathrm{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'})$. Hence, the trace of a linear map is well-defined.

Proof. *1.* $\mathrm{Tr}(A \cdot B) = \sum_{i=1}^{n} (A \cdot B)_{ii} = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} A_{ij} \cdot B_{ji} \right) = \sum_{ij} A_{ij} \cdot B_{ji} = \sum_{ij} A_{ji} \cdot B_{ij} = \sum_{i=1}^{n} \sum_{j=1}^{n} B_{ij} \cdot A_{ji} = \sum_{i=1}^{n} (B \cdot A)_{ii} = \mathrm{Tr}(B \cdot A)$.

2. If $B = P^{-1}AP$, then

$$\mathrm{Tr}(B) = \mathrm{Tr}(P^{-1}AP) = \mathrm{Tr}\big(P^{-1}(AP)\big) = \mathrm{Tr}\big((AP)P^{-1}\big) = \mathrm{Tr}(A),$$

by (1).

□

Lecture 37: Notes pending

20 Mar 2026

Recap

Characteristic polynomial. For $A \in M_{n\times n}(K)$, we define

$$p_A(x) := \det(A - xI) = (-1)^n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0.$$

Trace. For $A = (A_{ij}) \in M_{n\times n}(K)$, we set

$$\mathrm{Tr}(A) := A_{11} + A_{22} + \cdots + A_{nn}.$$

Moreover,

$$\mathrm{Tr}(AB) = \mathrm{Tr}(BA) \qquad \text{and} \qquad \mathrm{Tr}(P^{-1}AP) = \mathrm{Tr}(A)$$

for every invertible P.

Endomorphisms. If $T\colon V \to V$ is a linear map on a finite-dimensional vector space and $A = [T]_{\mathcal{B}}^{\mathcal{B}}$ for a basis $\mathcal{B}$ of V, then

$$p_T(x) := p_A(x), \qquad \mathrm{Tr}(T) := \mathrm{Tr}(A).$$

Both quantities are independent of the choice of basis $\mathcal{B}$.

Somehow nonlinear algebra enters when you start to study linear algebra.

Claim. $\mathrm{Tr}(T)$ is independent of $\mathcal{B}$. Indeed, if $\mathcal{B}'$ is another basis of V, then

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$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = \underbrace{[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}}_{=P^{-1}}[T]_{\mathcal{B}}^{\mathcal{B}}\underbrace{[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}}}_{=P}$$

So $\mathrm{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'}) = \mathrm{Tr}[T]_{\mathcal{B}}^{\mathcal{B}}$.

Lemma (21.a.11). $C_{n-1} = (-1)^{n-1}\,\mathrm{Tr}(A)$ and $c_0 = \det A$.

Proof.

$$p_A(x) = \det\begin{pmatrix} A_{11}-x & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22}-x & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nn}-x \end{pmatrix}$$

Denote this matrix by B. Then

$$p_A(x) = (A_{11}-x)(A_{22}-x)\cdots(A_{nn}-x) + \sum_{\substack{\sigma\neq \mathrm{id}\\ \sigma\in S_n}} \underbrace{(\operatorname{sgn}\sigma)\cdot B_{1,\sigma 1}\cdots B_{n,\sigma n}}_{(*)}.$$

If $\sigma \neq \mathrm{id}$, then $\exists$ at least two indices $i \neq j$ s.t. $\sigma i \neq i$, $\sigma j \neq j$. Therefore, in the product $(*)$, $\exists$ at least two factors

$$(B_{i,\sigma i} \,\&\, B_{j,\sigma j})$$

that are not from the diag. of B. So these two factors do NOT contain x. Hence

$$\deg(\circledast) \leq n-2.$$

So

$$\begin{aligned} p_A(x) &= (A_{11}-x)(A_{22}-x)\cdots(A_{nn}-x) + (\text{polyn. of deg } \leq n-2) = \\ &= (-1)^n x^n + A_{11}\cdot(-1)^{n-1}\cdot x^{n-1} + A_{22}(-1)^{n-1}x^{n-1} \\ &\quad + \cdots + A_{nn}(-1)^{n-1}x^{n-1} + \big(\text{another polyn. of deg } \leq n-2\big) = \\ &= (-1)^n x^n + \underbrace{(-1)^{n-1}\operatorname{Tr}(A)}_{=c_{n-1}} x^{n-1} + c_{n-2}x^{n-2} + \cdots + c_1 x + c_0. \end{aligned}$$

$$c_0 = p_A(0) = \det(A - 0\cdot I) = \det(A).$$

Substitute $x=0$. ☐

Definition (21,2,12). Let V be a vect. sp. over K, and $U_1,\ldots,U_r \leq V$ lin. subspaces. Denote $W := U_1+\cdots+U_r$. We say that W is a direct sum of $U_1,\ldots,U_r$ if every $w\in W$ has a unique representation as a sum

$$w = u_1+\cdots+u_r \qquad \text{with } u_i\in U_i\ \forall i.$$

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Remark. Exercise. Show that $U_1+\cdots+U_r$ is a direct sum iff any of the following conditions hold:

1) The only way to write 0 as a sum

$$0 = u_1+\cdots+u_r, \text{ with } u_i\in U_i$$

is as

$$0 = 0+\cdots+0.$$

2) $\forall\ 2\leq j\leq r$ we have

$$U_j \cap (U_1+\cdots+U_{j-1}) = \{0\}.$$

Under the additional assumption that $U_1,\ldots,U_r$ are f. dimensional, we have:

$$U_1+\cdots+U_r \text{ is a direct sum iff } \dim(U_1+\cdots+U_r) = \dim U_1+\cdots+\dim U_r.$$

$$\Leftrightarrow$$

wherever $B_1,\ldots,B_r$ are bases for $U_1,\ldots,U_r$ respectively, then $B_1,\ldots,B_r$ forms a basis for $U_1+\ldots+U_r$.

Remark. Exercise. If $U_1, \ldots, U_r$ is a direct sum, then $\exists$ a canonical isomorphism.

$$\begin{array}{ccc} U_1 \oplus \cdots \oplus U_r & \xrightarrow{\cong} & U_1 + \cdots + U_r \\ \Psi & & \Psi \\ (u_1, \ldots, u_r) & \longmapsto & u_1 + \cdots + u_r \end{array}$$

Proposition (21.a.13)**.** Let V be a vector space over K, and let $T \in \mathrm{End}(V)$. Let $\lambda_1, \ldots, \lambda_r \in K$ be pairwise distinct eigenvalues of T. Then $Eig_T(\lambda_1), \ldots, Eig_T(\lambda_r)$ is a direct sum.

Proof. Assume $u_1 + \ldots + u_r = u_1' + \ldots + u_r'(*)$ where $u_i, u_i' \in Eig_T(\lambda_i) \quad \forall i$. Rewrite $(*)$ as:

$$(u_1 - u_1') + \ldots + (u_r - u_r') = 0(**).$$

Remove/ignore in the sum $(**)$ all the summands $u_j - u_j'$ that are zero. $(**)$ becomes:

$$(u_{i_1} - u_{i_1}') + \ldots + (u_{i_s} - u_{i_s}') = 0(***),$$

where $0 \leq \ell, 1 \leq i_k \leq r \forall k$ and $u_{i_k} - u_{i_k}' \neq 0 \forall k$. Note that $0 \neq u_{i_k} - u_{i_k}' \in Eig_T(\lambda_{i_k}$. $\Rightarrow u_{i_k} - u_{i_k}'$ is an eigenvector λ_{i_k}. But $\lambda_{i_1}, \ldots, \lambda_{i_\ell}$ are pairwise distinct. By Proposition 21.a.4, $u_{i_1} - u_{i_\ell}'$ must be linear independent. This contradicts the $(***)$, unless $\ell = 0$, id est $u_j = u_j' \forall j$. □

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Corollary (21.a.14)**.** Assume V is finite-dimensional, $T : V \to V$ linear. Let $\lambda_1, \ldots, \lambda_r$ be all the eigenvalues of T. (we assume that $\lambda_i \neq \lambda_j \forall i \neq j$) Then T is diagonalizable iff

$$\sum_{i=1}^{r} \dim(\mathrm{Eig}_T(\lambda_i)) = \dim V(*).$$

Proof. Assume $(*) \Rightarrow T$ is diagonalizable. Choose a basis $\mathcal{B}_i$ for $\mathrm{Eig}_T(\lambda_i)$. We have seen that

$$W := \mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_r)$$

is a direct sum. $\Rightarrow (\mathcal{B}_1, \mathcal{B}_2, \ldots, \mathcal{B}_r)$ is a basis for W. $\Rightarrow \dim W = \sum_{i=1}^{r} |\mathcal{B}_i| = \sum_{i=1}^{r} \dim(\mathrm{Eig}_T(\lambda_i)) \underbrace{=}_{\text{by } (*)} \dim V$. $\Rightarrow W = V$. V has a basis consisting of eigenvectors of T. $\Rightarrow T$ is diagonalizable. T is diagonalizable $\Rightarrow (*)$. This is left as an exercise to the reader. □

Example.

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

$$p_A(x) = (1-x)^2 = (x-1)^2.$$

$$p_B(x) = \ldots = (x-1)^2.$$

Example. Pick $\lambda \in K$, consider

$$J := J_{\lambda,n} = \begin{pmatrix} \lambda & 1 & & & 0 \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & 1 \\ 0 & & & & \lambda \end{pmatrix}.$$

Jordan block.

$$p_{J_{\lambda,n}}(x) = (\lambda - x)^n.$$

The only eigenvalue of $J_{\lambda,n}$ is λ,

Note (Eigenvectors).

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$$J - \lambda \cdot I = \begin{pmatrix} 0 & 1 & & & 0 \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ 0 & & & & 0 \end{pmatrix}$$

The cols. $\#2, \ldots, \#n$ of $\uparrow$ are lin. indep. And we get that $\operatorname{rank}(J - \lambda I_n) = n - 1$.

$$\dim \ker(J - \lambda I) = 1,$$

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \text{ is an eigenvector of } J_{\lambda,n}.$$

$$\operatorname{Eig}_J(\lambda) = \operatorname{Sp}(e_1).$$

Proof. By assumption $\exists$ an invertible matrix P, s.t.

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} =: \Lambda$$

$$\Rightarrow \; p_A(x) = p_\Lambda(x) = \det \begin{pmatrix} \lambda_1 - x & & \\ & \ddots & \\ & & \lambda_n - x \end{pmatrix} =$$

$$= (\lambda_1 - x) \cdots (\lambda_n - x).$$

□

Remark.

$$\cancel{\det(xI - A)}$$
$$\det(A - xI)$$

Example. $K = \mathbb{R}, \quad A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$

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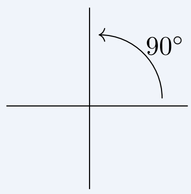
$$\begin{aligned} T_A : \mathbb{R}^2 &\longrightarrow \mathbb{R}^2 \\ (x, y) &\longmapsto (-y, x), \end{aligned} \qquad \nexists \text{ eigenvalues/vects,}$$

$$p_A(x) = \det \begin{pmatrix} -x & -1 \\ 1 & -x \end{pmatrix} = x^2 + 1, \qquad \nexists \text{ zeroes for } p_A(x) \text{ in } K = \mathbb{R}.$$

Hence $\nexists$ eigenvalues.



Lecture 38: Notes pending

25 Mar 2026

4.4 Geometric and algebraic multiplicities

We first need a quick digression on polynomials. Let $0 \neq p(x) \in K[x]$ and $\lambda \in K$. By dividing $p(x)$ by $(x - \lambda)$, we get can write

$$p(x) = (x - \lambda)q(x) + R,$$

where $q(x) \in K[x]$ with $\deg g = \deg p - 1$ and $R \in K$. The process by which one is able
38 to compute this manually is similar to integer long division. Notice that in that case, $p(\lambda) = R$.

Using this, we can conclude that

$$p(\lambda) = 0 \Leftrightarrow p(x) = (x - \lambda)q(x).$$

We say that λ is a zero (or root) of $p(x)$.

Sometimes λ is a zero also of $q(x)$, which means that we can write $q(x) = (x - \lambda)h(x)$ with $h(x)$ such that $\deg h = \deg p - 2$. This motivates the following definition.

Definition (Multiplicity)**.** Let $0 \neq p(x) \in K[x]$ and $\lambda \in K$. We say that λ is a zero of multiplicity $m \in \mathbb{Z}_{\geq 1}$ if $p(x) = (x-\lambda)^m g(x)$, where $g(x) \in K[x]$ and $g(\lambda) \neq 0$.

If $m = 1$, we say λ is a simple zero of $p(x)$. If $m \geq 2$ we say that λ is a multiple zero.

Definition (Geometric and algebraic multiplicity)**.** Let V be a vector space over K and $T : V \to V$ linear and $\lambda \in K$ an eigenvalue of T. We define

$$m_g(T, \lambda) := \dim \mathrm{Eig}_T(\lambda).$$

Assume, in addition, that $\dim V < \infty$ and define the algebraic multiplicity of λ to be multiplicity of the zero λ in $P_T(x)$. We denote this number $m_a(T, \lambda)$.

Proposition (Inequality for multiplicities)**.** If V is finite dimensional, it always holds $m_g(T, \lambda) \leq m_a(T, \lambda)$.

Proof. Let $v_1, \dots, v_k$ be a basis for $\mathrm{Eig}_T(\lambda)$, so $k = m_g(T, \lambda)$. Extend this to a basis $\mathcal{B} = (v_1, \dots, v_k, v_{k+1}, \dots, v_n)$ of V.

It's not hard to realise that $[T]_{\mathcal{B}}^{\mathcal{B}}$ will be a block matrix

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|c} \Lambda & * \\ \hline 0 & C \end{array}\right)$$

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where $\Lambda = \lambda \cdot I_k$ and C is $n-k \times n-k$ and could be anything. Then, we have

$$P_T(x) = P_{[T]_{\mathcal{B}}^{\mathcal{B}}} = P_\Lambda(x) \cdot P_C(x) = (\lambda - x)^k \cdot P_C(x).$$

Hence $m_a(T, \lambda) \geq k = m_g(T, \lambda)$. $\square$

Example. Consider the Jordan block $J = J_{\lambda,n}$. We saw that $P_J(x) = (\lambda - x)^n$. We also saw that $\mathrm{Eig}_J(\lambda)$ is 1-dimensional. This gives $m_a(J, \lambda) = n$ and $m_g(J, \lambda) = 1$. In a way, this is the most extreme example, m_g being minimal and m_a maximal.

4.5 Diagonalisability

Now the goal is to find clear criteria for diagonalisability.

Theorem (Broader criteria for diagonalisation)**.** Let V be an n-dimensional vector space over K and $T \in \mathrm{End}(V)$. The following statements are equivalent.

1. T is diagonalisable.
2. $\exists$ basis for V whose elements are eigenvectors.
3. $P_T(x)$ splits into a product of linear factors and $m_g(T, \lambda) = m_a(T, \lambda)$.

4. $\sum_{i=1}^k \dim \mathrm{Eig}_T(\lambda_i) = \dim V$.

5. $V \cong \bigoplus_{i=1}^k \mathrm{Eig}_T(\lambda_i)$.

Proof. (1) $\Leftrightarrow$ (2) has been proven already.

(1) $\Rightarrow$ (3). Let $\mathcal{B}$ be a basis in which $[T]_{\mathcal{B}}^{\mathcal{B}} = \Lambda$ is diagonal. By changing the order of the elements of $\mathcal{B}$, we can arrange that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \mathrm{diag}\big(\underbrace{\lambda_1, \dots, \lambda_1}_{\times \ell_1}, \underbrace{\lambda_2, \dots, \lambda_2}_{\times \ell_2}, \dots, \underbrace{\lambda_k, \dots, \lambda_k}_{\times \ell_k}\big)$$

where $\lambda_1, \dots, \lambda_k$ are the eigenvalues of T, and

$$\lambda_i \neq \lambda_j \qquad \forall\, i \neq j.$$

Write $\mathcal{B} = (v_1^{(1)}, \dots, v_{\ell_1}^{(1)}, v_1^{(2)}, \dots, v_{\ell_2}^{(2)}, \dots, v_1^{(k)}, \dots, v_{\ell_k}^{(k)})$,

$$\Rightarrow \; p_T(x) = \det\big([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I\big) = (\lambda_1 - x)^{\ell_1} \cdots (\lambda_k - x)^{\ell_k}. \qquad (*)$$

So $p_T(x)$ splits as a product of linear terms. It remains to show that $m_g(T, \lambda_i) = m_a(T, \lambda_i)$ for all i. Clearly, $v_1^{(i)}, \dots, v_{\ell_i}^{(i)} \in \mathrm{Eig}_T(\lambda_i)$ are linearly independent, so

$$\ell_i \leq \dim \mathrm{Eig}_T(\lambda_i) = m_g(T, \lambda_i).$$

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But $\ell_i = m_a(T, \lambda_i)$, so

$$m_a(T, \lambda_i) \leq m_g(T, \lambda_i).$$

But we also always have $m_g(T, \lambda_i) \leq m_a(T, \lambda_i)$. Therefore

$$m_g(T, \lambda_i) = m_a(T, \lambda_i).$$

(3) $\Rightarrow$ (4). Assume

$$p_T(x) = (\lambda_1 - x)^{\ell_1} \cdots (\lambda_k - x)^{\ell_k},$$

and

$$m_g(T, \lambda_i) = m_a(T, \lambda_i) \qquad \forall\, i.$$

Clearly $\lambda_1, \dots, \lambda_k$ are all the eigenvalues of T because these are all the zeros of $p_T(x)$. Also by definition $\ell_i = m_a(T, \lambda_i)$, so

$$n = \dim V = \deg p_T(x) = \ell_1 + \cdots + \ell_k = \sum_{i=1}^k m_a(T, \lambda_i) \overset{\text{by assumption}}{=} \sum_{i=1}^k m_g(T, \lambda_i) = \sum_{i=1}^k \dim \mathrm{Eig}_T(\lambda_i).$$

(4) $\Rightarrow$ (5). Recall that

$$\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_k) \subseteq V$$

is a direct sum.

$$\sum_{i=1}^{k} \dim \operatorname{Eig}_T(\lambda_i) = \dim\big(\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k)\big)$$

By assumption, the left-hand side equals $\dim V$. Hence

$$\dim\big(\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k)\big) = \dim V.$$

Since

$$\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k)$$

is a subspace of V, we get that

$$\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k) = V.$$

But since it is also isomorphic to

$$\operatorname{Eig}_T(\lambda_1) \oplus \cdots \oplus \operatorname{Eig}_T(\lambda_k),$$

it is a direct sum. Hence

$$V \cong \bigoplus_{i=1}^{k} \operatorname{Eig}_T(\lambda_i).$$

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(5) $\Rightarrow$ *(1).* If

$$V \cong \operatorname{Eig}_T(\lambda_1) \oplus \cdots \oplus \operatorname{Eig}_T(\lambda_k),$$

then, since

$$\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k) \subseteq V$$

is a direct sum, we must have that

$$\operatorname{Eig}_T(\lambda_1) + \cdots + \operatorname{Eig}_T(\lambda_k) = V.$$

So, if

$$v_1^{(i)}, \ldots, v_{\ell_i}^{(i)} \text{ is a basis for } \operatorname{Eig}_T(\lambda_i),$$

then

$$\big(v_1^{(1)}, \ldots, v_{\ell_1}^{(1)}, \ldots, v_1^{(k)}, \ldots, v_{\ell_k}^{(k)}\big)$$

is a basis for V whose elements are eigenvectors of T. □

Corollary (21.C.2)**.** If $p_T(x)$ splits as a product of linear factors and each eigenvalue λ has $m_a(T,\lambda) = 1$, then T is diagonalizable.

Proof. We have $1 \leq m_g(T,\lambda) \leq m_a(T,\lambda) = 1$.

$$\Rightarrow \forall \text{ eigenvalue } \lambda, \qquad m_g(T,\lambda) = m_a(T,\lambda),$$

The corollary follows from the previous theorem. □

4.6 Trigonalization

This means bringing a matrix/endomorphism to upper triangular form. For example,

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad p_A(x) = (x-1)^2.$$

$$m_a(A,1) = 2, \qquad m_g(A,1) = 1.$$

Definition (21.C.3). We say that $T \in \mathrm{End}(V)$ is trigonalizable if $\exists$ a basis $\mathcal{B}$ for V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix} \qquad (n = \dim V).$$

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Theorem (21.C.4). Let V be a vector space over K and $T \in \mathrm{End}(V)$. Then T is trigonalizable iff

$$p_T(x) \text{ splits as } p_T(x) = (\lambda_1 - x)^{\ell_1} \cdots (\lambda_k - x)^{\ell_k}.$$

Note. Digression.

Theorem (Fundamental theorem of algebra (21.C.5)). Every $p(z) \in \mathbb{C}[x]$ (polynomial with $\mathbb{C}$-coefficients) splits as a product of linear factors and a constant factor, i.e.

$$p(z) = c(z-a_1)^{n_1} \cdots (z-a_r)^{n_r},$$

with $c \in \mathbb{C}$, $r \geq 0$, $a_j \in \mathbb{C}$, $n_j \in \mathbb{Z}_{\geq 1}$.

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Corollary. Let V be a finite-dimensional vector space over $\mathbb{C}$. Then every $T \in \mathrm{End}(V)$ is trigonalizable.

Proof of Theorem (21.C.4). $\Rightarrow$. Suppose T is trigonalizable. Let $\mathcal{B}$ be a basis for V such that

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$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ & \ddots & \ddots & \vdots \\ 0 & & \ddots & * \\ & & & \lambda_n \end{pmatrix}.$$

Then

$$p_T(x) = \det\big([T]_{\mathcal{B}}^{\mathcal{B}} - xI\big) = \det \begin{pmatrix} \lambda_1 - x & * & \cdots & * \\ & \ddots & \ddots & \vdots \\ 0 & & \ddots & * \\ & & & \lambda_n - x \end{pmatrix} = (\lambda_1 - x)\cdots(\lambda_n - x).$$

$\Leftarrow$. Induction on $n := \dim V$.

$\underline{n = 1.}$ In this case $[T]_{\mathcal{B}}^{\mathcal{B}} = (\lambda)$ (for any basis), and this is diagonal. (Nothing to prove).

$\underline{\text{Let } n \geq 1.}$ Assume that the theorem holds for all vector spaces of dimension $\leq n$ and all their endomorphisms. Let V be an $(n+1)$-dimensional vector space, and $T \in \operatorname{End}(V)$, s.t. $p_T(x) =$ product of linear factors.

$$\Rightarrow \exists \text{ one eigenvalue } \lambda_1 \in K \text{ of } T.$$

Choose an eigenvector, $0 \neq u_1 \in \ker(T - \lambda_1 \operatorname{id}_V)$. Extend u_1 to a basis $\mathcal{B}' = (u_1, u_2, \_, u_{n+1})$ of V. We have

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = \begin{pmatrix} \lambda_1 & \\ 0 & \\ \vdots & \boxed{\;*\;} \\ 0 & \end{pmatrix} = \left(\begin{array}{c|ccc} \lambda_1 & * & \cdots & * \\ \hline 0 & & & \\ \vdots & & A & \\ 0 & & & \end{array}\right) =: M,$$

with A the $n \times n$ block.

We have

$$p_T(x) = (\lambda_1 - x) \cdot p_A(x).$$

Since $p_T(x)$ splits as a product of linear factors, so does $p_A(x)$.

Consider $S_A : K^n \to K^n$, $S_A(u) = Au$. By the induction hypothesis $\exists$ a basis $\mathcal{E}$ of K^n s.t.

$$[S_A]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} \lambda_2 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_{n+1} \end{pmatrix}.$$

Denote by $\mathcal{E}$ the standard basis of K^n.

We have

$$p_T(x) = (\lambda_1 - x) \cdot p_A(x).$$

Since $p_T(x)$ splits as a product of linear factors, so does $p_A(x)$.

Consider $S_A : K^n \to K^n,\ S_A(u) = Au$. By the induction hypothesis $\exists$ a basis $\mathcal{E}$ of

K^n s.t.

$$[S_A]^{\mathcal{E}}_{\mathcal{E}} = \begin{pmatrix} \lambda_2 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_{n+1} \end{pmatrix}.$$

Denote by $\mathcal{C}$ the standard basis of K^n.

$$[S_A]^{\mathcal{E}}_{\mathcal{E}} = [\mathrm{id}]^{\mathcal{E}}_{\mathcal{C}}\,[S_A]^{\mathcal{C}}_{\mathcal{C}}\,[\mathrm{id}]^{\mathcal{C}}_{\mathcal{E}} = \underbrace{[\mathrm{id}]^{\mathcal{E}}_{\mathcal{C}}}_{Q^{-1}}\,\underbrace{[S_A]^{\mathcal{C}}_{\mathcal{C}}}_{A}\,\underbrace{[\mathrm{id}]^{\mathcal{C}}_{\mathcal{E}}}_{\text{denote by } Q}.$$

So

$$Q^{-1}AQ = \begin{pmatrix} \lambda_2 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_{n+1} \end{pmatrix}.$$

Define

$$P := \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & Q & \\ 0 & & & \end{array}\right) \in M_{(n+1)\times(n+1)}(K).$$

39 **Claim.**

1) P is invertible and

$$P^{-1} = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & Q^{-1} & \\ 0 & & & \end{array}\right)$$

2)

$$P^{-1}MP = \left(\begin{array}{c|ccccc} \lambda_1 & *' & \cdots & \cdots & *' \\ \hline 0 & \lambda_2 & * & \cdots & * \\ \vdots & 0 & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & * \\ 0 & 0 & \cdots & 0 & \lambda_{n+1} \end{array}\right)$$

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Proof of claim. For (1),

$$P\cdot\left(\begin{array}{c|ccc}1&0&\cdots&0\\\hline 0&&&\\\vdots&&Q^{-1}&\\0&&&\end{array}\right)=\left(\begin{array}{c|ccc}1&0&\cdots&0\\\hline 0&&&\\\vdots&&Q\cdot Q^{-1}&\\0&&&\end{array}\right)=I.$$

Therefore

$$\left(\begin{array}{c|ccc}1&0&\cdots&0\\\hline 0&&&\\\vdots&&Q^{-1}&\\0&&&\end{array}\right)=P^{-1}.$$

For (2),

$$\begin{aligned}P^{-1}MP&=\left(\begin{array}{c|ccc}1&0&\cdots&0\\\hline 0&&&\\\vdots&&Q^{-1}&\\0&&&\end{array}\right)\left(\begin{array}{c|cccc}\lambda_1&*'&\cdots&\cdots&*'\\\hline 0&&&&\\\vdots&&A&&\\0&&&&\end{array}\right)\left(\begin{array}{c|ccc}1&0&\cdots&0\\\hline 0&&&\\\vdots&&Q&\\0&&&\end{array}\right)\\&=\left(\begin{array}{c|ccc}1&0&\cdots&0\\\hline 0&&&\\\vdots&&Q^{-1}&\\0&&&\end{array}\right)\left(\begin{array}{c|cccc}\lambda_1&*'&\cdots&\cdots&*'\\\hline 0&&&&\\\vdots&&AQ&&\\0&&&&\end{array}\right)\\&=\left(\begin{array}{c|cccc}\lambda_1&*'&\cdots&\cdots&*'\\\hline 0&&&&\\\vdots&&Q^{-1}AQ&&\\0&&&&\end{array}\right)\\&=\left(\begin{array}{c|cccc}\lambda_1&*'&\cdots&\cdots&*'\\\hline 0&\lambda_2&*&\cdots&*\\\vdots&0&\ddots&\ddots&\vdots\\\vdots&\vdots&\ddots&\ddots&*\\0&0&\cdots&0&\lambda_{n+1}\end{array}\right).\end{aligned}$$

Claim. $\exists$ a basis for V, s.t.

$$[\mathrm{id}]^{\mathcal{B}}_{\mathcal{B}'}=P.$$

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Proof of claim. Write

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1k} & \cdots & p_{1m} \\ \vdots & \vdots & & \vdots & & \vdots \\ \vdots & & \ddots & & & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mk} & \cdots & p_{mm} \end{pmatrix}, \qquad (m := n+1 = \dim V).$$

Write $\mathcal{B}'$ as $\mathcal{B}' = (u_1, \ldots, u_m)$. Define

$$\begin{aligned} v_1 &= p_{11}u_1 + p_{21}u_2 + \cdots + p_{m1}u_m, \\ v_2 &= p_{12}u_1 + p_{22}u_2 + \cdots + p_{m2}u_m, \\ &\vdots \\ v_k &= p_{1k}u_1 + p_{2k}u_2 + \cdots + p_{mk}u_m, \\ &\vdots \\ v_m &= p_{1m}u_1 + p_{2m}u_2 + \cdots + p_{mm}u_m. \end{aligned} \tag{*}$$

Exercise. $\mathcal{B} := (v_1, \ldots, v_m)$ is a basis for V.

Hint. Since $\dim V = m$, it is enough to show $v_1, \ldots, v_m$ are linearly independent. Suppose

$$c_1v_1 + \cdots + c_mv_m = 0. \tag{**}$$

Substitute $(*)$ into $(**)$, and obtain

$$P \cdot \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

But P is invertible,

$$\Rightarrow \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

This will prove claim B.

We now have

$$[T]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} [T]_{\mathcal{B}'}^{\mathcal{B}'} [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} = \underbrace{[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}}}_{P^{-1}} \underbrace{[T]_{\mathcal{B}'}^{\mathcal{B}'}}_{M} \underbrace{[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}}_{P} = P^{-1}MP.$$

By claim A,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_{n+1} \end{pmatrix}.$$

□

4.7 Minimal polynomial & Cayley–Hamilton theorem

Definition (Polynomial in a matrix). Let $A \in M_{n\times n}(K)$. For $\forall\, k \geq 1$, define

$$A^k := \underbrace{A \cdot \ldots \cdot A}_{\times k}.$$

Let

$$g(x) = a_d x^d + \cdots + a_1 x + a_0 \in K[x].$$

Define

$$g(A) := a_d A^d + a_{d-1} A^{d-1} + \cdots + a_1 A + a_0 I,$$

where $I = I_n$ is the identity matrix.

Claim.

$$\forall\, A \in M_{n\times n}(K),\ \exists\, g \in K[x],\ g \neq 0,\ \text{s.t. } g(A) = 0_{n\times n}.$$

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Proof. Consider

$$I_n, A, A^2, \ldots, A^r \in M_{n\times n}(K).$$

But

$$\dim M_{n\times n}(K) = n^2.$$

So, if $r > n^2$, the list

$$I_n, A, A^2, \ldots, A^r$$

must be linearly dependent. Therefore

$$\exists\, a_0, \ldots, a_r \in K,\ \text{not all zero, s.t.}\ \ a_0 I_n + a_1 A + \cdots + a_r A^r = 0.$$

So take

$$g(x) = a_r x^r + \cdots + a_1 x + a_0.$$

Then $g \neq 0$ and $g(A) = 0_{n\times n}$. □

Note. Everything here applies also for $T \in \mathrm{End}(V)$. Define

$$T^0 := \mathrm{id}_V, \qquad T^k := \underbrace{T \circ \cdots \circ T}_{\times k}.$$

Then we can define

$$g(T) := a_d T^d + a_{d-1} T^{d-1} + \cdots + a_1 T + a_0\, \mathrm{id}_V\,.$$

Also, if V is finite-dimensional, then

$$\dim \mathrm{End}(V) = (\dim V)^2.$$

Definition (21.d.1)**.** Let $T \in \mathrm{End}(V)$. A minimal polynomial for T is a polynomial

$$0 \neq g(x) \in K[x]$$

of minimal possible degree, s.t. $g(T) = 0$.

Theorem (Cayley–Hamilton, 21.d.2)**.** 1) Let $A \in M_{n\times n}(K)$. Then

$$p_A(A) = 0.$$

2) Let V be a finite-dimensional vector space over K, and $T \in \mathrm{End}(V)$. Then

$$p_T(T) = 0.$$

Example. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then

$$p_A(x) = \det \begin{pmatrix} a-x & b \\ c & d-x \end{pmatrix} = (a-x)(d-x) - bc = x^2 - (a+d)x + (ad-bc),$$

where $a + d = \mathrm{Tr}(A)$ and $ad - bc = \det(A)$. Hence

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$$p_A(A) = A^2 - (a+d)A + (ad-bc)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

"Easy proof" of CH[1] (21.d.2):

[1]:set theorists hate this trick

$$p_A(x) = \det(A - xI),$$

so

$$p_A(A) = \det(A - A \cdot I) = \det(0) = 0.$$

Analytic proof. Assume first that $K = \mathbb{C}$. Prove 1) first for $A \in M_{n\times n}(\mathbb{C})$ which is diagonalizable. Indeed, if $\exists$ an invertible matrix Q, s.t.

$$Q^{-1}AQ = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix} =: \Lambda,$$

and $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ are the eigenvalues of A. Write

$$p_A(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0.$$

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Then

$$\begin{aligned}
Q^{-1}p_A(A)Q &= c_n \underbrace{Q^{-1}A^nQ}_{(Q^{-1}AQ)^n} + c_{n-1}\underbrace{Q^{-1}A^{n-1}Q}_{(Q^{-1}AQ)^{n-1}} + \cdots + c_1Q^{-1}AQ + c_0I \\
&= c_n(Q^{-1}AQ)^n + c_{n-1}(Q^{-1}AQ)^{n-1} + \cdots + c_1(Q^{-1}AQ) + c_0I \\
&= p_A(Q^{-1}AQ) = p_A(\Lambda) \qquad \text{(check)} \\
&= \begin{pmatrix} p_A(\lambda_1) & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & p_A(\lambda_n) \end{pmatrix} = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 \end{pmatrix},
\end{aligned}$$

because $p_A(\lambda_i) = 0 \ \forall\, i$. $\square$

TODO: Add biran quote rambling rant here!!!

Lecture 40: Notes pending

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4.8 Lecture 40 Placeholder

Notes for this lecture will be added here.

Lecture 41: Notes pending

13 Apr 2026

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4.9 Lecture 41 Placeholder

Notes for this lecture will be added here.

Lecture 42: Notes pending

15 Apr 2026

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4.10 Lecture 42 Placeholder

Notes for this lecture will be added here.